MATHEMATICS

CONTINUITY & DIFFERENTIBILITY

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CONTINUITY & DIFFERENTIBILITY

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JEE EXERCISES

Exercise J-1

CONTINUITY CHECK

1. f is a continuous function on the real line. Given that

$x^2 + (f(x) - 2)\,x - \sqrt{3}\cdot f(x) + 2\sqrt{3} - 3 = 0$. Then the value of $f(\sqrt{3})$:

(A) can not be determined (B) $2(1-\sqrt{3})$

(C) zero (D) $\dfrac{2\left(\sqrt{3}-2\right)}{\sqrt{3}}$

यदि प्रत्येक वास्तविक संख्या के लिये f एक सतत् फलन हो तथा दिया है कि

$x^2 + (f(x) - 2)\,x - \sqrt{3}\cdot f(x) + 2\sqrt{3} - 3 = 0$ तो $f(\sqrt{3})$ का मान होगा :

(A) ज्ञात नही किया जा सकता (B) $2(1-\sqrt{3})$

(C) शून्य (D) $\dfrac{2\left(\sqrt{3}-2\right)}{\sqrt{3}}$

Ans. B

Sol. $(x-\sqrt{3})\,f(x) = -x^2 + 2x - 2\sqrt{3} + 3$

$$f(x) = \frac{-x^2 + 2x - 2\sqrt{3} + 3}{x - \sqrt{3}}$$

$$= \frac{(x-\sqrt{3})(2-\sqrt{3}-x)}{x-\sqrt{3}} = 2 - \sqrt{3} - x$$

2. The function $f(x) = [x]^2 - [x^2]$ (where $[y]$ is the greatest integer less than or equal to y), is discontinuous at :

(A) all integers (B) all integers except 0 & 1

(C) all integers except 0 (D) all integers except 1

फलन $f(x) = [x]^2 - [x^2]$ कहाँ – कहाँ असतत् होगा (जहाँ $[y]$ महत्तम पूर्णांक फलन है y) :

(A) सभी पूर्णांको पर (B) 0 व 1 को छोड़कर सभी पूर्णांको पर

(C) 0 के छोड़कर सभी पूर्णांको पर (D) 1 को छोड़कर सभी पूर्णांको पर

Ans. D

Sol. $f(x) = [x]^2 - [x^2]$

$(-1)^2 - 0 = 1$ $x = 0 - h$

$$f(x)=\begin{cases}0 & x=0 & [x]=-1\\ 0-0=0 & 0<x<1 & [x^2]=0\\ 1^2-1=0 & 1\le x<\sqrt{2} & 1\le x^2<2\\ 1-2=-1 & \sqrt{2}\le x<\sqrt{3} & [x]=1\\ 1-3=-2 & \sqrt{3}\le x<2 & 2\le x2<3\end{cases}$$

i.e. it is continuous at x = 1 only and discontinuous at all other points]

3. Let $g(x)=\tan^{-1}|x|-\cot^{-1}|x|$, $f(x)=\frac{[x]}{[x+1]}\{x\}$, $h(x)=|g(f(x))|$ where {x} denotes fractional part and [x] denotes the integral part then which of the following holds good :

(A) h is continuous at x = 0 (B) h is discontinuous at x = 0

(C) $h(0^-)=\pi/2$ (D) $h(0^+)=-\pi/2$

माना कि $g(x)=\tan^{-1}|x|-\cot^{-1}|x|$, $f(x)=\frac{[x]}{[x+1]}\{x\}$, $h(x)=|g(f(x))|$ जहाँ {x} एक अपूर्णांश फलन है तथ्य [x] एक महत्तम पूर्णांक फलन हो, तो निम्न में से कौन सा विकल्प सत्य है :

(A) x = 0 पर h सतत् है। (B) x = 0 पर h असतत् फलन है।

(C) $h(0^-)=\pi/2$ (D) $h(0^+)=-\pi/2$

Ans. A

Sol. $g(x)=\tan^{-1}|x|-\cot^{-1}|x|$

$$f(x)=\frac{[x]}{[x+1]}\{x\};\ x\in[-1,0)$$

$g(x)=|g(f(x))|$

$$h(0^+)=\lim_{\alpha\to0}h(0+\alpha)=\lim_{\alpha\to0}|g(f(\alpha))|$$

$$|g(0)|=\left|0-\frac{\pi}{2}\right|=\frac{\pi}{2}$$

$$h(0)=|g(f(0))|=|g(0)|=\frac{\pi}{2}$$

$\Rightarrow$ h is 0 at x = 0

4. The function $f(x)=[x].\cos\frac{2x-1}{2}\pi$, where [•] denotes the greatest integer function, is discontinuous at :

(A) all x (B) all integer points (C) no x (D) x which is not an integer

दिया है कि $f(x)=[x].\cos\frac{2x-1}{2}\pi$, जहाँ [•] एक महत्तम पूर्णांक फलन हो, तो फलन f(x) असतत् होगा :

(A) सभी x पर (B) सभी पूर्णांक संख्या पर

(C) किसी भी x पर नहीं (D) सभी x पर जो पूर्णांक नही हो

Ans. C

Sol. $f(x)=[x]\cdot\cos\left(\left(\frac{2x-1}{2}\right)\pi\right)$

$\Rightarrow f(x)=[x]\cdot\cos\left(\pi x-\frac{\pi}{2}\right)$

$f(x)=[x]\sin\pi x$

Let $x=k\in Z$

$\lim\limits_{x\to k} f(x)$

$\text{L.H.L.}=\lim\limits_{h\to 0} f(k-h)=\lim\limits_{h\to 0}[k-h]\sin(\pi k-\pi h)=0$

$\text{RHL}=\lim\limits_{h\to 0} f(k+h)=\lim\limits_{h\to 0}[k+h]\sin(\pi k+\pi h)=0$

$f(k)=[k]\sin\pi k=0$

$\therefore$ f(x) is continuous $\forall\ x\in R$

5. Consider the functions

$f(x)=\text{sgn}(x-1)$ and $g(x)=\cot^{-1}[x-1]$

where [ ] denotes the greatest integer function :

Statement-1 : The function $F(x)=f(x)\cdot g(x)$ is discontinuous at $x=1$.

because

Statement-2 : If f(x) is discontinuous at $x=a$ and g(x) is also discontinuous at $x=a$ then the product function f(x) · g(x) is discontinuous at $x=a$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true

दिया है

$f(x)=\text{sgn}(x-1)$ और $g(x)=\cot^{-1}[x-1]$

जहाँ [ ] महत्तम पूर्णांक फलन है :

कथन-1 : फलन $F(x)=f(x)\cdot g(x)$, $x=1$ असतत् होगा

क्योंकि

कथन-2 : यदि $x=a$ पर फलन f(x) असतत् है तथा g(x) भी $x=a$ पर असतत् है तो उनका गुणनफल

$f(x)\cdot g(x)$ भी $x=a$ पर असतत् होगा

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. C

Sol. $f(x) = \text{sgn}(x-1)$ $\quad g(x) = \cot^{-1}[x-1]$

$F(x) = f(x) \cdot g(x)$

$= \text{sgn}(x-1) \cdot \cot^{-1}[x-1]$

$F(1) = 0$

$$\text{L.H.L.} = \lim_{h\to 0} F(1-h) = \lim_{h\to 0} \text{sgn}(-h)\cot^{-1}[-h]$$

$= -\pi$

$$\text{R.H.L.} = \lim_{h\to 0} F(1+h) = \lim_{h\to 0} \text{sgn}(h)\cot^{-1}[h] = \frac{\pi}{2}$$

$\therefore$ F(x) is discontinuous at x = 1.

Statement - 2

$$F(x) = \left\{\begin{matrix} \frac{1}{2} & 0 \le x \le 1 \\ 2 & 1 < x \le 2 \end{matrix}\right\}$$

$$g(x) = \left\{\begin{matrix} 2 & 0 \le x \le 1 \\ \frac{1}{2} & 1 < x \le 2 \end{matrix}\right\}$$

$\rightarrow$ F(x) & g(x) are discontinuous at x = 1 but $f(x) \cdot g(x) = 1 \ \forall\ x \in [0,2]$

$\Rightarrow$ F(x) is discontinuous at x = 1

TYPES OF DISCONTINUITY

6. $$f(x) = \begin{cases} \sin\left(\frac{a-x}{2}\right)\tan\left[\frac{\pi x}{2a}\right] & \text{for } x > a \\ \frac{\left[\cos\left(\frac{\pi x}{2a}\right)\right]}{a-x} & \text{for } x < a \end{cases}$$

where [x] is the greatest integer function of x, and $a > 0$, then :

(A) $f(a^-) < 0$ (B) f has a removable discontinuity at x = a

(C) f has an irremovable discontinuity at x=a (D) $f(a^+) < 0$

$$f(x)=\begin{cases}\sin\left(\dfrac{a-x}{2}\right)\tan\left[\dfrac{\pi x}{2a}\right] & \text{for } x>a\\ \dfrac{\left[\cos\left(\dfrac{\pi x}{2a}\right)\right]}{a-x} & \text{for } x<a\end{cases}$$

जहाँ [x] एक महत्तम पूर्णांक फलन हो, तथा $a>0$ है तब :

(A) $f(a^-)<0$

(B) $x=a$ पर f विस्थापनीय बिन्दु असत्ता प्रदर्शित करता है

(C) $x=a$ पर f अविस्थापनीय बिन्दु असत्ता प्रदर्शित करता है

(D) $f(a^+)<0$

Ans. B

Sol. $$f(x)=\begin{cases}\sin\left(\dfrac{a-x}{2}\right)\tan\left[\dfrac{\pi x}{2}\right] & \text{For } x>a\\ \dfrac{\left[\cos\left(\dfrac{\pi x}{2a}\right)\right]}{a-x} & \text{For } x<a\end{cases}$$

$$\lim_{h\to 0} f(a+h)$$

$$=\lim_{h\to 0}\sin\left(\frac{a-(a+h)}{2}\right)\tan\left[\frac{\pi a}{2a}+\frac{\pi h}{2a}\right]$$

$$\lim_{h\to 0}-\sin\left(\frac{h}{2}\right)\cdot\tan\left[\frac{\pi}{2}+\frac{h}{2a}\right]=0$$

$$\lim_{h\to 0} f(a-h)=\lim_{h\to 0}\sin\left(\frac{h}{2}\right)\tan\left[\frac{\pi}{2}-\frac{\pi h}{2a}\right]$$

$=0$

$\Rightarrow$ f has a removbable discontinuity at $x=a$.

DIFFERENTIABILITY CHECK

7. Let $f: R\to R$ be a function defined by $f(x)=\text{Min}\{x+1, |x|+1\}$. Then which of the following is true :

(A) $f(x)\geq 1$ for all $x\in R$ (B) $f(x)$ is not differentiable at $x=1$

(C) $f(x)$ is differentiable everywhere (D) $f(x)$ is not differentiable at $x=0$

माना $f: R\to R$ एक फलन है जो $f(x)=\text{Min}\{x+1, |x|+1\}$ से परिभाषित है, तब निम्न में से कौनसा सही है :

(A) $f(x)\geq 1$ सभी $x\in R$ के लिये (B) $f(x)$, $x=1$ पर अवकलनीय नहीं है

(C) $f(x)$, प्रत्येक बिन्दु पर अवकलनीय है (D) $f(x)$, $x=0$ पर अवकलनीय नहीं है

Ans. C

Sol. $f(x) = \text{Min}\left[x+1, |x|+1\right]$

$$= \begin{cases} x+1 & \text{if} \quad x \geq 0 \\ x+1 & \text{if} \quad x < 0 \end{cases}$$

$\Rightarrow f(x) = x + 1$

$\therefore$ f(x) is differentiable everywhere

8. Let $f(x) = \begin{cases} (x-1)\sin\dfrac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$. Then which one of the following is true :

(A) f is differentiable at x = 0 and at x = 1 (B) f is differentiable at x = 0 but not at x = 1

(C) f is differentiable at x = 1 but not at x = 0 (D) f is neither differentiable at x = 0 nor at x = 1

माना $f(x) = \begin{cases} (x-1)\sin\dfrac{1}{x-1} & \text{if } x \neq 1 \\ 0 & \text{if } x = 1 \end{cases}$ तो निम्न में से कौनसा सत्य है :

(A) f , x = 0 तथा x = 1 पर अवकलनीय है (B) f , x = 0 पर अवकलनीय है लेकिन x = 1 पर नहीं

(C) f , x = 1 पर अवकलनीय है लेकिन x = 0 पर नहीं (D) f , न तो x = 0 पर तथा न ही x = 1 पर अवकलनीय है

Ans. B

Sol. $f(x) = \begin{cases} (x-1)\sin\dfrac{1}{(x-1)} & \text{if} \quad x \neq 1 \\ 0 & x = 1 \end{cases}$

$\lim\limits_{x\to 1} f(x) = 0 = f(1)$

$\therefore$ f(x) will be continuous at x = 1.

$$\text{L.H.D.} = \lim_{h\to 0} \frac{f(1-h)-f(1)}{-h} = \lim_{h\to 0} \frac{\text{HSIN}\dfrac{1}{h}}{-h} = \text{D.N.E.}$$

$$\text{R.H.D.} = \lim_{h\to 0} \sin\left(\frac{1}{h}\right) = \text{DNE}$$

$\therefore$ f(x) is not differentiable at x = 1.

but f(x) is differentiable at x = 0.

9. Let $g(x) = \dfrac{(x-1)^n}{\ell n \cos^m(x-1)}$; $0 < x < 2$, m and n are integers, $m \neq 0$, $n > 0$ and let p be the left hand derivative of $|x-1|$ at x = 1. If $\lim\limits_{x\to 1^+} g(x) = p$, then :

(A) n = 1, m = 1 (B) n = 1, m = –1 (C) n = 2, m = 2 (D) n > 2, m = n

यदि $g(x)=\dfrac{(x-1)^n}{ln\cos^m(x-1)}$; $0<x<2$, m तथा n पूर्णांक है , $m\neq 0$, $n>0$. माना कि $|x-1|$ का $x=1$ पर वाम पक्ष अवकलज (LHD) p है, तो यदि $\lim\limits_{x\to 1^+} g(x)=p$, तो :

(A) $n=1, m=1$ (B) $n=1, m=-1$ (C) $n=2, m=2$ (D) $n>2, m=n$

Ans. C

Sol. $p=-1$ Now

$$\lim_{x\to 1^+}\frac{(x-1)^n}{\log\cos^m(x-1)}=\lim_{x\to 1^+}\frac{(x-1)^n}{\log[\cos^m(x-1)-1+1]}$$

$$=\lim_{x\to 1^+}\frac{(x-1)^n}{\cos^m(x-1)-1}$$

$$=\lim_{x\to 1^+}\frac{n(x-1)^{n-1}}{m\cos^{m-1}(x-1)\sin(x-1)}$$

$$=\frac{-n}{m}\lim_{x\to 1^+}\frac{(x-1)}{\sin(x-1)}\cdot\frac{1}{\cos^{m-1}(x-1)}\times(x-1)^{n-2}$$

$$=\frac{-n}{m}\lim_{x\to 1^+}(x-1)^{n-2}=-1 \text{ (Given)}$$

$\Rightarrow$ $n=2$ and $m=2$

10. For what triplets of real numbers (a, b, c) with $a\neq 0$ the function

$f(x)=\begin{cases} x & x\leq 1 \\ ax^2+bx+c & \text{otherwise}\end{cases}$ is differentiable for all real x :

यदि फलन $f(x)=\begin{cases} x & x\leq 1 \\ ax^2+bx+c & \text{otherwise}\end{cases}$ x के प्रत्येक मान के लिये अवकलनीय है। जहाँ $a\neq 0$ तो (a, b, c) का मान होगा :

(A) $\{(a, 1-2a, a) \mid a\in R, a\neq 0\}$ (B) $\{(a, 1-2a, c) \mid a, c\in R, a\neq 0\}$

(C) $\{(a, b, c) \mid a, b, c\in R, a+b+c=1\}$ (D) $\{(a, 1-2a, 0) \mid a\in R, a\neq 0\}$

Ans. A

Sol. $f(x)=\begin{cases} x & , \quad x\leq 1 \\ ax^2+bx+c & , \quad \text{otherwise}\end{cases}$

$f(x)$ should be continous at $x=1$

it gives $a+b+c=1$

$f(x)$ should be differentiable at $x=1$

it gives $2a+b=1$ $\Rightarrow$ $b=1-2a$ $c=1-a-b=a$

11. Number of points of non-differentiability of the function

$g(x)=[x^2]\{\cos^2 4x\}+\{x^2\}[\cos^2 4x]+x^2\sin^2 4x+[x^2][\cos^2 4x]+\{x^2\}\{\cos^2 4x\}$ in $(-50, 50)$ where $[x]$ and $\{x\}$ denote the greatest integer function and fractional part function of x respectively, is equal to :

फलन $g(x) = [x^2]\{\cos^2 4x\} + \{x^2\}[\cos^2 4x] + x^2 \sin^2 4x + [x^2]\,[\cos^2 4x] + \{x^2\}\{\cos^2 4x\}$ के $(-50, 50)$ में उन बिन्दुओं की संख्या जहाँ g (x) अवकलनीय नही है ([x] तथा {x} महत्तम पूर्णांक फलन तथा भिन्नांश भाग फलन का दर्शाता है) :

(A) 98 (B) 99 (C) 100 (D) 0

Ans. D

Sol. $g(x) = ([x^2] + \{x^2\})\left(\{\cos^2 4x\} + [\cos^2 4x]\right) + x^2 \sin^2 4x = x^2 \cos^2 4x + x^2 \sin^2 4x = x^2$

Clearly g(x) is continuous and differenitable $\forall\, x \in R$.]

12. If $f(x) = \begin{cases} x + \{x\} + x\sin\{x\} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ where {x} denotes the fractional part function, then :

(A) 'f' is continuous & diff. at x = 0

(B) 'f' is continuous but not diff. at x = 0

(C) 'f' is continuous & diff. at x = 2

(D) none of these

यदि $f(x) = \begin{cases} x + \{x\} + x\sin\{x\} & \text{for } x \neq 0 \\ 0 & \text{for } x = 0 \end{cases}$ जहाँ {x} एक भिन्नांक फलन हो, तो :

(A) x = 0 पर 'f' सतत् तथा अवकलनीय दोनों होगा।

(B) x = 0 पर 'f' सतत् होगा लेकिन अवकलनीय नहीं होगा।

(C) x = 2 पर फलन 'f' सतत् तथा अवकलनीय दोनों होगा।

(D) इनमें से कोई नहीं

Ans. D

Sol. $f(0) = \lim\limits_{x\to 0} f(x) = 0 - 1 + 0.\sin(-1) = -1$

$f(0^+) = \lim\limits_{x\to 0} f(x) = 0 + 0 + 0.\sin 0 = 0 = f(0)$

f(x) is not continuous at x = 0 at x = 2,

$f(2^+) = 2 + 2 + 2\sin 2 = 4 + 2\sin 2$

$f(2^-) = 2 + 1 + 2\sin 1 = 3 + 2\sin 1$

f(x) is not continuous at x = 2

13. The set of all points where the function $f(x) = \frac{x}{1+|x|}$ is differentiable is :

(A) $(-\infty, \infty)$ (B) $[0, \infty)$ (C) $(-\infty, 0) \cup (0, \infty)$ (D) $(0, \infty)$ (E) none

ऐसे सारे बिन्दुओं का समुच्चय, जहाँ $f(x) = \frac{x}{1+|x|}$ अवकलनीय होगा :

(A) $(-\infty, \infty)$ (B) $[0, \infty)$ (C) $(-\infty, 0) \cup (0, \infty)$ (D) $(0, \infty)$ (E) इनमें से कोई नहीं

Ans. A

Sol. $f(x) = \begin{cases} \frac{x}{1-x}, & x < 0 \\ \frac{x}{1+x}, & x \geq 0 \end{cases}$

f(x) is continous and differntiable for $x \in R$

14. Consider the function $f(x)=\begin{cases} x\{x\}+1 & 0\le x<1 \\ 2-\{x\} & 1\le x\le 2 \end{cases}$ where $\{x\}$ denotes the fractional part function. Which one of the following statements is NOT correct :

(A) $\lim\limits_{x\to 1} f(x)$ exists (B) $f(0)\neq f(2)$

(C) $f(x)$ is continuous in $[0, 2]$ (D) None

माना कि $f(x)=\begin{cases} x\{x\}+1 & 0\le x<1 \\ 2-\{x\} & 1\le x\le 2 \end{cases}$ जहाँ $\{x\}$ एक भिन्नात्मक फलन हो, तो निम्न में से कौन सा विकल्प सही नहीं है :

(A) $\lim\limits_{x\to 1} f(x)$ का अस्तित्व है। (B) $f(0)\neq f(2)$

(C) $[0, 2]$ के अन्तराल में $f(x)$ सतत् है। (D) इनमें से कोई नहीं

Ans. C

Sol. $f(1^+)=f(1^-)=f(1)=2$

$f(0)=1, \quad f(2)=2$

$f(2^-)=1; \quad f(2)=2 \quad \Rightarrow \quad f$ is not continuous at $x=2$

15. Let $g:R\to R$ be a differentiable function such that $g(2)=-40$ and $g'(2)=-5$.

Then $\lim\limits_{x\to 0}\left(\dfrac{g(2-x^2)}{g(2)}\right)^{\frac{4}{x^2}}$ is equal to :

माना $g:R\to R$ एक अवकलनीय फलन इस प्रकार है कि $g(2)=-40$ तथा $g'(2)=-5$.

तो $\lim\limits_{x\to 0}\left(\dfrac{g(2-x^2)}{g(2)}\right)^{\frac{4}{x^2}}$ का मान होगा :

(A) e^{32} (B) $\sqrt{e}$ (C) $\dfrac{1}{\sqrt{e}}$ (D) e^{-5}

Ans. C

Sol. $\lim\limits_{x\to 0}\left(\dfrac{g(2-x^2)}{g(2)}\right)^{\frac{4}{x^2}}$ $(1^\infty \text{ form}) = e^{\lim\limits_{x\to 0} 4\left(\frac{g(2-x^2)-g(2)}{x^2}\right)\times\frac{1}{g(2)}} = e^{-4\frac{g'(2)}{g(2)}} = e^{\frac{-4\times(-5)}{-40}} = e^{\frac{-1}{2}} = \dfrac{1}{\sqrt{e}}$.

16. Let $f:(0, 1)\to R$ be defined by $f(x)=\dfrac{b-x}{1-bx}$, where b is a constant such that $0<b<1$. Then : **[JEE - 2011]**

(A) f is not invertible on $(0, 1)$ (B) $f\neq f^{-1}$ on $(0, 1)$ and $f'(b)=\dfrac{1}{f'(0)}$

(C) $f=f^{-1}$ on $(0, 1)$ and $f'(b)=\dfrac{1}{f'(0)}$ (D) f^{-1} is differentiable on $(0, 1)$

माना f : (0, 1) $\to$ R इस प्रकार परिभाषित है $f(x) = \frac{b - x}{1 - b\,x}$, जहाँ b एक स्थिरांक है इस प्रकार कि 0 < b < 1 तो :

[JEE - 2011]

(A) f, अन्तराल (0, 1) में व्युत्क्रमणीय नहीं है (B) $f \neq f^{-1}$ on (0, 1) तथा $f'(b) = \frac{1}{f'(0)}$

(C) $f = f^{-1}$ on (0, 1) तथा $f'(b) = \frac{1}{f'(0)}$ (D) f^{-1} , अन्तराल (0, 1) में अवकलनीय है

Ans. A

Sol. $f : (0,1) \to R$

$$l(x) = \frac{b - x}{1 - bx} \qquad b \in (0,1)$$

$$\Rightarrow \quad l'(x) = \frac{b^2 - 1}{(bx - 1)^2}$$

$\Rightarrow$ $f'(x) < 0 \ \forall \ x \in (0, 1)$

hence $f(x)$ is decreasing function

hence its range (–1, b)

$\Rightarrow$ co-domain $\neq$ range

$\Rightarrow$ $f(x)$ is non-invertible function

17. Let S = {t $\in$ R: f(x) = |x – π| . ($e^{|x|}$ – 1) sin |x| is not differentiable at t}. Then the set S is equal to :

[JEE (Main) 2018]

(A) {π} (B) {0, π} (C) ϕ (an empty set) (D) {0}

माना S = {t $\in$ R: f(x) = |x – π| . ($e^{|x|}$ – 1) sin |x| जो पर अवकलनीय नहीं है}, तो समुच्चय S बराबर है :

[JEE (Main) 2018]

(A) {π} (B) {0, π} (C) ϕ (एक रिक्त समुच्चय) (D) {0}

Ans. C

Sol. We will check at x = 0, π

$$f(x) \begin{cases} \to (x - \pi)(e^x - 1)\sin x & x \geq \pi \\ \to (\pi - x)(e^x - 1)\sin x & 0 \leq x < \pi \\ \to -(\pi - x)(e^x - 1)\sin x & x < 0 \end{cases}$$

at x = 0

$f'(0^+) = 0$

$f'(0^-) = 0$

at $x = \pi$

$f'(\pi^+) = 0$

$f'(\pi^-) = 0$

It is differentiable at $x = 0, \pi$

$S = \{\} = \phi$

DIFFERENTIABILITY CHECK

18. Let f be an injective and differentiable function such that $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy)$ for all non negative real x and y with $f'(0) = 0, f'(1) = 2 \neq f(0)$, then :

(A) $x f'(x) - 2 f(x) + 2 = 0$ (B) $x f'(x) + 2 f(x) - 2 = 0$

(C) $x f'(x) - f(x) + 1 = 0$ (D) $2 f(x) = f'(x) + 2$

f एक एकैक तथा अवकलनीय फलन हो तथा यदि $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy)\ \forall\ x, y \in R^+ \cup \{0\}$ तथा $f'(0) = 0, f'(1) = 2 \neq f(0)$, तो :

Ans. A

Sol. $f(x) \cdot f(y) + 2 = f(x) + f(y) + f(xy); \quad x = 0, y = 0 \qquad \Rightarrow f(0) = 1$

$$f'(0) = \lim_{h\to 0} \frac{f(0+h) - f(0)}{h}$$

$$f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h} = \lim_{h\to 0} \frac{f\left(x\left(1+\frac{h}{x}\right)\right) - f(x)}{h}$$

$$= \lim_{h\to 0} \frac{f(x)\cdot f\left(1+\frac{h}{x}\right) + 2 - 2f(x) - f\left(1+\frac{h}{x}\right)}{h}$$

$$= \lim_{h\to 0} \frac{\left(f\left(1+\frac{h}{x}\right) - 2\right)}{h}\left(f(x) - 1\right) \qquad \{\because\ f(x) \text{ be an injective} \Rightarrow f(1) = 2\}$$

$$= \frac{1}{x}\lim_{h\to 0}\left(\frac{f\left(1+\frac{h}{x}\right) - f(1)}{\frac{h}{x}}\right)\left(f(x) - 1\right)$$

$$f'(x) = \frac{1}{x} f'(1)\left(f(x) - 1\right) \Rightarrow x f'(x) = 2f(x) - 2$$

19. Let $f(x)$ be continuous and differentiable function for all reals.

$f(x+y)=f(x)-3xy+f(y)$. If $\lim\limits_{h\to 0}\dfrac{f(h)}{h}=7$, then the value of $f'(x)$ is :

फलन $f(x)$, x के प्रत्येक मान के लिये सतत् तथा अवकलनीय हो तथा

$f(x+y)=f(x)-3xy+f(y)$. यदि $\lim\limits_{h\to 0}\dfrac{f(h)}{h}=7$, तो $f'(x)$ का मान होगा :

(A) $-3x$ (B) 7 (C) $-3x+7$ (D) $2f(x)+7$

Ans. C

Sol. $f'(x)=\dfrac{f(x+h)-f(x)}{h}=\dfrac{f(h)-3xh}{h}=7-3x$

20. If $f(x+y)=f(x)+f(y)+c$, for all real x and y and f(x) is continuous at $x=0$ and $f'(0)=1$ then $f'(x)$ equals to:
यदि $f(x+y)=f(x)+f(y)+c, \forall\ x,\ y \in R$ और $x=0$ पर f(x) एक सतत् फलन है तथा $f'(0)=1$ तो $f'(x)$ का मान होगा :

(A) c (B) –1 (C) 0 (D) 1

Ans. D

Sol. $f'(0)=\lim\limits_{h\to 0}\dfrac{f(0+h)-f(0)}{h}=1$ also $f(0)=-c$

$$f'(x)=f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}=\lim_{h\to 0}\frac{f(x)+f(h)+c-f(x)}{h}$$

$$=\lim_{h\to 0}\frac{f(h)-f(0)}{h}=f'(0)=1$$

$\therefore\quad f'(x)=1$

21. If for all values of x & y; $f(x+y)=f(x).f(y)$ and $f(5)=2$, $f'(0)=3$, then $f'(5)$ is :
यदि x तथा y के सभी मानों के लिऐ $f(x+y)=f(x).f(y)$ तथा $f(5)=2$, $f'(0)=3$, तो $f'(5)$ है :

(A) 3 (B) 4 (C) 5 (D) 6

Ans. D

Sol. $f(x+y)=f(x).f(y) \quad \forall\ x, y$

$\therefore\quad f(5+0)=f(5).f(0) \quad \{\because f(5)=2$

$\therefore\quad f(0)=1$

$$\text{Now } f'(5)=\lim_{h\to 0}\frac{f(5+h)-f(5)}{h}$$

$$=\lim_{h\to 0}\frac{f(5)f(h)-f(5)}{h}$$

$$=f(5)\lim_{h\to 0}\frac{f(h)-f(0)}{h}$$

$= f(5)\, f'(0) = 2 \times 3 = 6$

22. If f is a real-valued differentiable function satisfying $|f(x) - f(y)| \le (x-y)^2$, x, $y \in R$ and $f(0) = 0$, then $f(1)$ equals
यदि f एक वास्तविक फलन है, जो अवकलनीय है तथा $|f(x) - f(y)| \le (x-y)^2$, x, $y \in R$ को सन्तुष्ट करता है तथा $f(0) = 0$ है, तो $f(1)$ का मान है :

(A) –1 (B) 0 (C) 2 (D) 1

Ans. B

Sol. $|f(x) - f(y)| \le |x-y|^2$

$$\Rightarrow \frac{|f(x)-f(y)|}{|x-y|} \le |x-y|$$

$$\Rightarrow \lim_{x\to y}\left|\frac{f(x)-f(y)}{x-y}\right| \le \lim_{x\to y}|x-y|$$

$\Rightarrow f'(x) \le 0 \Rightarrow f'(x) = 0$

$\Rightarrow f(x)$ is continuous function

$\therefore f(1) = 0 = f(0)$

23. Suppose f(x) is differentiable at x = 1 and $\lim_{h\to 0} \frac{1}{h} f(1+h) = 5$, then f' (1) equals :
माना f(x) बिन्दु x = 1 पर अवकलनीय है तथा $\lim_{h\to 0} \frac{1}{h} f(1+h) = 5$ तो f' (1) का मान है :

(A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. Apply L Hospital rule

$$\lim_{h\to 0}\frac{f'(1+h)}{1} = 5$$

$\Rightarrow f'(1) = 5$

JEE EXERCISES

Exercise J-2

CONTINUITY CHECK

1. If the function $f(x)=\dfrac{3x^2+ax+a+3}{x^2+x-2}$ is continuous at $x=-2$. Find $f(-2)$:

यदि फलन $f(x)=\dfrac{3x^2+ax+a+3}{x^2+x-2}$, $x=-2$ पर सतत् हो, तो $f(-2)$ का मान ज्ञात करो :

Ans. -1

Sol. $f(x)=\dfrac{3x^2+ax+a+3}{x^2+x-2}$ is continuous at $x=-2$

$\Rightarrow \lim\limits_{x\to -2} f(x)=f(-2)$

$\Rightarrow \lim\limits_{x\to -2} \dfrac{3x^2+a(1+x)+3}{x^2+x-2}$ $\left(\dfrac{0}{0}\text{form}\right) \Rightarrow a=15$

$=\lim\limits_{x+2\to 0} \dfrac{3x^2+15x+18}{(x+2)(x-1)}$

$=3\lim\limits_{x\to -2} \dfrac{(x+2)(x+3)}{(x+2)(x-1)}=\dfrac{3\cdot 1}{-3}=-1$

$\therefore f(-2)=-1$

2. Find all possible values of a and b so that $f(x)$ is continuous for all $x \in R$ if

$$f(x)=\begin{cases} |ax+3| & \text{if } x\le -1 \\ |3x+a| & \text{if } -1<x\le 0 \\ \dfrac{b\sin 2x}{x}-2b & \text{if } 0<x<\pi \\ \cos^2 x-3 & \text{if } x\ge \pi \end{cases} :$$

यदि फलन $f(x)$, x के प्रत्येक वास्तविक मान के लिये सतत् हो, जहाँ

$$f(x)=\begin{cases} |ax+3| & \text{if } x\le -1 \\ |3x+a| & \text{if } -1<x\le 0 \\ \dfrac{b\sin 2x}{x}-2b & \text{if } 0<x<\pi \\ \cos^2 x-3 & \text{if } x\ge \pi \end{cases}$$ तो a व b के सभी सम्भव मान ज्ञात करो :

Ans. $a=0,\ b=1$

Sol. $$f(x)=\begin{cases} |ax+3| & x\le 1 \\ |3x+a| & -1\le x\le 0 \\ \dfrac{b\sin 2x}{x}-2b & \text{if } 0<x<\pi \\ \cos^2 x-3 & \text{if } x\ge\pi \end{cases}$$

Given f(x) is continuous $\forall\ x\in R$

At $x=0$

L.H.L. $=|a|=f(0)$

$$\text{R.H.L.}=\lim_{h\to 0}\left(\frac{b\sin 2h}{h}-2b\right)=0$$

$\Rightarrow |a|=0$

$\Rightarrow a=0$

at $x=+\pi$

$f(\pi^+)=f(\pi)=-2$

$$\text{L.H.L.}=\lim_{h\to 0}\frac{\sin(2\pi-2h)}{\pi-h}-2b=-2b$$

$\Rightarrow b=1$

3. The function $f(x)=\begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0<x<\frac{\pi}{2} \\ b+2 & \text{if } x=\frac{\pi}{2} \\ \left(1+|\cos x|\right)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2}<x<\pi \end{cases}$

Determine the values of 'a' & 'b' , if f is continuous at $x=\pi/2$:

दिया गया फलन $f(x)=\begin{cases} \left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if } 0<x<\frac{\pi}{2} \\ b+2 & \text{if } x=\frac{\pi}{2} \\ \left(1+|\cos x|\right)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if } \frac{\pi}{2}<x<\pi \end{cases}$

यदि फलन $x=\pi/2$ पर सतत् हो, तो 'a' व 'b' का मान ज्ञात करो :

Ans $a=0\ ;\ b=-1$

Sol.

$$f(x)=\begin{cases}\left(\frac{6}{5}\right)^{\frac{\tan 6x}{\tan 5x}} & \text{if} \quad 0<x<\frac{\pi}{2}\\ b+2 & \text{if} \quad x=\frac{\pi}{2}\\ \left(1+|\cos x|\right)^{\left(\frac{a|\tan x|}{b}\right)} & \text{if} \quad \frac{\pi}{2}<x<\pi\end{cases}$$

Given f is continuous at $x=\frac{\pi}{2}$

$$\text{L.H.L.} = \lim_{h\to 0} f\left(\frac{\pi}{2}-h\right)$$

$$= \lim_{h\to 0}\left(\frac{6}{5}\right)^{\frac{\tan\left(\frac{6\pi}{2}-6h\right)}{\tan\left(\frac{5\pi}{2}-5h\right)}}$$

$$= \lim_{h\to 0}\left(\frac{6}{5}\right)^{-\frac{\tan 6h}{\cot 5h}}$$

$$= \lim_{h\to 0}\left(\frac{6}{5}\right)^{-\tan 5h\cdot\tan 6h} = 1$$

$\Rightarrow b + 2 = 1$

$\Rightarrow b = -1$

$$\text{R.H.L.} = \lim_{h\to 0} f\left(\frac{\pi}{2}+h\right) = \lim_{h\to 0}\left(1+|-\sinh|\right)^{\left|\frac{\bar{a}\coth}{b}\right|}$$

$$= \lim_{h\to 0}\left(1-\sinh\right)^{a\coth} = 1$$

if $a = 0$

4. Suppose that $f(x) = x^3 - 3x^2 - 4x + 12$ and $h(x) = \begin{cases}\frac{f(x)}{x-3} & , x \neq 3\\ K & , x = 3\end{cases}$ then

(a) find all zeros of $f(x)$

(b) find the value of K that makes h continuous at $x = 3$

(c) using the value of K found in (b), determine whether h is an even function :

माना कि $f(x) = x^3 - 3x^2 - 4x + 12$ और $h(x) = \begin{cases} \frac{f(x)}{x-3} & , x \neq 3 \\ K & , x = 3 \end{cases}$ तो,

(a) $f(x) = 0$ के मूल बताइये

(b) यदि फलन $h(x)$, $x = 3$ पर सतत् हो, तो K का मान बताओ।

(c) (b) में प्राप्त K के मान का उपयोग करते हुए बताइयें कि क्या $h(x)$ एक सम फलन होगा :

Ans. (a) $-2, 2, 3$ (b) $K = 5$ (c) even

Sol. $f(x) = (x + 2)(x - 2)(x - 3)$

$h(x) = \begin{cases} (x+2)(x-2) & , x \neq 3 \\ k & , x = 3 \end{cases}$ for continuity $k = \lim\limits_{x \to 3} h(x) = 5$

$h(x) = (x + 2)(x - 2) = x^2 - 4$ which is even $\forall\, x \in R$

5. Let $f(x) = \begin{cases} \frac{1-\sin \pi x}{1+\cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases}$. Determine the value of p, if possible, so that the function is continuous at x=1/2:

माना कि $f(x) = \begin{cases} \frac{1-\sin \pi x}{1+\cos 2\pi x}, & x < \frac{1}{2} \\ p, & x = \frac{1}{2} \\ \frac{\sqrt{2x-1}}{\sqrt{4+\sqrt{2x-1}}-2}, & x > \frac{1}{2} \end{cases}$. यदि फलन x=1/2 पर सतत् हो, तो यदि संभव है तो p का मान बताइये :

Ans p not possible

Sol. $f\left(\frac{1}{2}\right) = P$

$f\left(\frac{1^+}{2}\right) = \lim\limits_{h \to 0} f\left(\frac{1}{2} + h\right)$

$= \lim\limits_{h \to 0} \frac{\sqrt{2h}}{\sqrt{4+\sqrt{2h}}-2} = \lim\limits_{h \to 0} \frac{\sqrt{2h}\left(\sqrt{4+\sqrt{2h}}+2\right)}{2+\sqrt{2h}} = 0$

$f\left(\frac{1^-}{2}\right) = \lim\limits_{h \to 0} f\left(\frac{1}{2} - h\right)$

$= \lim\limits_{h \to 0} \frac{1-\sin\left(\frac{\pi}{2} - \pi h\right)}{1+\cos(\pi - 2\pi h)}$

$$= \lim_{h\to 0} \frac{1-\cos \pi h}{1-\cos(2\pi h)} = \frac{2\sin^2 \frac{\pi}{h}}{2\sin^2 \pi h} \cdot \frac{\pi^2 h^2}{4 \cdot \frac{\pi^2 h^2}{4}}$$

$$= \frac{1}{4}$$

$\Rightarrow$ f(x) is not continuous at $x = \frac{1}{2}$.

No value of P is possible for which f(x) is continuous.

6. Given the function $g(x) = \sqrt{6-2x}$ and $h(x) = 2x^2 - 3x + a$. Then :

(a) evaluate $h(g(2))$ (b) If $f(x) = \begin{cases} g(x), & x \le 1 \\ h(x), & x > 1 \end{cases}$, find 'a' so that f is continuous.

फलन $g(x) = \sqrt{6-2x}$ तथा $h(x) = 2x^2 - 3x + a$. दिया गया है

(a) $h(g(2))$ का मान बताओ। (b) यदि $f(x) = \begin{cases} g(x), & x \le 1 \\ h(x), & x > 1 \end{cases}$, तो 'a' का मान बताओ जबकि फलन f सतत् हो।

Ans. (a) $4 - 3\sqrt{2} + a$, (b) $a = 3$

Sol. (a) $h(g(2)) = h(\sqrt{2}) = 2\cdot 2 - 3\sqrt{2} + a$

$= 4 - 3\sqrt{2} + a$

(b) $f(x) = \begin{cases} \sqrt{6-2x} & x \le 1 \\ 2x^2 - 3x + a & x > 1 \end{cases}$

If f is continuous at x = 1 then $f(1) = \sqrt{4} = 2$ & $f(1^+) = 2 - 3 + a = a - 1$

$\Rightarrow a = 2 + 1 = 3$

7. Determine the values of a, b & c for which the function $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x<0 \\ c & \text{for } x=0 \\ \frac{\left(x+bx^2\right)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x>0 \end{cases}$

is continuous at x = 0 :

यदि फलन $f(x) = \begin{cases} \frac{\sin(a+1)x + \sin x}{x} & \text{for } x<0 \\ c & \text{for } x=0 \\ \frac{\left(x+bx^2\right)^{1/2} - x^{1/2}}{bx^{3/2}} & \text{for } x>0 \end{cases}$, x = 0 पर सतत् हो, तो a, b व c के मान ज्ञात करो :

Ans. $a = -3/2, b \neq 0, c = 1/2$

Sol. $\lim\limits_{x\to 0^-} \dfrac{\sin(a+1)x+\sin x}{x} = a+2$

and $\lim\limits_{x\to 0^+} \dfrac{x+bx^2-x}{bx^{3/2}(\sqrt{x+bx^2}+\sqrt{x})} = \dfrac{1}{2}$ as $b \neq 0$

according to question

$c = \dfrac{1}{2}$ & $a+2 = \dfrac{1}{2} \Rightarrow a = \dfrac{-3}{2}$

8. If $f(x) = \dfrac{\sin 3x + A\sin 2x + B\sin x}{x^5}$ $(x \neq 0)$ is cont. at $x = 0$. Find A & B. Also find f(0).

यदि फलन $f(x) = \dfrac{\sin 3x + A\sin 2x + B\sin x}{x^5}$ $(x \neq 0)$ पर सतत् हो, तो A व B का मान ज्ञात करो तथा f(0) का भी मान बताओ।

Ans. $A = -4$, $B = 5$, $f(0) = 1$

Sol. $f(x) = \dfrac{\sin 3x + A\sin 2x + B\sin x}{x^5}; x \neq 0$

Given f(x) is continuous at $x = 0$

$\Rightarrow f(0) = \lim\limits_{x\to 0} f(x)$

$$= \lim_{x\to 0} \frac{\left(3x - \frac{(3x)^3}{\lfloor 3} + ...\right) + A\left(2x - \frac{(2x)^3}{\lfloor 3} + \frac{(2x)^5}{\lfloor 5} + ...\right) + B\left(x - \frac{x^3}{\lfloor 3} + \frac{x^5}{\lfloor 5} + ...\right)}{x^5}$$

$$= \lim_{x\to 0} \frac{(3+2A+B) - \frac{x^3}{\lfloor 3}(27+8A+B) + \frac{x^5}{\lfloor 5}(32A+243B) + ...}{x^5}$$

$\Rightarrow 3 + 2A + B = 0$

$27 + 8A + B = 0$

$\Rightarrow A = -4$ & $B = 5$

$\Rightarrow f(0) = \dfrac{32A + 243 + B}{\lfloor 5}$

$= \dfrac{120}{120} = 1$

9. Let $f(x)=\begin{cases}\dfrac{\left(\frac{\pi}{2}-\sin^{-1}\left(1-\{x\}^2\right)\right)\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2}\left(\{x\}-\{x\}^3\right)} & \text{for } x\neq 0\\ \dfrac{\pi}{2} & \text{for } x=0\end{cases}$ where $\{x\}$ is the fractional part of x.

Consider another function g(x) ; such that $g(x)=\begin{cases} f(x) & \text{for } x\geq 0\\ 2\sqrt{2}f(x) & \text{for } x<0\end{cases}$.

Discuss the continuity of the functions f(x) & g(x) at x = 0 :

दिया हो कि $f(x)=\begin{cases}\dfrac{\left(\frac{\pi}{2}-\sin^{-1}\left(1-\{x\}^2\right)\right)\sin^{-1}\left(1-\{x\}\right)}{\sqrt{2}\left(\{x\}-\{x\}^3\right)} & \text{for } x\neq 0\\ \dfrac{\pi}{2} & \text{for } x=0\end{cases}$ जहाँ $\{x\}$ x का भिनांक फलन है एक दूसरा फलन g(x) इस

प्रकार से परिभाषित है कि $g(x)=\begin{cases} f(x) & \text{for } x\geq 0\\ 2\sqrt{2}f(x) & \text{for } x<0\end{cases}$

तो f(x) तथा g(x) की x = 0 पर सतता की जाँच करिये :

Ans. $f(0^+)=\dfrac{\pi}{2}$; $f(0^-)=\dfrac{\pi}{4\sqrt{2}}$ $\Rightarrow$ f is discont. at x = 0 ; $g(0^+)=g(0^-)=g(0)=\pi/2$ $\Rightarrow$ g is cont. at x = 0

Sol. $\lim\limits_{x\to 0^+} f(x)=\lim\limits_{h\to 0}\dfrac{(\frac{\pi}{2}-\sin^{-1}(1-\{h\}^2)).\sin^{-1}(1-\{h\})}{\sqrt{2}(\{h\}-\{h\}^3)}$

$=\lim\limits_{h\to 0}\dfrac{(\frac{\pi}{2}-\sin^{-1}(1-h^2))\sin^{-1}(1-h)}{\sqrt{2}(h-h^3)}$

$=\lim\limits_{h\to 0}\dfrac{\cos^{-1}(1-h^2)}{\sqrt{2}(1-h^2)}\times\dfrac{\sin^{-1}(1-h)}{h}=\dfrac{\pi}{2}$

$\lim\limits_{h\to 0^-} f(x)=\lim\limits_{h\to 0}\dfrac{(\frac{\pi}{2}-\sin^{-1}(1-(1-h)^2)\sin^{-1}(1-(1-h))}{\sqrt{2}((1-h)-(1-h)^3)}$

$=\lim\limits_{h\to 0}\dfrac{\frac{\pi}{2}\sin^{-1}h}{\sqrt{2}(1-h)(2-h)h}=\dfrac{\pi}{4\sqrt{2}}$

so f(x) is discontinuous at x = 0

$$\text{Now } g(x) = \begin{cases} \dfrac{\pi}{2} & ; \quad x \geq 0 \\ 2\sqrt{2}\,\dfrac{\pi}{4\sqrt{2}} & ; \quad x < 0 \end{cases}$$

$$g(x) = \begin{cases} \dfrac{\pi}{2} & ; \quad x \geq 0 \\ \dfrac{\pi}{2} & ; \quad x < 0 \end{cases}$$

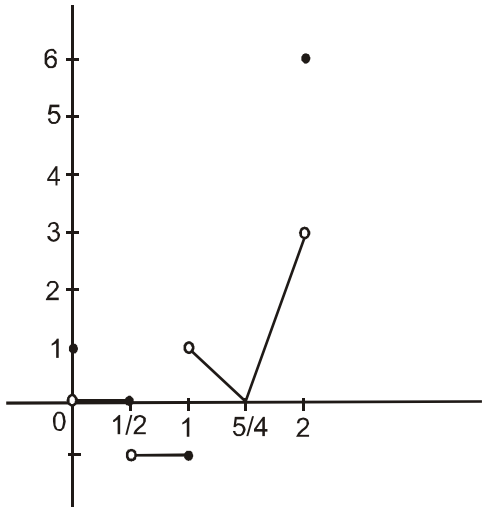
so g(x) is continuous at x = 0

10. Discuss the continuity of f in [0,2] where $f(x) = \begin{bmatrix} |4x-5|\,[x] & \text{for} \quad x > 1 \\ [\cos \pi x] & \text{for} \quad x \leq 1 \end{bmatrix}$; where [x] is the greatest integer not greater than x. Also draw the graph :

यदि $f(x) = \begin{bmatrix} |4x-5|\,[x] & \text{for} \quad x > 1 \\ [\cos \pi x] & \text{for} \quad x \leq 1 \end{bmatrix}$ जहाँ [x] महत्तम पूर्णांक फलन हो, तो f की [0,2] अन्तराल में सतता की जाँच करये तथा y = f(x) का लेखा चित्र भी खीचियें :

Ans. the function f is continuous everywhere in [0 , 2] except for $x = 0 , \frac{1}{2}$, 1 & 2.

Sol. $$f(x) = \begin{cases} 1 & , \quad x = 0 \\ 0 & , \quad 0 < x \leq 1/2 \\ -1 & , \quad 1/2 < x \leq 1 \\ 5-4x & , \quad 1 < x < 5/4 \\ 4x-5 & , \quad 5/4 \leq x < 2 \\ 6 & , \quad x = 2 \end{cases}$$



f(x) (असत्त है ।)is discontinous at x = 0, 1/2, 1, 2 in [0, 2]

11. If $f(x) = x + \{-x\} + [x]$, where [x] is the integral part & {x} is the fractional part of x. Discuss the continuity of f in $[-2, 2]$:

यदि $f(x) = x + \{-x\} + [x]$, जहाँ [x] महत्तमपूर्णांक फलन तथा {x} भिन्नांश फलन हो, तो f(x) की $[-2, 2]$ के अन्तराल में सतता की जाँच करिये :

Ans. discontinuous at all integral values in $[-2, 2]$

Sol. $f(x) = x + \{-x\} + [x]$

$= x - x - [-x] + [x]$

$= [x] - [-x]$

$$f(x) = \begin{cases} -4 & x = -2 \\ -3 & -2 < x < -1 \\ & x = -1 \\ -2 & -1 < x < 0 \\ -1 & x = 0 \\ 0 & 0 < x < 1 \\ 2 & x = 1 \\ 3 & 1 < x < 2 \\ 4 & x = 2 \end{cases}$$

$\Rightarrow$ f(x) is discontinuous at x = –2, –1, 0, 1, 2.

12. Find the locus of (a, b) for which the function $f(x) = \begin{cases} ax - b & \text{for} \quad x \le 1 \\ 3x & \text{for} \quad 1 < x < 2 \\ bx^2 - a & \text{for} \quad x \ge 2 \end{cases}$

is continuous at x = 1 but discontinuous at x = 2 :

यदि फलन $f(x) = \begin{cases} ax - b & \text{for} \quad x \le 1 \\ 3x & \text{for} \quad 1 < x < 2 \\ bx^2 - a & \text{for} \quad x \ge 2 \end{cases}$, x = 1 पर सतत् तथा x = 2 पर असतत् हो, तो बिन्दु (a, b) का बिन्दुपथ ज्ञात करिये :

Ans. locus (a, b) $\rightarrow$ x, y is y = x – 3 excluding the points where y = 3 intersects it

Sol. $$f(x) = \begin{cases} ax - b & x \le 1 \\ 3x & 1 < x < 2 \\ bx^2 - a & x \ge 2 \end{cases}$$

at x = 1 | at x = 2

$f(1) = a - b$ | $f(2^-) = 6$

$f(1^+) = 3$ | $f(2^+) = f(2) = 4b - a$

$f(1^-) = a - b$ | $\Rightarrow 4b - a \neq 6$

$\Rightarrow a - b = 3$ | $\Rightarrow 4b - (3 + b) \neq 6$

$\Rightarrow 3b \neq 9$

$\Rightarrow b \neq 3$

Locus of (a, b) is

$y = x - 3$; excluding the points where $y = 3$ intersects it.

13. A function $f: R \to R$ is defined as $f(x) = \underset{n\to\infty}{\text{Lim}} \dfrac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ where f is continuous on R. Find the values of a, b and c :

एक फलन $f: R \to R$ इस प्रकार से परिभाषित होता है कि $f(x) = \underset{n\to\infty}{\text{Lim}} \dfrac{ax^2 + bx + c + e^{nx}}{1 + c \cdot e^{nx}}$ यदि f(x) सभी वास्तविक संख्या के लिये सतत् हो, तो a, b तथा c का मान ज्ञात करो :

Ans. $c = 1,\ a, b \in R$

Sol. $f(x) = \lim\limits_{n\to\infty} \dfrac{ax^2 + bx + c + e^{nx}}{1 + ce^{nx}}$

$$= \lim_{n\to\infty} \frac{ax^2 + bx + c + 1 + nx + \dfrac{n^2x^2}{\lfloor 2} + ...}{1 + c\left(1 + nx + \dfrac{n^2x^2}{\lfloor 2} + ...\right)}$$

$$= \lim_{n\to\infty} \frac{ax^2 + bx + c + 1 + nx + \dfrac{n^2x^2}{\lfloor 2} + ...}{1 + c + ncx + \dfrac{n^2x^2}{\lfloor 2} + ...}$$

14. Let $f(x) = x^3 - x^2 - 3x - 1$ and $h(x) = \dfrac{f(x)}{g(x)}$ where *h* is a rational function such that

(a) it is continuous every where except when $x = -1$, (b) $\underset{x\to\infty}{\text{Lim}}\ h(x) = \infty$ and (c) $\underset{x\to -1}{\text{Lim}}\ h(x) = \dfrac{1}{2}$.

Find $\underset{x\to 0}{\text{Lim}} \left(3h(x) + f(x) - 2g(x)\right)$:

यदि $f(x) = x^3 - x^2 - 3x - 1$ तथा $h(x) = \dfrac{f(x)}{g(x)}$. जहाँ पर h(x) एक परिमेय फलन हो तथा निम्नलिखित बाते दी गयी है

(a) $x = -1$ के अतिरिक्त सभी बिन्दुओं पर सतत् है (b) $\underset{x\to\infty}{\text{Lim}}\ h(x) = \infty$ तथा (c) $\underset{x\to -1}{\text{Lim}}\ h(x) = \dfrac{1}{2}$.

$\lim_{x\to 0}\left(3h(x)+f(x)-2g(x)\right)$ का मान बताओ :

Ans. $g(x)=4(x+1)$ and limit $=-\frac{39}{4}$

Sol. $h(x)=\frac{x^3-x^2-3x-1}{g(x)}$

$=\frac{(x+1)\left(x^2-2x-1\right)}{g(x)}$

$=\frac{(x+1)\left(x^2-2x-1\right)}{g(x)}$

$\lim_{x\to 1^-} h(x)=\frac{1}{2}$

$\Rightarrow \lim_{x\to 1^-}\frac{(x+1)\left(x^2-2x-1\right)}{(x+1)Q(x)}=\frac{1}{2}$

$\Rightarrow \frac{+2}{Q(-1)}=\frac{1}{2}$ $\qquad$ $\lim_{x\to\infty} h(x)=\infty$

$\Rightarrow Q(-1)=4$ $\qquad$ $\Rightarrow \lim_{x\to\infty}\frac{x^2-2x-1}{Q(x)}=\infty$

$\therefore g(x)=(x+1)Q(x)$ $\qquad$ $\therefore Q(x)=ax+b$

$\Rightarrow b-a=4$

15. Find the number of points of discontinuity of the function $f(x)=[5x]+\{3x\}$ in $[0, 5]$. where $[y]$ and $\{y\}$ denote largest integer less than or equal to y and fractional part of y respectively :

फलन $f(x)=[5x]+\{3x\}$ के $[0, 5]$ में असतत् बिन्दुओं की संख्या क्या होगी जहाँ $[y]$ तथा $\{y\}$ महत्तम पूर्णांक फलन एवं भिन्नांश फलन को क्रमशः निरूपित करते है :

Ans. 30

Sol. $f(x)=[5x]-[3x]+3x$

Now in $[0, 1]$ we have to check at $x=\frac{1}{5},\ \frac{2}{5},\frac{3}{5},\frac{4}{5},\frac{1}{3},\frac{2}{3}$

and $f(x)$ is discontinuous at $x=\frac{1}{5},\frac{2}{3},\frac{3}{5},\frac{4}{5},\frac{1}{3},\frac{2}{3}$ i.e. 6 points.

$\therefore$ Total points $6\times 5=30$ **Ans.**

Note : f(x) is continuous at all integers in $[0, 5]$

16. Let $f(x) = x^3 - 3x^2 + 6$ and $g(x) = \begin{cases} \max\{f(t), x+1 \le t \le x+2 & , \quad -3 \le x < 0 \\ 1-x & , \quad x \ge 0 \end{cases}$. test the continuity of g(x) for $x \in [-3, 1]$:

माना $f(x) = x^3 - 3x^2 + 6$ तथा $g(x) = \begin{cases} \max\{f(t), x+1 \le t \le x+2 & , \quad -3 \le x < 0 \\ 1-x & , \quad x \ge 0 \end{cases}$. तो फलन g(x) की अन्तराल $x \in [-3, 1]$ में सांतत्यता की जाँच कीजिये :

Ans. Discont. at x = 0

Sol. $f(x) = x^3 - 3x^2 + 6$

$$g(x) = \begin{cases} \max\{f(t), x+1 \le t \le x+2 & , \quad -3 \le x < 0 \\ 1-x & , \quad x \ge 0 \end{cases}$$

$$= \{\max (t^3 - 3t^2 + 6; x + 1 \le t \le x + 2\}; -3 \le x < 0$$

TYPES OF DISCONTINUITY

17. Let $f(x) = \begin{cases} \dfrac{l\text{n}\cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x > 0 \\ \dfrac{e^{\sin 4x}-1}{l\text{n}(1+\tan 2x)} & \text{if } x < 0 \end{cases}$

Is it possible to define f(0) to make the function continuous at x = 0. If yes what is the value of f(0), if not then indicate the nature of discontinuity :

माना कि $f(x) = \begin{cases} \dfrac{l\text{n}\cos x}{\sqrt[4]{1+x^2}-1} & \text{if } x > 0 \\ \dfrac{e^{\sin 4x}-1}{l\text{n}(1+\tan 2x)} & \text{if } x < 0 \end{cases}$

क्या ये सम्भव है कि f(0) इस प्रकार से परिभाषित हो, कि फलन f(0), x = 0 पर सतत हो जाये? यदि हाँ, तो f(0) का मान क्या होगा, यदि ना हो तो असतता का स्वभाव बताइये :

Ans. $f(0^+) = -2$; $f(0^-) = 2$ hence f(0) not possible to define

Sol. $$f(x) = \begin{cases} \dfrac{\ln\cos x}{\left(1+x^2\right)^{\frac{1}{4}}-1} & \text{if} \quad x > 0 \\ \dfrac{e^{\sin 4x}-1}{\ln\left(1+\tan 2x\right)} & \text{if} \quad x < 0 \end{cases}$$

$$\text{L.H.L.} = f(0^-) = \lim_{h\to 0} \frac{e^{-\sin 4h}-1}{\ln\left(1-\tan 2h\right)}$$

$$= \lim_{h\to 0} \frac{\left(e^{\sin 4h}-1\right)}{\sin 4h}\cdot\frac{\sin 4h}{e^{\sin 4h}}\frac{\tan 2h}{\ln\left(1-\tan 2h\right)}\cdot\frac{1}{\tan 2h}$$

$$= \lim_{h\to 0} \frac{2\cdot\sin 2h\cdot\cos^2 2h}{\sin 2h} = 2$$

$$f\left(0^+\right) = \lim_{h\to 0} \frac{\ln\cosh}{\left(1+h^2\right)^{\frac{1}{4}}-1}\times\frac{\left(1+h^2\right)^{\frac{1}{4}}+1}{\left(1+h^2\right)^{\frac{1}{4}}+1}$$

$$= \lim_{h\to 0} \frac{\left(1+h^2\right)^{\frac{1}{4}}+1}{\left(\sqrt{1+h^2}-1\right)}\ln\left(1+\cosh-1\right)$$

$$= \lim_{h\to 0} \frac{\left(\left(1+h^2\right)^{\frac{1}{4}}+1\right)\left(\sqrt{1+h^2}+1\right)}{h^2}\cdot\frac{\ln\left(1+\cosh-1\right)}{\cosh-1}\cdot\cosh-1$$

$$= \lim_{h\to 0} -\frac{4\cdot 2\sin^2\frac{h}{2}}{4\cdot\frac{h^2}{4}} = -2$$

$\therefore$ f(x) is discontinuous at x = 0.

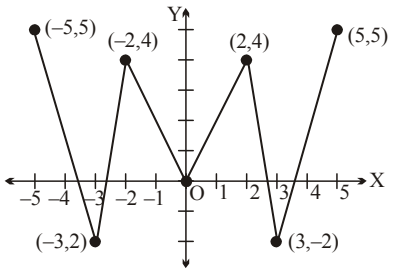
INTERMEDIATE VALUE THEOREM

18. Let f be a real valued continuous function on R and satisfying $f(-x)-f(x)=0\ \forall\ x\in R$. If $f(-5)=5$, $f(-2)=4$, $f(3)=-2$ and $f(0)=0$ then find the minimum number of zero's of the equation f(x) = 0 :

माना f, R में परिभाषित वास्तविक मान वाला फलन है जो $f(-x)-f(x)=0\ \forall\ x\in R$ को संतुष्ट करता है। यदि $f(-5)=5$, $f(-2)=4$, $f(3)=-2$ तथा $f(0)=0$ तो f(x) = 0 के न्यूनतम मूलों की संख्या बताइयें :

Ans. 5

Sol. One of the possible graph of f(x)



having minimum number of zeroes.

Clearly f(x) = 0 has atleast 5 solutions.]

19. If g : [a, b] onto [a, b] is continous show that there is some $c \in [a, b]$ such that $g(c) = c$:
यदि $g : [a, b] \to [a, b]$ सतत् आच्छादक है तो सिद्ध करो कि कोई $c \in (a, b)$ ऐसा होगा कि $g(c) = c$:

Ans.

Sol. $g:[a,b] \xrightarrow[\text{continuous}]{\text{onto}} [a,b]$

$$f(x) = g\underbrace{(x) - x}_{\substack{\text{continuous} \\ \forall x \in [a,b]}}$$

$$\left.\begin{aligned} f(a) &= g(a) - a \geq 0 \\ f(b) &= g(b) - b \leq 0 \end{aligned}\right\} \{\because a \leq g(x) \leq b\}$$

By IMVT $\exists\ c \in (a, b)$ s.t.
$f(c) = 0$
$\Rightarrow g(c) = c$

20. Let f be continuous on the interval [0, 1] to R such that $f(0) = f(1)$. Prove that there exists a point c in $\left[0, \frac{1}{2}\right]$ such that $f(c) = f\left(c + \frac{1}{2}\right)$:

f एक [0, 1] अन्तराल में एक सतत् फलन हो तथा यदि $f(0) = f(1)$ तो सिद्ध करो कि $\left[0, \frac{1}{2}\right]$ में c एक ऐसा बिन्दु होगा कि $f(c) = f\left(c + \frac{1}{2}\right)$ होगा :

Ans.

Sol. $f:[0,1] \xrightarrow{\text{continuous}} R$

$$\text{Let } h(x) = \underbrace{f(x) - f\left(x + \frac{1}{2}\right)}_{\substack{\text{continuous} \\ \text{in } \left[0, \frac{1}{2}\right]}}$$

$$h(0) = f(0) - f\left(\frac{1}{2}\right)$$

$$h\left(\frac{1}{2}\right) = f\left(\frac{1}{2}\right) - f(1)$$

$$\Rightarrow h(0) + h\left(\frac{1}{2}\right) = 0$$

$\Rightarrow h(0)$ & $h\left(\frac{1}{2}\right)$ are of opposite sign.

$\therefore \exists c \in \left(0, \frac{1}{2}\right)$ s.t. $h(c) = 0$

$\Rightarrow f(c) = f\left(c + \frac{1}{2}\right)$

DIFFERENTIABILITY CHECK

21. If the function $f(x)$ defined as $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \le 0 \\ x^n \sin\frac{1}{x} & \text{for } x > 0 \end{cases}$ is continuous but not derivable at $x = 0$ then find the range of n :

यदि फलन $f(x) = \begin{cases} -\frac{x^2}{2} & \text{for } x \le 0 \\ x^n \sin\frac{1}{x} & \text{for } x > 0 \end{cases}$, $x = 0$ पर सतत् तो है लेकिन अवकलनीय नही हो, तो n का परिसर ज्ञात करो :

Ans. $0 < n \le 1$

Sol. $f(x) = \begin{cases} -\frac{x^2}{2} & x \le 0 \\ x^n \sin\frac{1}{x} & x > 0 \end{cases}$

at $x = 0$

$f(0) = 0$

$f(0^-) = 0$

$f(0^+) = 0$ if $n > 0$

$$\text{L.H.D.} = \lim_{h\to 0} \frac{-\frac{h^2}{2} - 0}{-h} = 0$$

$$\text{R.H.D.} = \lim_{h\to 0} \frac{h^n \sin\frac{1}{h}}{h} = \lim_{h\to 0} h^{n-1} \sin\frac{1}{h} \qquad = \begin{cases} 0 & \text{if} \quad n > 1 \\ \text{DNE} & \text{if} \quad n \le 1 \end{cases}$$

$\Rightarrow 0 < n \le 1$

22. Examine the origin for continuity & derrivability in the case of the function f defined by
$f(x) = x\tan^{-1}(1/x)$, $x \neq 0$ and $f(0) = 0$:
यदि फलन $f(x) = x\tan^{-1}(1/x)$, $x \neq 0$ तथा $f(0) = 0$ दिया गया हो, तो $x = 0$ पर फलन की सतत्ता तथा अवकलनीयता की जाँच करो :

Ans. cont. but not diff. at $x = 0$

Sol. $$f(x) = \begin{cases} x\tan^{-1}\left(\frac{1}{x}\right) & ; \quad x \neq 0 \\ 0 & ; \quad x = 0 \end{cases}$$

$$f\left(0^+\right) = \lim_{h\to 0} f(0+h) = \lim_{h\to 0} h\tan^{-1}\left(\frac{1}{h}\right) = 0$$

$$f\left(0^-\right) = \lim_{h\to 0} f(0-h) = \lim_{h\to 0} h\tan^{-1}\left(\frac{1}{h}\right) = 0$$

$\therefore$ at $x = 0$; $f(x)$ is continuous.

$$\text{L.H.D.} = \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h} = \lim_{h\to 0} \frac{+h\tan^{-1}\left(\frac{1}{h}\right)}{-h} = -\infty$$

$$\text{R.H.D.} = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = +\infty$$

23. If $f(x) = |x-1|.([x]-[-x])$, then find $f'(1^+)$ & $f'(1^-)$ where [x] denotes greatest integer function :
यदि $f(x) = |x-1|.([x]-[-x])$ तो $f'(1^+)$ तथा $f'(1^-)$ का मान ज्ञात करों जहाँ [x] महत्तम पूर्णांक फलन हो :

Ans. $f'(1^+) = 3$, $f'(1^-) = -1$

Sol. $f(x) = |x-1|([x]-[-x])$

$$f'\left(1^+\right) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h}$$

$$= \lim_{h\to 0} \frac{h(1+2)}{h} = 3$$

$$f'\left(1^-\right) = \lim_{h\to 0} \frac{f(1-h)-f(1)}{-h}$$

$$= \lim_{h\to 0} \frac{h(0+1)}{-h} = -1$$

24. If $f(x) = \begin{cases} ax^2-b & \text{if } |x|<1 \\ -\frac{1}{|x|} & \text{if } |x|\geq 1 \end{cases}$ is derivable at $x = 1$. Find the values of a & b :

यदि $f(x)=\begin{cases} ax^2-b & \text{if } |x|<1 \\ -\frac{1}{|x|} & \text{if } |x|\geq 1 \end{cases}$, x = 1 पर अवकलनीय हो, तो a व b का मान ज्ञात करो :

Ans. a= 1/2 , b = 3/2

Sol. $f(x)=\begin{cases} ax^2-b & ; \quad -1<x<1 \\ \dfrac{-1}{x} & ; \quad x\geq 1 \\ \dfrac{1}{x} & ; \quad x\leq -1 \end{cases}$

Now f(x) is differentiable at x = 1

$\Rightarrow \quad 2ax=\dfrac{1}{x^2} \qquad$ at x = 1

$\Rightarrow \quad a=\dfrac{1}{2}$

Also f(x) is continuous at x = 1

$\Rightarrow \quad a(1)^2-b=-1 \qquad \Rightarrow \quad b=\dfrac{3}{2}$

25. Let f(x) be defined in the interval [-2, 2] such that $f(x)=\begin{cases} -1 & , \ -2\leq x\leq 0 \\ x-1 & , \ 0<x\leq 2 \end{cases}$ &

$g(x)=f(|x|)+|f(x)|$. Test the differentiability of g(x) in $(-2, 2)$:

अन्तराल [-2, 2] में फलन f(x) इस प्रकार से परिभाषित है कि $f(x)=\begin{cases} -1 & , \ -2\leq x\leq 0 \\ x-1 & , \ 0<x\leq 2 \end{cases}$ तथा

$g(x)=f(|x|)+|f(x)|$ तो g(x) की $(-2, 2)$ के अन्तराल में अवकलनीयता की जाँच करो :

Ans. not derivable at x = 0 & x = 1

Sol. $f(x)=\begin{cases} -1 & -2\leq x\leq 0 \\ x-1 & 0<x\leq 2 \end{cases}$

$g(x)=f(|x|)+|f(x)|$

$=\begin{cases} x-1+1 & -2\leq x\leq 0 \\ x-1+1-x & 0\leq x\leq 1 \\ 2x-2 & 1<x\leq 2 \end{cases}$

$=\begin{cases} x & -2\leq x\leq 0 \\ 0 & 0<x\leq 1 \\ 2(x-1) & 1<x\leq 2 \end{cases}$

g(x) is continuous $\forall\ x \in [-2,2]$

$$g'(x)=\begin{cases}1 & -2 \le x < 0\\ 0 & 0 < x < 1\\ 2 & 1 < x < 2\end{cases}$$

g′(x) is not differentiable at x = 0 & x = 1 in (–2, 2).

26. $f(x) = x.\left(\dfrac{e^{[x]+|x|}-2}{[x]+|x|}\right)$, $x \neq 0$ & $f(0) = -1$ where [x] denotes greatest integer less than or equal to x. Test the differentiability of f(x) at x = 0 :

यदि $f(x) = x.\left(\dfrac{e^{[x]+|x|}-2}{[x]+|x|}\right)$, $x \neq 0$ और $f(0) = -1$ जहाँ [x] महत्तम पूर्णांक फलन हो, तो f(x) की x = 0 पर अवलकनीयता की जाँच करो :

Ans. not derivable at x = 0

Sol. $$f(x)=\begin{cases}x\cdot\left(\dfrac{e^{[x]+|x|}-2}{[x]+|x|}\right) & x \neq 0\\ -1 & x = 0\end{cases}$$

$$f'(x)=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$$

$$=\lim_{h\to 0}\frac{h\cdot\left(e^{h}-2\right)+h}{h^{2}}$$

$$=\lim_{h\to 0}\frac{e^{h}-2}{h}=1$$

$$f'(0)=\lim_{h\to 0}\frac{f(0-h)-f(0)}{h}$$

$$=\lim_{h\to 0}\frac{-h\left(e^{h-1}-2\right)+h-1}{(-1+h)h}$$

$$=\lim_{h\to 0}\frac{h\left(3-e^{h-1}\right)-1}{h(-1+h)}=\text{DNE}$$

$\therefore$ f(x) is not differentiable at x = 0.

27. Discuss the continuity & the derivability in [0 , 2] of $f(x)=\begin{cases}|2x-3|\,[x] & \text{for } x\geq 1\\ \sin\frac{\pi x}{2} & \text{for } x<1\end{cases}$

where [ ] denote greatest integer function :

फलन $f(x)=\begin{cases}|2x-3|\,[x] & \text{for } x\geq 1\\ \sin\frac{\pi x}{2} & \text{for } x<1\end{cases}$ की [0 , 2] के अन्तराल में सतत्ता तथा अवकलनीयता की जाँच करो।

जहाँ [ ] महत्तम पूर्णांक फलन है :

Ans. f is conti. at x = 1 , 3/2 & disconti. at x = 2, f is not diff. at x =1, 3/2 , 2

Sol. $f(x)=\begin{cases}|2x-3|[x] & 1\leq x\leq 2\\ \sin\dfrac{\pi x}{2} & 0\leq x<1\end{cases}$

$$=\begin{cases}0 & x=0\\ \sin\dfrac{\pi x}{2} & 0<x<1\\ |2x-3| & 1\leq x<2\\ 2 & x=2\end{cases}$$

$$=\begin{cases}0 & x=0\\ \sin\dfrac{\pi x}{2} & 0<x<1\\ -2x+3 & 1\leq x<\dfrac{3}{2}\\ 2x-3 & \dfrac{3}{2}\leq x<2\\ 2 & x=2\end{cases}$$

$f(0)=0 \qquad f\left(0^+\right)=\lim\limits_{h\to 0}\sin\left(\dfrac{\pi}{2}h\right)=0$

$f\left(1^-\right)=\lim\limits_{h\to 0}\sin\left(\dfrac{\pi}{2}-\dfrac{\pi}{2}h\right)=+\lim\limits_{h\to 0}\cos\left(\dfrac{\pi}{2}h\right)=1$

$f(1)=1,\ f(1^+)=1$

$f(2)=2,\ f\left(2^-\right)=\lim\limits_{h\to 0}4-2h-3=1$

$f\left(\dfrac{3}{2}\right)=0, \qquad f\left(\dfrac{3^+}{2}\right)=0, \qquad f\left(\dfrac{3^-}{2}\right)=0$

$\therefore$ f(x) is continuous at $x = 1, 0 \& \frac{3}{2}$

discontinuous at $x = 2$ only $\rightarrow$ not differentiable at $x = 2$

$$f'(x) = \begin{cases} \cos\frac{\pi}{2}x \cdot \frac{\pi}{2} & 0 < x < 1 \\ -2 & 1 < x < \frac{3}{2} \\ 2 & \frac{3}{2} < x < 2 \end{cases}$$

$\therefore$ f(x) is not differentiable at $x = 1, \frac{3}{2} \& 2$.

28. Let f (x) = [3 + 4 sin x] (where [ ] denotes the greatest integer function). If sum of all the values of 'x' in $[\pi, 2\pi]$ where f(x) fails to be differentiable, is $\frac{k\pi}{2}$, then find the value of k :

माना कि f (x) = [3 + 4 sin x] (जहाँ [ ] महत्तम पूर्णांक फलन है।). यदि अन्तराल $[\pi, 2\pi]$ में ऐसे सारे 'x' का योग, जहाँ फलन अवकलनीय नहीं है, $\frac{k\pi}{2}$ है। तो k का मान बताओं :

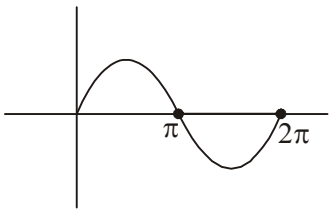
Ans. 24

Sol. f(x) = [3 + 4 sin x]

If $x \in [\pi, 2\pi]$ then $\sin x \in [-1, 0]$

$\Rightarrow 4 \sin x \in [-4, 0]$

$\Rightarrow 3 + 4 \sin x \in [-1, 3]$



$$f(x) = \begin{cases} -1 & -1 \le 3 + 4\sin x < 0 \\ 0 & 0 \le 3 + 4\sin x < 1 \\ 1 & 1 \le 3 + 4\sin x < 2 \\ 2 & 2 \le 3 + 4\sin x < 3 \\ 3 & 3 + 4\sin x = 3 \end{cases}$$

$$f(x) = \begin{cases} -1 & -1 \le \sin x < \frac{-3}{4} \\ 0 & -\frac{3}{4} \le \sin x < -\frac{1}{2} \\ 1 & -\frac{1}{2} \le \sin x < 0 \\ 2 & -\frac{1}{4} \le \sin x < 0 \\ 3 & \sin x = 0 \end{cases}$$

clearly f(x) is discontinuous at $\sin x = \frac{-3}{4}, \frac{-1}{2}, \frac{-1}{4}, 0$

$\sin x = 0 \rightarrow x = \pi, 2\pi$

$\sin x = \frac{-1}{2} \rightarrow x = \pi + \frac{\pi}{6}, 2\pi - \frac{\pi}{6}$

$\sin x = \frac{-1}{4} \rightarrow x = \pi + \alpha$ or $2\pi - \alpha$

$\sin x = \frac{-3}{4}$

$x = \pi + \beta$ or $2\pi - \beta$

$\Rightarrow$ sum of al values of $x = 12\pi = \frac{24\pi}{2}$

$\Rightarrow k = 24$

29. The function $f(x) = \begin{cases} ax(x-1)+b & \text{when } x < 1 \\ x-1 & \text{when } 1 \le x \le 3 \\ px^2+qx+2 & \text{when } x > 3 \end{cases}$

Find the values of the constants a, b, p, q so that

(i) f(x) is continuous for all x (ii) f' (1) does not exist (iii) f'(x) is continuous at x = 3 :

दिया है कि $f(x) = \begin{cases} ax(x-1)+b & \text{when } x < 1 \\ x-1 & \text{when } 1 \le x \le 3 \\ px^2+qx+2 & \text{when } x > 3 \end{cases}$

a, b, p, q का मान ज्ञात करो, यदि

(i) फलन f(x), x के प्रत्येक मान के लिये सतत् है। (ii) f'(1) विद्यमान नही है (iii) x = 3 पर f'(x) सतत् हो, तो :

Ans. $a \neq 1, b = 0, p = \frac{1}{3}$ and $q = -1$

Sol. $f(x) = \begin{cases} ax^2 - ax + b & x < 1 \\ x-1 & 1 \le x \le 3 \\ px^2+qx+2 & x > 3 \end{cases}$

(i) Given f(x) is continuous for all x

at x = 1

$f(1^-) = b = 0 = f(1) = f(1^+)$

$\Rightarrow b = 0$

at x = 3

$f(3^-) = f(3) = 2 = f(3^+) = 9p + 3q + 2$ ...(ii)

Now given $f'(1)$ does not exist

$$f'(1^+) = \lim_{h\to 0} \frac{f(1+h)-f(1)}{h} = \lim_{h\to 0} \frac{h-0}{h} = 1$$

$$f'(1^-) = \lim_{h\to 0} \frac{f(1-h)-f(1)}{h}$$

$$= \lim_{h\to 0} \frac{a(1-h)(1-h-1)+b}{-h}$$

$$= \lim_{h\to 0} a(1-h) = a$$

$\therefore a \neq 1$

$$f'(x) = \begin{cases} 2ax - a & x < 1 \\ 1 & 1 < x < 3 \\ 2px + q & x > 3 \end{cases}$$

Given $f'(x)$ is continuous at $x = 3$

$\Rightarrow f'(3^-) = 1$

$f'(3^+) = 6p + q$

$\Rightarrow 6p + q = 1 = 6p + q$ ...(iii)

$$f'(3) = \lim_{h\to 0} \frac{f(3+h)-f(3)}{h}$$

$$= \frac{p(3+h)^2 + 3q + qh}{h}$$

$$= \lim_{h\to 0} \frac{9p + 3q + ph^2 + 6ph + qh}{h} = 6p + q$$

By (ii) & (iii)

$$p = \frac{1}{3} \text{ \& } q = -1$$

30. Examine the function, $f(x) = x.\ \frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ $(a > 0)$ and $f(0) = 0$ for continuity and existence of the derivative at the origin :

फलन $f(x) = x.\ \frac{a^{1/x} - a^{-1/x}}{a^{1/x} + a^{-1/x}}$, $x \neq 0$ $(a > 0)$ और $f(0) = 0$ की $x = 0$ सतत्ता व अवकलनीयता का परिक्षण करो :

Ans. If $a \in (0, 1)$ $f'(0^+) = -1$; $f'(0^-) = 1 \Rightarrow$ continuous but not derivable

$a = 1$; $f(x) = 0$ which is constant $\Rightarrow$ continuous and derivable

If $a > 1$ $f'(0^-) = -1$; $f'(0^+) = 1 \Rightarrow$ continuous but not derivable

Sol.
$$f(x)=\begin{cases} x\cdot\dfrac{a^{\frac{1}{x}}-a^{-\frac{1}{x}}}{a^{\frac{1}{x}}+a^{-\frac{1}{x}}} & x\neq 0(a>0) \\ 0 & x=0 \end{cases}$$

$$f(x)=\begin{cases} x\left(\dfrac{a^{\frac{2}{x}}-1}{a^{\frac{2}{x}}+1}\right) & a>0, a\neq 1, x\neq 0 \\ 0 & a=1, x\neq 0 \\ 0 & x=0 \end{cases}$$

C-I $a=1$

$\underbrace{f(x)=0}_{\text{constant}} \rightarrow$ cont. and derivable

C-II $a>1$

$$f(0^-)=\lim_{h\to 0}\frac{-h\left(a^{-\frac{2}{h}}-1\right)}{a^{-\frac{2}{h}}+1}=0$$

$$f(0)^+=\lim_{h\to 0}\frac{h\left(a^{\frac{2}{h}}-1\right)}{a^{\frac{2}{h}}+1}=0$$

$$f'(0^+)=\lim_{h\to 0}\frac{f(0+h)-f(0)}{h}$$

$$=\lim_{h\to 0}\frac{a^{\frac{2}{h}}-1}{a^{\frac{2}{h}}+1}=+1$$

$$f'(0^-)=\lim_{h\to 0}\frac{f(0-h)-f(0)}{-h}=\lim_{h\to 0}\left(\frac{a^{-\frac{2}{h}}-1}{a^{-\frac{2}{h}}+1}\right)=1$$

$\Rightarrow$ f(x) is continuous at x = 0

Not differentiable at x = 0

C-III $0 < a < 1$

$f(0^-) = 0 = f(0^+)$

$$f'(0^-) = \lim_{h\to 0} \frac{-h\left(a^{-\frac{2}{h}} - 1\right)}{-h\left(a^{-\frac{2}{h}} + 1\right)} = 1$$

$$f'(0^+) = \lim_{h\to 0} \frac{h\left(a^{\frac{2}{h}} - 1\right)}{h\left(a^{-\frac{2}{h}} + 1\right)} = -1$$

$\Rightarrow$ f(x) is continuous at x = 0

not differentiable at x = 0.

31. Discuss the continuity & the derivability of 'f' where f(x) = degree of $(u^{x^2} + u^2 + 2u - 3)$ at $x = \sqrt{2}$:

यदि f(x) = समीकरण $(u^{x^2} + u^2 + 2u - 3)$ की डिग्री है जहाँ $x = \sqrt{2}$ है, तो फलन f(x) की सतत्ता तथा अवकलनीयता की जाँच करो :

Ans. continuous but not derivable at $x = \sqrt{2}$

Sol. continuous but not derivable at $x = \sqrt{2}$

32. If $f(x) = \begin{cases} \dfrac{\sin[|x|]\pi}{a x^4} + a x^2 + b; & -2 \le x < -1 \\ \\ 3\sec \pi x + \cot^{-1} x; & -1 \le x \le \dfrac{-3}{4} \end{cases}$

is differentiable in $\left(-2, \frac{-3}{4}\right)$ then $a = \frac{1}{\lambda_1}$ and $b = \frac{3\pi - \lambda_2}{4}$. Find the value of $(\lambda_1 + \lambda_2)$.

Note: [y] denotes largest integer $\le y$:

यदि $f(x)=\begin{cases}\dfrac{\sin[|x|]\pi}{a\,x^4}+a\,x^2+b; & -2\le x<-1\\ \\ 3\sec\pi x+\cot^{-1}x; & -1\le x\le\dfrac{-3}{4}\end{cases}$

Note: [y] महत्तम पूर्णांक फलन है :

$\left(-2,\dfrac{-3}{4}\right)$ में अवकलनीय फलन है तो $a=\dfrac{1}{\lambda_1}$ तथा $b=\dfrac{3\pi-\lambda_2}{4}$ है। $(\lambda_1+\lambda_2)$ का मान ज्ञात कीजिये।

Ans. 17

Sol. Clearly $f(x)=\begin{cases}a\,x^2+b; & -2\le x<-1\\ \\ 3\sec\pi x+\cot^{-1}x; & -1\le x\le\dfrac{-3}{4}\end{cases}$

As the function is differentiable in $\left(-2,\dfrac{-3}{4}\right)$, so function is differentiable at $x=-1$.

$\therefore \quad f'(-1^-)=f'(-1^+)$ .....(1)

$$f'(-1^-)=\lim_{h\to0}\frac{f(-1-h)-f(-1)}{-h}=\lim_{h\to0}\frac{\left(a(-1-h)^2+b\right)-(a+b)}{-h}$$

$$=\lim_{h\to0}\frac{a\left(1+h^2+2h\right)-a}{-h}=\lim_{h\to0}\frac{ah(h+2)}{-h}=-2a \quad ....(2)$$

$$f'(-1^+)=\lim_{h\to0}\frac{f(-1+h)-f(-1)}{h}=\lim_{h\to0}\frac{3\sec\pi(-1+h)+\cot^{-1}(-1+h)-3\sec(-\pi)-\cot^{-1}(-1)}{h}$$

$$=\lim_{h\to0}\frac{-3\sec\pi h+\cot^{-1}(-1+h)+3-\frac{3\pi}{4}}{h}=\lim_{h\to0}\frac{3(1-\sec\pi h)}{h}+\lim_{h\to0}\frac{\cot^{-1}(-1+h)-\frac{3\pi}{4}}{h}$$

$$=\lim_{h\to0}\frac{3(\cos\pi h-1)}{h\cos\pi h}+\lim_{\theta\to\frac{3\pi^-}{4}}\frac{\theta-\frac{3\pi}{4}}{1+\cot\theta}\quad(\text{Let }\cot^{-1}(-1+h)=\theta)$$

$$=(3\times0)+\lim_{\theta\to0^-}\frac{\alpha}{1+\cot\left(\frac{3\pi}{4}+\alpha\right)}\qquad\left(\text{Put}\quad\theta=\frac{3\pi}{4}+\alpha\right)$$

$$=0+\lim_{\alpha\to0^-}\frac{\alpha}{1+\tan\left(\frac{\pi}{4}+\alpha\right)}=\lim_{\alpha\to0^-}\frac{\alpha}{1-\left(\frac{1+\tan\alpha}{1-\tan\alpha}\right)}=\lim_{\alpha\to0}\frac{\alpha(1-\tan\alpha)}{(-2\tan\alpha)}=\frac{-1}{2}\qquad.....(3)$$

$\therefore$ Using equation (2) and equation (3) in equation (1), we get

$$-2a = \frac{-1}{2} \Rightarrow a = \frac{1}{4} = \frac{1}{\lambda_1} \text{ (Given)}$$

$$\therefore \quad \lambda_1 = 4$$

Also function will be continuous at $x = -1$.

$$\therefore \quad \lim_{x \to -1^-} f(x) = \lim_{x \to -1^+} f(x)$$

$$\Rightarrow \quad a + b = -3 + \frac{3\pi}{4}$$

$$\therefore \quad b = \frac{-1}{4} - 3 + \frac{3\pi}{4} = \frac{3\pi - 13}{4} = \frac{3\pi - \lambda_2}{4}$$

$$\Rightarrow \quad \lambda_2 = 13$$

Hence $(\lambda_1 + \lambda_2) = 4 + 13 = 17$

33. Let $f(x) = \begin{cases} a\cot^{-1}\left(\frac{b+x}{4}\right), & \frac{-2}{3} < x < 0 \\ 2, & x = 0 \\ \frac{\ell n(1-cx)}{x}, & 0 < x < \frac{2}{3} \end{cases}$

If the function $f(x)$ is differentiable at $x = 0$, then find the value of $(b^2 - 2a + c^6)$:

माना $f(x) = \begin{cases} a\cot^{-1}\left(\frac{b+x}{4}\right), & \frac{-2}{3} < x < 0 \\ 2, & x = 0 \\ \frac{\ell n(1-cx)}{x}, & 0 < x < \frac{2}{3} \end{cases}$

यदि फलन $f(x)$, $x = 0$ पर अवकलनीय है तो $(b^2 - 2a + c^6)$ का मान ज्ञात कीजिये :

Ans. 48

Sol. As $f(x)$ is derivable at $x = 0$, so $f(x)$ is also continuous at $x = 0$.

$$\therefore \quad f(0^+) = \lim_{h \to 0} \frac{\ell n(1-ch)}{h} \left(\frac{0}{0}\right) = \lim_{h \to 0} \frac{-c \times \ell n(1-ch)}{-ch} = -c$$

$$\Rightarrow \quad -c = 2 \quad \Rightarrow \quad c = -2 \qquad \text{.....(1)}$$

$$\text{Now } f'(0^+) = \lim_{h \to 0} \frac{\frac{\ell n(1+2h)}{h} - 2}{h} = \lim_{h \to 0} \frac{\ell n(1+2h) - 2h}{h^2} = \lim_{h \to 0} \frac{\left(2h - \frac{(2h)^2}{2} + \text{.....}\right) - 2h}{h^2} = -2$$

Also $f'(0^-) = \lim_{h\to 0} \dfrac{a\cot^{-1}\left(\frac{b-h}{4}\right)-2}{-h} \left(\dfrac{0}{0}\right) = \lim_{h\to 0} \dfrac{\frac{-a}{1+\left(\frac{b-h}{4}\right)^2}\times\frac{-1}{4}}{-1} = \dfrac{-4a}{b^2+16}$

As $f'(0^-) = f'(0^+)$, so $\dfrac{-4a}{b^2+16} = -2$

$\Rightarrow$ $2a = b^2 + 16$

$\therefore$ $b^2 - 2a = -16$ .....(2)

Hence $(b^2 - 2a + c^6) = -16 + 64 = 48$ (Using equation (1) and equation (2))

DETERMINATION OF FUNCTION

34. If $|f(x_1) - f(x_2)| \le (x_1 - x_2)^2$, for all $x_1, x_2 \in R$. Find the equation of tangent to the curve $y = f(x)$ at the point (1, 2) :

यदि $|f(x_1) - f(x_2)| \le (x_1 - x_2)^2$, $x_1, x_2 \in R$ तो वक्र $y = f(x)$ का (1, 2) पर स्पर्श रेखा का समीकरण बताओं :

Ans. $y - 2 = 0$

Sol. Put $x_1 = x + h$ & $x_2 = x$

$|f(x+h) - f(x)| \le h^2$

$\lim_{h\to 0} \left|\dfrac{f(x+h) - f(x)}{h}\right| \le \lim_{h\to 0} h$

$|f'(x)| \le 0$

Possible only if $f'(x) = 0$

$f(x) = c$

at point (1, 2) $f(x) = 2$

$y = 2$

35. Let f be a function that is differentiable every where and that has the following properties:

(i) $f(x+h) = f(x) \cdot f(h)$ (ii) $f(x) > 0$ for all real x. (iii) $f'(0) = -1$

Use the definition of derivative to find $f'(x)$ in terms of $f(x)$:

यदि फलन f के सभी जगहों पर अवकलनीय है तथा निम्नलिखित गुण रखता है :

(i) $f(x+h) = f(x) \cdot f(h)$ (ii) $f(x) > 0 \ \forall\ x \in R$ (iii) $f'(0) = -1$

तो अवकलनीयता की परिभाषा से $f'(x)$ का मान $f(x)$ के पदों में ज्ञात करो।

Ans. $f'(x) = -f(x)$

Sol. $f'(x) = \lim_{h\to 0} \dfrac{f(x+h) - f(x)}{h}$

$= \lim_{h\to 0} \dfrac{f(x)(f(h) - 1)}{h}$

$= f(x)\lim_{h\to 0}\frac{f(h)-1}{h}$

$f(0+0) = (f(0))^2$

$\Rightarrow f(0) = 0 \quad \text{or} \quad f(0) = 1$

$\because f(x) > 0 \ \forall\ x$

$\Rightarrow f(0) = 1$

$\Rightarrow f'(x) = f(x)\cdot f'(0) = -f(x)$

$\Rightarrow \ln(f(x)) = -x + c$

$\because f(0) = 1$

$\Rightarrow c = 0$

$\Rightarrow f(x) = e^{-x}$

36. A derivable function $f: R^+ \to R$ satisfies the condition $f(x) - f(y) \geq l\text{n}(x/y) + x - y$ for every $x, y \in R^+$. If g denotes the derivative of f then compute the value of the sum $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$:

एक अवकलनीय फलन $f: R^+ \to R$ समीकरण $f(x) - f(y) \geq l\text{n}(x/y) + x - y$ जहाँ $x, y \in R^+$ को सन्तुष्ट करता है यदि f के अवकलन का मान g हो, तो $\sum_{n=1}^{100} g\left(\frac{1}{n}\right)$ का मान ज्ञात करो :

Ans. 5150

Sol. $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$

$\geq \lim_{h\to 0}\frac{\ell n\left(\frac{x+h}{x}\right)+x+h-x}{h}$

$\geq \lim_{h\to 0}\frac{\ell n\left(1+\frac{h}{x}\right)}{h}+1 \quad \geq \frac{1}{x}+1 \qquad \text{.....(i)}$

$f'(x) = \lim_{h\to 0}\frac{f(x-h)-f(x)}{-h}$

$\leq \lim_{h\to 0}\frac{\ell n\left(\frac{x-h}{x}\right)+x-h-x}{-h}$

$\leq \lim_{h\to 0}\frac{\ell n\left(1-\frac{h}{x}\right)}{-h}+1 \leq \frac{1}{x}+1 \quad \text{.....(ii)}$

from (i) and (ii)

$$\Rightarrow \quad f'(x) = \frac{1}{x} + 1$$

$$\therefore \sum_{n=1}^{100} g\left(\frac{1}{n}\right) = g\left(\frac{1}{1}\right) + g\left(\frac{1}{2}\right) + ... + g\left(\frac{1}{100}\right)$$

$= (1 + 2 + 3 + ... 100) + 100 = 5150$

Paragraph for Question nos. 37 to 38

Consider a function f(x) in $[0, 2\pi]$ defined as :

$$f(x) = \begin{cases} [\sin x] + [\cos x] & ; \quad 0 \le x \le \pi \\ [\sin x] - [\cos x] & ; \quad \pi < x \le 2\pi \end{cases}$$

where [.] denotes greatest integer function then

माना $[0, 2\pi]$ में फलन f(x) इस प्रकार परिभाषित है

$$f(x) = \begin{cases} [\sin x] + [\cos x] & ; \quad 0 \le x \le \pi \\ [\sin x] - [\cos x] & ; \quad \pi < x \le 2\pi \end{cases}$$

जहां [.] महत्तम पूर्णांक फलन को दर्शाता है

37. Number of points where f(x) is non-derivable :

बिन्दुओं की संख्या जहां पर f(x) अवकलनीय नहीं है, होगी :

(A) 2 (B) 3 (C) 4 (D) 5

Ans. C

Sol.

$$f(x) = \begin{cases} 1 & x = 0 \\ 0 & 0 < x < \frac{\pi}{2} \\ 1 & x = \frac{\pi}{2} \\ -1 & \frac{\pi}{2} < x \le \pi \\ 0 & \pi < x < \frac{3\pi}{2} \\ -1 & x = \frac{3\pi}{2} \\ -1 & \frac{3\pi}{2} < x < 2\pi \\ -1 & x = 2\pi \end{cases}$$

f(x) is not differentiable at $x = 0, \frac{\pi}{2}, \pi, \frac{3\pi}{2}$

38. $\lim\limits_{x\to\left(\frac{3\pi}{2}\right)^+} f(x)$ equals

$\lim\limits_{x\to\left(\frac{3\pi}{2}\right)^+} f(x)$ का मान है –

(A) 0 (B) 1 (C) –1 (D) 2

Ans. C

Sol. $\lim\limits_{x\to\left(\frac{3\pi}{2}\right)^+} f(x) = -1$

Paragraph for Question nos. 39 to 41

Let $f(x) = \begin{cases} x[x] & 0 \le x < 2 \\ (x-1)[x] & 2 \le x \le 3 \end{cases}$ where [x] = greatest integer less than or equal to x, then

माना $f(x) = \begin{cases} x[x] & 0 \le x < 2 \\ (x-1)[x] & 2 \le x \le 3 \end{cases}$ है जहां [x] = महत्तम पूर्णांक फलन है तो

39. The number of values of x for $x \in [0, 3]$ where f(x) is discontinuous is :
$x \in [0, 3]$ में बिन्दुओं की संख्या जहां पर f(x) असतत है, होगी :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. C

Sol.

40. The number of values of x for $x \in [0, 3]$ where f(x) is non-differentiable is :
$x \in [0, 3]$ में बिन्दुओं की संख्या जहां पर f(x) अवकलनीय नहीं है, होगी :

(A) 0 (B) 1 (C) 2 (D) 3

Ans. D

Sol.

41. The number of integers in the range of y = f(x) is :
y = f(x) के परिसर में पूर्णांकों की संख्या होगी :

(A) 3 (B) 4 (C) 5 (D) 6

Ans. C

Sol. $f(x)=\begin{cases} 0 & 0\le x<1 \\ x & 1\le x<2 \\ 2(x-1) & 2\le x<3 \\ 6 & x=3 \end{cases}$

$\therefore$ f(x) is discontinuous at x= 1, 3. $\rightarrow$ not differentiable at 1 & 3.

$f'(x)=\begin{cases} 0 & 0<x<1 \\ 1 & 1<x<2 \\ 2 & 2<x<3 \end{cases}$

$\therefore$ f(x) is not differentiable at x = 1,2,3.

$R_f=[1,4)\cup\{0\}\cup\{6\}$

No. of integers = 5.

ONE OR MORE THAN ONE CORRECT TYPE

42. If $f(x)=\begin{cases} -x-\frac{\pi}{2}, & x\le\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2}<x\le 0 \\ x-1, & 0<x\le 1 \\ \ln x & x>1 \end{cases}$ then : **[JEE 2011]**

(A) f(x) is continuous at $x=-\frac{\pi}{2}$ (B) f(x) is not differentiable at x = 0

(C) f(x) differentiable at x = 1 (D) f(x) is differentiable at $x=-\frac{3}{2}$

यदि $f(x)=\begin{cases} -x-\frac{\pi}{2}, & x\le\frac{\pi}{2} \\ -\cos x, & -\frac{\pi}{2}<x\le 0 \\ x-1, & 0<x\le 1 \\ \ln x & x>1 \end{cases}$ तो : **[JEE 2011]**

(A) f(x), $x=-\frac{\pi}{2}$ पर सतत् होगा (B) f(x), x = 0 पर अवकलनीय नहीं होगा

(C) f(x), $x = 1$ पर अवकलनीय होगा (D) f(x), $x = -\frac{3}{2}$ पर अवकलनीय होगा

Ans. ABCD

Sol. $l\left(-\frac{\pi^-}{2}\right) = 0$, $l\left(-\frac{\pi^+}{2}\right) = 0$

$$f'(x) = \begin{cases} -1 & x \le \frac{\pi}{2} \\ \sin x & -\frac{\pi}{2} < x \le 0 \\ 1 & 0 < x \le 1 \\ \frac{1}{x} & x > 1 \end{cases}$$

$f'(0^-) = 0$, $f'(0^+) = 1$ $\therefore$ not differentiable at $x = 0$

$f'(1^-) = 1$, $f'(1^+) = 1$ $\therefore$ differentiable at $x = 1$

as $-\frac{3}{2} \in \left(-\frac{\pi}{2}, 0\right)$

$f'(x) = \sin x$ which is differentiable at $x = -\frac{3}{2}$

43. Which one of the following statements is NOT CORRECT :

(A) The derivative of a differentiable periodic function is a periodic function with the same period

(B) If f(x) and g(x) both are defined on the entire number line and are aperiodic then the function $F(x) = f(x) \cdot g(x)$ can not be periodic

(C) Derivative of an even differentiable function is an odd function and derivative of an odd differentiable function is an even function

(D) Every function f(x) can be represented as the sum of an even and an odd function

निम्न में से कौनसा कथन सही नहीं है :

(A) किसी अवकलनीय आवर्ती फलन का अवकलज उसी आवर्त में अवकलनीय होता है।

(B) यदि f(x) तथा g(x) समपूर्ण संख्या रेखा पर परिभाषित है तथा आवर्ती नहीं है तो $F(x) = f(x) \cdot g(x)$ आवर्ती नहीं हो सकते

(C) सम अवकलनीय फलन का अवकलज विषम होता है तथा विषम अवकलनीय फलन का अवकलज सम होता है।

(D) प्रत्येक फलन f(x) एक सम तथा विषम फलन के योग के रूप में व्यक्त किया जा सकता है।

Ans. BD

Sol. (A) $f(x+T) = f(x) \forall x \in R$

$$f'(x) = \lim_{h \to 0} \frac{f(x+h) - f(x)}{h}$$

$$\Rightarrow f'(x+T)=\lim_{h\to 0}\frac{f(x+h+T)-f(x+T)}{h}$$

$=f'(x)$

(B) $\left.\begin{matrix}f(x)=e^{x}\\ g(x)=e^{-x}\end{matrix}\right]\rightarrow$ aperiodic

$\Rightarrow f(x)\cdot g(x)=1\rightarrow$ Periodic

(C) f(–x) = f(x)

$$f'(x)=\lim_{h\to 0}\frac{f(x+h)-f(x)}{h}$$

$$f'(-x)=\lim_{h\to 0}\frac{f(-x+h)-f(-x)}{h}$$

$$=\lim_{h\to 0}\frac{f(x-h)-f(x)}{h}$$

$=-f'(x)$

(D) Domain should be symmetric about origin.

Take $f(x)=\log_e x$.

44. If $f(x)=\begin{cases}\frac{x\,.\,\ell n(\cos x)}{\ell n(1+x^2)} & x\neq 0\\ 0 & x=0\end{cases}$ then :

(A) f is continuous at $x=0$

(B) f is continuous at $x=0$ but not differentiable at $x=0$

(C) f is differentiable at $x=0$

(D) f is not continuous at $x=0$

यदि $f(x)=\begin{cases}\frac{x\,.\,\ell n(\cos x)}{\ell n(1+x^2)} & x\neq 0\\ 0 & x=0\end{cases}$ तो :

(A) f, $x=0$ पर संत्त है

(B) f, $x=0$ पर संत्त है परन्तु अवकलनीय नहीं है :

(C) f, $x=0$ अवकलनीय है

(D) f, $x=0$ संत्त नहीं है

Ans. AC

Sol. $f'(0^+)=\underset{h\to 0}{\text{Lim}}\frac{h\,ln(\cos h)}{h\,ln(1+h^2)}=\underset{h\to 0}{\text{Lim}}\frac{ln(\cos h)^{1/h^2}}{\frac{ln(1+h^2)}{h^2}}=\underset{h\to 0}{\text{Lim}}\frac{1}{h^2}(\cos h-1)=-\frac{1}{2}$; |||ly $f'(0^-)=-\frac{1}{2}$

hence f is continuous and derivable at $x=0$

45. Let $f(x)$ be defined as follows: $f(x)=\begin{cases} x^6 & , \ x^2>1 \\ x^3 & , \ x^2 \le 1 \end{cases}$. Then $f(x)$ is:

(A) continuous everywhere (B) differentiable everywhere

(C) discontinuous at $x=-1$ (D) not differentiable at $x=1$

यदि $f(x)=\begin{cases} x^6 & , \ x^2>1 \\ x^3 & , \ x^2 \le 1 \end{cases}$. तो $f(x)$ होगा :

(A) सभी बिन्दुओं पर संत्त है (B) सभी बिन्दुओं पर अवकलनीय है

(C) $x=-1$ पर असंत्त है (D) $x=1$ पर अवकलनीय नहीं है

Ans. CD

Sol. $f(x)=\begin{cases} x^6 & x^2>1 \\ x^3 & x^2 \le 1 \end{cases}$

$$f(x)=\begin{cases} x^6 & x<-1 \\ x^3 & -1 \le x \le 1 \\ x^6 & x>1 \end{cases}$$

$\therefore$ f(x) is discontinuous at $x=1$.

at $x=1$

$$f'(x)=\begin{cases} 3x^2 & x<1 \\ 6x^5 & x>1 \end{cases}$$

$f'(1^+)=6$

$f'(I^-)=3$

$\Rightarrow$ f(x) is not differentiable at $x=1$.

46. Which of the following functions f has/have a removable discontinuity at the indicated point :

(A) $f(x)=\dfrac{x^2-2x-8}{x+2}$ at $x=-2$ (B) $f(x)=\dfrac{x-7}{|x-7|}$ at $x=7$

(C) $f(x)=\dfrac{x^3+64}{x+4}$ at $x=-4$ (D) $f(x)=\dfrac{3-\sqrt{x}}{9-x}$ at $x=9$

दिये गये बिन्दु पर निम्नलिखित फलन में कौन सा फलन ऐसा जिसमें विस्थापनीय असतत्ता है :

(A) $f(x)=\dfrac{x^2-2x-8}{x+2}$, $x=-2$ पर (B) $f(x)=\dfrac{x-7}{|x-7|}$, $x=7$ पर

(C) $f(x)=\dfrac{x^3+64}{x+4}$, $x=-4$ पर (D) $f(x)=\dfrac{3-\sqrt{x}}{9-x}$, $x=9$ पर

Ans. ACD

Sol. (A) $f(x)=\dfrac{x^2-2x-8}{x+2}=\dfrac{(x+2)(x-4)}{(x+2)}$

LHL = –6 = RHL

f(–2) = Not defined → Removable discontinuity

(B) $f(x)=\dfrac{x-7}{|x-7|}$

at x = 7

LHL = –1 → Non removable

RHL = 1

(C) $f(x)=\dfrac{x^3+64}{x+4}$ $\quad x=-4$

$$\left.\begin{array}{c} LHL=16=RHL \\ f(-4)=\text{not defined} \end{array}\right] \rightarrow \text{Removable discontinuity}$$

(D) $f(x)=\dfrac{3-\sqrt{x}}{9-x}$

at x = 9

$$\left.\begin{array}{c} LHL=\dfrac{1}{6}=RHL \\ f(9)=\text{not defined} \end{array}\right] \rightarrow \text{Removable discontinuity}$$

47. In which of the following cases the given equations has atleast one root in the indicated interval :

(A) x – cos x = 0 in $(0, \pi/2)$

(B) x + sin x = 1 in $(0, \pi/6)$

(C) $\dfrac{a}{x-1}+\dfrac{b}{x-3}=0$, a, b > 0 in (1, 3)

(D) f(x) – g(x) = 0 in (a, b) where f and g are continuous on [a, b] and f(a) > g(a) and f(b) < g(b)

निम्न में से कौन से फलन का दिये गये संगत अन्तराल में कम से कम एक मूल होगा :

(A) x – cos x = 0, $(0, \pi/2)$ में (B) x + sin x = 1, $(0, \pi/6)$ में

(C) $\dfrac{a}{x-1}+\dfrac{b}{x-3}=0$, a, b > 0 , (1, 3) में

(D) f(x) – g(x) = 0, (a, b) में जबकि f तथा g अन्तराल [a, b] में सतत् हो तथा f(a) > g(a) और f(b) < g(b).

Ans. ABCD

Sol. f(x) = x – cos x

$x \in \left(0, \frac{\pi}{2}\right) \rightarrow$ f(x) is continuous & differentiable

$f(0) = -1$

$f\left(\frac{\pi}{2}\right) = \frac{\pi}{2}$

$\Rightarrow f(0) \cdot f\left(\frac{\pi}{2}\right) < 0$

$\Rightarrow \exists c \in \left(0, \frac{\pi}{2}\right)$ s.t. $f(c) = 0$

$\Rightarrow c - \cos c = 0$

(B) $f(x) = x + \sin x - 1$

$x \in \left(0, \frac{\pi}{6}\right) \rightarrow$ f(x) is continuous & differentiable

$f(0) = -1$

$f\left(\frac{\pi}{6}\right) = \frac{\pi}{6} - \frac{1}{2} = \frac{\pi - 3}{6}$

$\Rightarrow f(0) \cdot f\left(\frac{\pi}{6}\right) < 0$

$\Rightarrow \exists c \in \left(0, \frac{\pi}{6}\right)$ s.t. $f(c) = 0$

(C) $f(x) = \frac{a}{x-1} + \frac{b}{x-3}$, $a, b > 0$

in $(1,3) \rightarrow$ f(x) is continuous & differentiable

$f\left(1^+\right) \rightarrow \infty$

$f\left(3^-\right) \rightarrow \infty$

$\Rightarrow \exists c \in (1,3)$ s.t. $\frac{a}{c-1} + \frac{b}{c-3} = 0$

(D) f(x) and g(x) are continuous on [a, b].

Let $h(x) = f(x) - g(x) \rightarrow$ continuous on [a, b].

$h(a) > 0$ & $h(b) < 0$

$\Rightarrow \exists c \in (a,b)$ s.t. $h(c) = 0$

48. Indicate all correct alternatives if, $f(x) = \frac{x}{2} - 1$, then on the interval $[0, \pi]$:

(A) $\tan(f(x))$ & $\frac{1}{f(x)}$ are both continuous (B) $\tan(f(x))$ & $\frac{1}{f(x)}$ are both discontinuous

(C) $\tan(f(x))$ & $f^{-1}(x)$ are both continuous (D) $\tan(f(x))$ is continuous but $\frac{1}{f(x)}$ is not

यदि $f(x) = \frac{x}{2} - 1, [0, \pi]$ तो सही विकल्प को चुनों :

(A) $\tan(f(x))$ तथा $\frac{1}{f(x)}$ दोनों सतत् है। (B) $\tan(f(x))$ तथा $\frac{1}{f(x)}$ दोनों असतत् है।

(C) $\tan(f(x))$ तथा $f^{-1}(x)$ दोनों सतत् है। (D) $\tan(f(x))$ सतत् है लेकिन $\frac{1}{f(x)}$ सतत् नहीं है।

Ans. CD

Sol. (i) $\tan f(x) = \tan\left(\frac{x}{2} - 1\right) \qquad x \in [0, p]$

$0 \le x \le \pi \Rightarrow -1 \le \frac{x}{2} - 1 \le \frac{\pi}{2} - 1$

By graph we say tan(f(x)) is continuous in $[0, \pi]$

(ii) $\frac{1}{f(x)} = \frac{2}{x-2}$ is not defined at $x = 2 \in [0, \pi]$

(iii) $y = \frac{x-2}{2}$

$\Rightarrow \quad f^{-1}(x) = 2x + 2$ is continuous in R.

49. Which of the following function(s) not defined at $x = 0$ has/have non-removable discontinuity at the point $x = 0$:
निम्न में से कौनसे फलन $x = 0$ पर परिभाषित नहीं है तथा $x = 0$ पर अविस्थापनीय असतत्ा है :

(A) $f(x) = \frac{1}{1 + 2^{\frac{1}{x}}}$ (B) $f(x) = \arctan\frac{1}{x}$ (C) $f(x) = \frac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1}$ (D) $f(x) = \frac{1}{\ell n|x|}$

Ans. ABC

Sol. (A) $f(x) = \frac{1}{1 + 2^{\frac{1}{x}}} \rightarrow$ not defined at $x = 0$

$$LHL = \lim_{h \to 0} f(0 - h) = \lim_{h \to 0} \frac{1}{1 + 2^{-\frac{1}{h}}} = 1$$

$$RHL = \lim_{h \to 0} \frac{1}{1 + 2^{\frac{1}{h}}} = 0$$

$\Rightarrow$ f(x) has non removable discontinuous at x = 0.

(B) $f(x) = \tan^{-1}\left(\frac{1}{x}\right) \rightarrow$ not defined at x = 0.

$$\left.\begin{aligned} LHL &= -\lim_{h\to 0} \tan^{-1}\left(\frac{1}{h}\right) = -\frac{\pi}{2} \\ RHL &= \lim_{h\to 0} \tan^{-1}\left(\frac{1}{h}\right) = \frac{\pi}{2} \end{aligned}\right] \rightarrow \text{non-removable discontinuity}$$

(C) $f(x) = \dfrac{e^{\frac{1}{x}} - 1}{e^{\frac{1}{x}} + 1} \rightarrow$ not defined at x = 0

$$\left.\begin{aligned} LHL &= \lim_{h\to 0} \frac{e^{\frac{1}{h}} - 1}{e^{\frac{1}{h}} + 1} = -1 \\ RHL &= \lim_{h\to 0} \frac{e^{-\frac{1}{h}} - 1}{e^{-\frac{1}{h}} + 1} = 1 \end{aligned}\right] \rightarrow \text{non removable discontinuity}$$

(D) $f(x) = \dfrac{1}{\ln|x|} \rightarrow$ not defined at x = 0

$$LHL = \lim_{h\to 0} \frac{1}{\ln|h|} = RHL = 0 \rightarrow \text{removable discontinuity}$$

50. The function, $f(x) = [|x|] - |[x]|$ where $[x]$ denotes greatest integer function *:*

(A) is continuous for all positive integers

(B) is discontinuous for all non positive integers

(C) has finite number of elements in its range

(D) is such that its graph does not lie above the x – axis

फलन $f(x) = [|x|] - |[x]|$ है जहां $[x]$ महत्तम पूर्णांक फलन को दर्शाता है *:*

(A) सभी धनात्मक पूर्णांक मानों के लिए सतत है

(B) सभी अधनात्मक पूर्णांक मानों के लिए असतत है

(C) इसके परिसर में परिमित संख्या में अवयव है

(D) इस प्रकार है कि इसका आरेख x – अक्ष के ऊपर विद्यमान नहीं है

Ans. ABCD

Sol. $f(x) = [|x|] - |[x]|$

Let $x = \alpha \in I^+$

$\lim_{h\to 0} f(\alpha+h) = [\alpha+h] - |[\alpha+h]|$

$= \alpha - |\alpha| = 0$

$\lim_{h\to 0} f(\alpha-h) = [|\alpha-h|] - |[\alpha-h]|$

$= (\alpha-1) - (\alpha-1)$

$f(\alpha) = 0$

$\therefore$ f(x) is continuous for all positive integer

Let $x = \alpha \in I^-$

$f(\alpha) = [|\alpha|] - |[\alpha]|$

$= |\alpha| - |\alpha| = 0$

$\lim_{h\to 0} f(\alpha-h) = \lim_{h\to 0} [|\alpha-h|] - |[\alpha-h]|$

$= \lim_{h\to 0} [h-\alpha] - |\alpha-1|$

$= (-\alpha) - (1-\alpha)$

$= -1$

$\lim_{h\to 0} f(\alpha+h) = \lim_{h\to 0} [|\alpha+h|] - |[\alpha+h]|$

$= \lim_{h\to 0} [-h-\alpha] - |\alpha|$

$= (-\alpha-1) + \alpha = -1$

$\therefore$ f(x) is discontinuous $\forall \alpha \in I$

Now $f(x) = [|x|] - |[x]|$

$$= \begin{cases} [x] - |[x]| & x \geq 0 \\ [-x] - |[x]| & x < 0 \end{cases}$$

$$= \begin{cases} 0 & x \geq 0 \\ [-x] + [x] & x < 0 \end{cases}$$

$$= \begin{cases} 0 & x \geq 0 \\ -1 & -1 < x < 0 \\ -1 & -2 < x < -1 \end{cases}$$

$\Rightarrow R_f = \{0, -1\}$

Graph of f(x) does not lie above the x-axis.

51. Given that the derivative f ¢ (a) exists. Indicate which of the following statement(s) is/are always True :
दिया है कि f ′ (a) का अस्तित्व है, तो निम्न में से कौन सा कथन हमेशा सत्य होगा :

(A) $f'(a) = \underset{h\to a}{\text{Limit}} \frac{f(h)-f(a)}{h-a}$ (B) $f'(a) = \underset{h\to 0}{\text{Limit}} \frac{f(a)-f(a-h)}{h}$

(C) $f'(a) = \underset{t\to 0}{\text{Limit}} \frac{f(a+2t)-f(a)}{t}$ (D) $f'(a) = \underset{t\to 0}{\text{Limit}} \frac{f(a+2t)-f(a+t)}{2t}$

Ans. AB

Sol. (C) is false and is True only if $f'(a) = 0$ limit is $2f'(a)$. In (D) same logic limit is $\frac{1}{2}f'(a)$]

52. The function $f(x) = \sqrt{1-\sqrt{1-x^2}}$:

(A) has its domain $-1 \le x \le 1$.

(B) has finite one sided derivates at the point $x = 0$.

(C) is continuous and differentiable at $x = 0$.

(D) is continuous but not differentiable at $x = 0$

दिया है $f(x) = \sqrt{1-\sqrt{1-x^2}}$ तो फलन f(x) :

(A) का प्रान्त $-1 \le x \le 1$ होगा।

(B) का $x = 0$ पर परिमित तथा एक तरफा अवकलन होगा।

(C) $x = 0$ पर सतत् तथा अवकलनीय होगा।

(D) $x = 0$ पर फलन सतत् तो होगा लेकिन अवकलनीय नहीं होगा।

Ans. ABD

Sol. $f'(0^+) = \frac{1}{\sqrt{2}}$; $f'(0^-) = -\frac{1}{\sqrt{2}}$; $f(x) = \frac{\sqrt{x^2}}{\sqrt{1+\sqrt{1-x^2}}} = \frac{|x|}{\sqrt{1+\sqrt{1-x^2}}}$]

53. Let $f: (0, \infty) \to [1, \infty)$ be defined as $f(x) = \begin{cases} \frac{1}{x}, & 0 < x < 1 \\ \frac{x}{[x]}, & 1 \le x < \infty \end{cases}$:

where $[x]$ denotes largest integer less than or equal to x. Then which of the following statements is (are) correct ?

(A) $f(x)$ is neither injective nor surjective.

(B) $f(x)$ is continuous but non-derivable at $x = 1$.

(C) $f(x)$ has non-removable discontinuity of finite type at $x = 2$ with jump equals $\frac{1}{2}$.

(D) $f'(e) = \frac{1}{2}$

माना $f:(0,\infty)\to[1,\infty)$ परिभाषित है $f(x)=\begin{cases}\dfrac{1}{x}, & 0<x<1\\ \dfrac{x}{[x]}, & 1\le x<\infty\end{cases}$:

जहाँ $[x]$ महत्तम पूर्णांक फलन है तो निम्न में से कौनसे कथन सत्य है ?

(A) $f(x)$ ना तो एकेकी है ना ही आच्छादक

(B) $f(x)$, $x=1$ पर सतत् है लेकिन अवकलनीय नही

(C) $f(x)$, $x=2$ पर एक निश्चित प्रकार के अविस्थापनीय असतत्ता रखता है जहाँ $\frac{1}{2}$ का उछाल है

(D) $f'(e)=\frac{1}{2}$.

Ans. BD

Sol. (A) Clearly $f\left(\frac{2}{3}\right)=\frac{3}{2}$ and $f\left(\frac{3}{2}\right)=\frac{3}{2}$.

So, $f(x)$ is many one function.

Also range of $f(x)=[1,\infty)$, so

$f(x)$ is onto also. $\Rightarrow$ (A) is incorrect]

(B) $f(1^-)=1=f(1^+)=f(1)$

But $f'(1^-)=-1$ and $f'(1^+)=1$

So, $f(x)$ is continuous but non-derviable at $x=1$ $\Rightarrow$ (B) is correct.

(C) $f(2^+)=1$, $f(2^-)=2$

$\therefore$ $\lim\limits_{x\to2} f(x)$ does not exist.

Also jump of discontinuity $=|2-1|=1\Rightarrow$ (C) is correct

(D) Clearly $f'(e)=\frac{1}{2}$ (verify)]

54. $f(x)=|x[x]|$ in $-1\le x\le 2$, where $[x]$ is greatest integer $\le x$ then $f(x)$ is :

(A) continuous at $x=0$ (B) discontinuous $x=0$

(C) not differentiable at $x=2$ (D) differentiable at $x=2$

यदि $f(x)=|x[x]|$, $-1\le x\le 2$, जहाँ $[x]$ महत्तम पूर्णांक फलन हो, तो $f(x)$:

(A) $x=0$ पर सतत् होगा। (B) $x=0$ पर असतत् होगा

(C) $x=2$ पर अवकलनीय नहीं होगा (D) $x=2$ पर अवकलनीय होगा

Ans. AC

Sol. $f(x)=|x[x]|$ $\quad -1\le x\le 2$

$$= \begin{cases} |-x| & -1x \le x < 0 \\ 0 & 0 \le x < 1 \\ |x| & 1 \le x < 2 \\ 4 & x = 2 \end{cases}$$

$$= \begin{cases} (-x) & -1 \le x < 0 \\ 0 & 0 \le x < 1 \\ x & 1 \le x < 2 \\ 4 & x = 2 \end{cases}$$

f(x) is continuous at x = 0.

$\therefore$ f(x) is not differentiable at x = 2 as f(x) is not continuous at x = 2.

55. f(x) =1 + x.[cosx] in $0 < x \le \pi/2$, where [ ] denotes greatest integer function then :

(A) It is continuous in $0 < x < \pi/2$ (B) It is differentiable in $0 < x < \pi/2$

(C) Its maximum value is 2 (D) It is not differentiable in $0 < x < \pi/2$

यदि f(x) =1 + x.[cosx] $0 < x \le \pi/2$, जहाँ [ ] महत्तम पूर्णांक फलन हो, तो :

(A) फलन $0 < x < \pi/2$ में सतत् है। (B) फलन $0 < x < \pi/2$ में अवकलनीय है।

(C) फलन का अधिकतम मान 2 होगा (D) फलन $0 < x < \pi/2$ में अवकलनीय नही होगा।

Ans. AB

Sol. $f(x) = 1 + x \cdot [\cos x]; 0 < x \le \frac{\pi}{2}$

$\Rightarrow f(x) = 1 \rightarrow$ continuous & differentiable in $\left(0, \frac{\pi}{2}\right)$

56. Select the correct statements :

(A) The function f defined by $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \le 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$ is neither differentiable nor continuous at $x = 1$.

(B) The function $f(x) = x^2 |x|$ is twice differentiable at $x = 0$.

(C) If f is continuous at x = 5 and $f(5) = 2$ then $\lim\limits_{x \to 2} f(4x^2 - 11)$ exists.

(D) If $\lim\limits_{x \to a} (f(x) + g(x)) = 2$ and $\lim\limits_{x \to a} (f(x) - g(x)) = 1$ then $\lim\limits_{x \to a} f(x) \cdot g(x)$ need not exist

सही कथन का चुनाव कीजिये :

(A) f द्वारा परिभाषित फलन $f(x) = \begin{cases} 2x^2 + 3 & \text{for } x \le 1 \\ 3x + 2 & \text{for } x > 1 \end{cases}$, x = 1 पर ना तो अवकलनीय और ना ही सतत

(B) फलन $f(x) = x^2 |x|$, x = 0 पर दो बार अवकलनीय है

(C) यदि f, $x = 5$ पर सतत है तथा $f(5) = 2$ है तो $\lim\limits_{x\to 2} f(4x^2 - 11)$ विद्यमान है

(D) यदि $\lim\limits_{x\to a}\big(f(x)+g(x)\big) = 2$ तथा $\lim\limits_{x\to a}\big(f(x)-g(x)\big) = 1$ है तो $\lim\limits_{x\to a} f(x)\cdot g(x)$ आवश्यक नहीं है कि विद्यमान हो

Ans. BC

Sol. (A) $f(x) = \begin{cases} 2x^2+3 & x \le 1 \\ 3x+2 & x > 1 \end{cases}$

at $x = 1$; $f(x)$ is continuous & not differentiable

(B) $f(x) = \begin{cases} -x^3 & x < 0 \\ x^3 & x \ge 0 \end{cases}$

$$f'(x) = \begin{cases} -3x^2 & x < 0 \\ 3x^2 & x \ge 0 \end{cases}$$

$$f''(x) = \begin{cases} -6x & x < 0 \\ 6x & x > 0 \end{cases}$$

$\therefore$ $f(x)$ is twice differentiable at $x = 0$.

(C) $\lim\limits_{x\to 2} f\left(4x^2-11\right) = f(5) = 2$

(D) $\lim\limits_{x\to a}\Big[\big(f(x)+g(x)\big)+f(x)-g(x)\Big] = 3$

$\lim\limits_{x\to a} f(x) = \dfrac{3}{2}$ & $\lim\limits_{x\to b} g(x) = \dfrac{1}{2}$

$\Rightarrow \lim\limits_{x\to a} f(x)\cdot g(x) = \dfrac{3}{4}$

57. Let $f(x) = |2x - 9| + |2x| + |2x + 9|$. Which of the following are true :

(A) $f(x)$ is not differentiable at $x = \dfrac{9}{2}$ (B) $f(x)$ is not differentiable at $x = \dfrac{-9}{2}$

(C) $f(x)$ is not differentiable at $x = 0$ (D) $f(x)$ is differentiable at $x = \dfrac{-9}{2}, 0, \dfrac{9}{2}$

माना $f(x) = |2x - 9| + |2x| + |2x + 9|$ है तो निम्न में से कौनसा सत्य है :

(A) $x = \dfrac{9}{2}$ पर $f(x)$ अवकलनीय नहीं है (B) $x = \dfrac{-9}{2}$ पर $f(x)$ अवकलनीय नहीं है

(C) $x = 0$ पर $f(x)$ अवकलनीय नहीं है (D) $x = \dfrac{-9}{2}, 0, \dfrac{9}{2}$ पर $f(x)$ अवकलनीय है

Ans. ABC

Sol. $f(x)=\underbrace{|2x-9|}_{x=\frac{9}{2}}+\underbrace{|2x|}_{x=0}+\underbrace{|2x+9|}_{x=-\frac{9}{2}}$

$$=\begin{cases}-2x+9-2x-2x-9 & x<-\frac{9}{2}\\ -2x+9-2x+2x+9 & -\frac{9}{2}\le x<0\\ -2x+9+2x+2x+9 & 0\le x<\frac{9}{2}\\ 6x-9+9 & x\ge\frac{9}{2}\end{cases}$$

$$=\begin{cases}-6x & x<-\frac{9}{2}\\ -2x+18 & -\frac{9}{2}\le x<0\\ 2x+18 & 0\le x<\frac{9}{2}\\ 6x & x\ge\frac{9}{2}\end{cases}$$

$\therefore$ f(x) is continuous $\forall\ x\in R$

$$f'(x)=\begin{cases}-6 & x<-\frac{9}{2}\\ -2 & -\frac{9}{2}<x<0\\ 2 & 0<x<\frac{9}{2}\\ 6 & x>\frac{9}{2}\end{cases}$$

$\therefore$ f(x) is not differentiable at $-\frac{9}{2},0,\frac{9}{2}$

MATCH THE COLUMN

58. **Column-I** **Column-II**

(A) If $f(x)$ is derivable at $x = 3$ & $f'(3) = 2$, then $\lim\limits_{h\to 0} \frac{f(3+h^2) - f((3-h^2)}{2h^2}$ equals (P) 0

(B) Let f(x) be a function satisfying the condition $f(-x) = f(x)$ for all real x. If $f'(0)$ exists, then its value is equal to (Q) 1

(C) For the function $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, x\neq 0 \\ 0, x=0 \end{cases}$, the derivative from the left, $f'(0^-)$ equals (R) 2

(D) The number of points at which the function $f(x) = \max. \{a-x, a+x, b\}, -\infty < x < \infty$, $0 < a < b$ cannot be differentiable is (S) 3

स्तम्भ -I **स्तम्भ-II**

(A) यदि $f(x)$, $x = 3$ पर अवकलनीय है तथा $f'(3) = 2$, (P) 0

तो $\lim\limits_{h\to 0} \frac{f(3+h^2) - f((3-h^2)}{2h^2}$ का मान होगा।

(B) यदि x के सभी वास्तविक मानो के लिये $f(x) = f(-x)$ है तो (Q) 1

$f'(0)$ का मान ज्ञात करें।

(C) फलन $f(x) = \begin{cases} \frac{x}{1+e^{1/x}}, x\neq 0 \\ 0, x=0 \end{cases}$, में $f'(0^-)$ का मान होगा (R) 2

(D) फलन $f(x) =$ अधिकतम $\{a-x, a+x, b\}, -\infty < x < \infty, 0 < a < b$ (S) 3

कितने बिन्दुओं पर अवकलनीय नहीं होगा।

Ans. (A) R, (B) P, (C) Q, (D) R

Sol. (A) $\lim\limits_{h\to 0} \frac{f(3+h)^2 - f(3-h)^2}{2h^2} = f'(3) = 2$

(B) $f'(0) = \lim\limits_{h\to 0} \frac{f(0-h) - f(0)}{-h}$

$= -\lim\limits_{h\to 0} \frac{f(h) - f(0)}{h}$

$= -f'(0)$

$\Rightarrow f'(0) = 0$

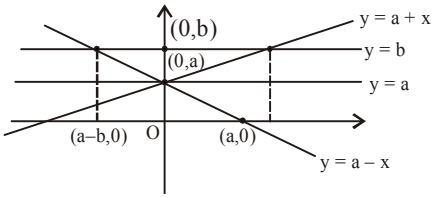
(C) $f'(0^-) = \lim\limits_{h\to 0} \frac{f(0-h) - f(0)}{-h}$

$= \lim\limits_{h\to 0} \frac{-h}{1+e^{-\frac{1}{h}\cdot(-h)}} = 1$

(D) $f(x) = \max\{a-x, a+x, b\}$ $\quad 0 < a < b$

$$= \begin{cases} a-x & x < a-b \\ b & a-b \le x < b-a \\ a+x & x \ge b-a \end{cases}$$

$$f'(x) = \begin{cases} -1 & x < a-b \\ 0 & a-b < x < b-a \\ 1 & x > b-a \end{cases}$$



$\therefore$ f(x) is not differentiable at $x = a-b, b-a$.

59. **Column-I** contains 4 functions and **column-II** contains comments w.r.t their continuity and differentiability at x = 0. Note that **column-I** may have more than one matching options in **column-II**.

	Column-I	**Column-II**
(A)	$f(x) = [x] + \lvert 1-x \rvert$ where [ ] denotes the greatest integer function	(P) continuous
(B)	$g(x) = \lvert x-2 \rvert + \lvert x \rvert$	(Q) differentiable
(C)	$h(x) = [\tan^2 x]$ where [ ] denotes the greatest integer function	(R) discontinuous
(D)	$l(x) = \begin{cases} \dfrac{x(3e^{1/x}+4)}{(2-e^{1/x})} & x \ne 0 \\ 0 & x = 0 \end{cases}$	(S) non derivable

स्तम्भ-I में चार फलन दिये है तथा **स्तम्भ-II** में फलन की x = 0 पर सतत्ता एवं अवकलनीय पर टिप्पणी की गयी है। ध्यान देने की बात है कि **स्तम्भ-I** के किसी भी विकल्प के लिये **स्तम्भ-II** में एक या एक से अधिक मिलान सम्भव है।

	स्तम्भ-I	**स्तम्भ-II**
(A)	$f(x) = [x] + \lvert 1-x \rvert$	(P) सतत्
	जहाँ [ ] महत्तम पूर्णांक फलन है।	
(B)	$g(x) = \lvert x-2 \rvert + \lvert x \rvert$	(Q) अवकलनीय
(C)	$h(x) = [\tan^2 x]$	(R) असतत्
	जहाँ [ ] महत्तम पूर्णांक फलन है।	
(D)	$l(x) = \begin{cases} \dfrac{x(3e^{1/x}+4)}{(2-e^{1/x})} & x \ne 0 \\ 0 & x = 0 \end{cases}$	(S) अवकलनीय नहीं है।

Ans. (A) R, S; (B) P, S; (C) P, Q; (D) P, S

Sol. $f(x) = [x] + |1 - x|$

$$(A) = \begin{cases} [x] + 1 - x & x < 0 \\ 1 - x & 0 \le x < 1 \\ 1 & x = 1 \\ x - 1 + [x] & x > 1 \end{cases}$$

$$= \begin{cases} [x] + 1 - x & x < -1 \\ -x & -1 \le x < 0 \\ 1 - x & 0 \le x < 1 \\ 1 & x = 1 \\ x - 1 + [x] & x > 1 \end{cases}$$

$\therefore$ f(x) is discontinuous at x = 0.

is not differentiable at x = 0.

(B) $g(x) = |x - 2| + |x|$

$$= \begin{cases} -2x^2 + 2 & x < 0 \\ 2 & x = 0 \\ 2 & 0 < x \le 2 \\ 2x - 2 & x > 2 \end{cases}$$

$\because$ g(x) is continuous at x = 0.

$$g'(x) = \begin{cases} -2 & x < 0 \\ 2 & x > 0 \end{cases}$$

$\therefore$ g(x) is jnot differentiable at x = 0.

(C) $h(x) = [\tan^2 x]$

$$= \begin{cases} 0 & -\frac{\pi}{4} < x < 0 \\ 0 & x = 0 \\ 0 & 0 < x < \frac{\pi}{4} \end{cases}$$

$\because$ h(x) is differentiable and continuous at x = 0.

$$(D) \lim_{x \to 0^+} \frac{x\left(3e^{\frac{1}{x}} + 4\right)}{2 - e^{\frac{1}{x}}} = 0 \text{ \& } f(0) = 0$$

$$\lim_{x\to 0^-} \frac{x\left(3e^{\frac{1}{x}}+4\right)}{2-e^{\frac{1}{x}}} = 0$$

$\because$ f(x) is continuous at x = 0.

LHD = –2

RHD = –3

$\therefore$ f(x) is not differentiable at x = 0.

60. Match the following column :

	Column-I	**Column-II**
(A)	$f(x) = \begin{cases} x+1 & \text{if } x<0 \\ \cos x & \text{if } x\geq 0 \end{cases}$ at x = 0 is	(P) continuous
(B)	For every $x \in R$ the function $g(x) = \frac{\sin(\pi[x-\pi])}{1+[x]^2}$ where [x] denotes the greatest integer function is	(Q) derivable (R) discontinuous
(C)	$h(x) = \sqrt{\{x\}^2}$ where {x} denotes fractional part function $\forall x \in I$, is	(S) non derivable
(D)	$k(x) = \begin{cases} x^{\frac{1}{\ln x}} & \text{if } x\neq 1 \\ e & \text{if } x=1 \end{cases}$ at x = 1 is	

निम्न कॉलम को सुमेलित कीजिए :

	Column-I	**Column-II**
(A)	$f(x) = \begin{cases} x+1 & \text{if } x<0 \\ \cos x & \text{if } x\geq 0 \end{cases}$, x = 0 पर	(P) सतत्
(B)	x के प्रत्येक मान के लिये $g(x) = \frac{\sin(\pi[x-\pi])}{1+[x]^2}$, $x \in R$ जहाँ [x] महत्तम पूर्णांक फलन है।	(Q) अवकलनीय (R) असतत्
(C)	$h(x) = \sqrt{\{x\}^2}$ जहाँ {x} भिन्नात्मक फलन हो तथा $x \in I$	(S) अवकलनीय नहीं है।
(D)	$k(x) = \begin{cases} x^{\frac{1}{\ln x}} & \text{if } x\neq 1 \\ e & \text{if } x=1 \end{cases}$, x = 1 पर	

Ans. (A) P, S; (B) P, Q; (C) R, S; (D) P, Q

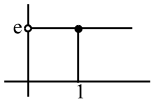
Sol. (A) $f'(0) = \lim_{h\to 0} \frac{\cos h - 0}{h}$ does not exist. Obviously $f(0) = f(0^-) = f(0^+) = 1$

Hence continuous and not derivable

(B) $g(x) = 0$ for all x, hence continuous and derivable

(C) as $0 \le \{f(x)\} < 1$, hence $h(x) = \sqrt{\{x\}^2} = \{x\}$ which is discontinuous hence non derivable all $x \in I$

(D) $\lim_{x\to 1} x^{\frac{1}{\ln x}} = \lim_{x\to 1} x^{\log_x e} = e = f(1)$



61. Let $f(x) = \begin{cases} [x], & -2 \le x < 0 \\ |x|, & 0 \le x \le 2 \end{cases}$ (where [.] denotes the greatest ineger function) $g(x) = \sec x$, $x \in R - (2n+1)\frac{\pi}{2}$, $n \in I$. Match the following statements in column I with their values in column II in the interval $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$.

Column - I	**Column-II**
(A) Abscissa of points where limit of fog(x) exist is/are	(P) –1
(B) Abscissa points in domain of gof(x), where limit of gof(x) does not exist is/are	(Q) π
(C) Abscissa of point of discontinuity of fog (x) is/are	(R) $\frac{5\pi}{6}$
(D) Abscissa of points of differentiability of fog(x) is/are	(S) $-\pi$
	(T) 0

माना $f(x) = \begin{cases} [x], & -2 \le x < 0 \\ |x|, & 0 \le x \le 2 \end{cases}$ है (जहां [·] महत्तम पूर्णांक फलन को दर्शाता है) $g(x) = \sec x$, $x \in R - (2n+1)\frac{\pi}{2}$, $n \in I$

अन्तराल $\left(-\frac{3\pi}{2}, \frac{3\pi}{2}\right)$ में स्तम्भ I के मानों को स्तम्भ II से सूमेलित कीजिये –

स्तम्भ-I	**स्तम्भ-II**
(A) x निर्देशांक जहां fog(x) की सीमा विद्यमान है, होगें	(P) –1
(B) gof(x) के प्रान्त में x निर्देशांक, जहां gof(x) की सीमा विद्यमान नहीं है	(Q) π
(C) x निर्देशांक जिस पर fog (x) असतत है	(R) $\frac{5\pi}{6}$
(D) x निर्देशांक जहां fog(x) अवकलनीय है	(S) $-\pi$
	(T) 0

Ans. (A) PQRST, (B) PT, (C) QS, (D) PRT

Sol. $f(x)=\begin{cases}-2 & -2\le x<-1\\ -1 & -1\le x<0\\ x & 0\le x\le 2\end{cases}$

$g(x)=\sec x;\quad x\in R-\left\{(2n+1)\frac{\pi}{2};n\in I\right\}$

$\therefore$ g(x) is continuous & differentiable $\forall x\in R-\left\{(2n+1)\frac{\pi}{2}\right\}$ & f(x) is continuous $\forall x\in R-\{-1,0\}$

$\Rightarrow$ fog(x) is continuous $\forall x\in R-\left\{-1,0,(2n+1)\frac{\pi}{2}\right\}$

gof(x) is continuous $\forall x\in R-\left\{-1,0,(2n+1)\frac{\pi}{2}\right\}$

at $x=-1$

$\lim\limits_{x\to-1} f(\sec x)=f(\sec 1)=\sec 1=fog(-1)$

at $x=0$

$\lim\limits_{x\to0} f(\sec x)=f(1)=1=fog(0)$

$\therefore$ fog(x) is continuous for $x=0,-1$.

(B) Domain is gof(x) $=[-2,2]-\left\{\frac{\pi}{2},-\frac{\pi}{2}\right\}$

$\because$ g(f(x)) is discontinuous at $x=-1,0$.

at $x=\pi$

$\lim\limits_{h\to0} fog(\pi+h)=\lim\limits_{h\to0} f(\text{sech})=-1$

$\lim\limits_{h\to0} fog(\pi-h)=\lim\limits_{h\to0} f(-\text{sech})=-2$

at $x=-\pi$

LHL $=-2$, RHL $=-1$

$\therefore$ f(x) is discontinuous at $x=-\pi,\ \pi 00$

Exercise J-M

1. Let a function $f: R \to R$ be defined as

$$f(x)=\begin{cases}\sin x - e^x & \text{if } x \le 0\\ a+[-x] & \text{if } 0 < x < 1\\ 2x-b & \text{if } x \ge 1\end{cases}$$

where [x] is the greatest integer less than or equal to x. If *f* is continuous on R, then (a + b) is equal to : **[JEE(Main)July2021]**

माना एक फलन $f: R \to R$

$$f(x)=\begin{cases}\sin x - e^x & \text{यदि } x \le 0\\ a+[-x] & \text{यदि } 0 < x < 1\\ 2x-b & \text{यदि } x \ge 1\end{cases}$$

द्वारा परिभाषित है, जहाँ [x] महत्तम पूर्णांक ≤ x है। यदि फलन *f*, R पर संतत है, तो (a + b) बराबर है : **[JEE(Main)July2021]**

(A) 2 (B) 5 (C) 4 (D) 3

Ans. D

Sol. (i) Continuous at x = 0

LHL = - 1

f(0) = -1

RHL = a - 1

a - 1 = - 1

$\Rightarrow$ a = 0

(ii) Continuous at x = 1

LHL = a – 1

f(1) = 2 – b

RHL = 2 – b

a - 1 = 2 – b

$\Rightarrow$ b = 3

2. Let $f: R \to R$ be defined as $f(x)=\begin{cases}\dfrac{x^3}{(1-\cos 2x)^2}\log_e\left(\dfrac{1+2xe^{-2x}}{(1-xe^{-x})^2}\right) & , x \ne 0.\\ \alpha & , x = 0\end{cases}$

If f is continuous at x = 0, then α is equal to : **[JEE(Main)July2021]**

माना $f: R \to R$ $f(x) = \begin{cases} \dfrac{x^3}{(1-\cos 2x)^2} \log_e\left(\dfrac{1+2xe^{-2x}}{(1-xe^{-x})^2}\right) & , x \neq 0. \\ \alpha & , x = 0 \end{cases}$

द्वारा परिभाषित है। यदि $x = 0$ पर f संतत है, तो α बराबर है : **[JEE(Main)July2021]**

(A) 0 (B) 1 (C) 2 (D) 3

Ans. B

Sol. If function is continuous at $x = 0$

RHL = f(0)

$$\lim_{h\to 0} \frac{h^3}{(1-\cos 2h)^2} \ln\left(\frac{1+2he^{-h}}{(1-he^{-h})^2}\right) = \alpha$$

$$\lim_{h\to 0} \frac{h^3}{4\sin^4 h}\left[\ln(1+2he^{-h}) - 2\ln(1-he^{-h})\right] = \alpha$$

$$\lim_{h\to 0} \frac{1}{4h}\left[\frac{\ln(1+2he^{-2h})}{2he^{-2h}}\left(2he^{-2h}\right) - \frac{2\ln(1-he^{-h})}{-he^{-h}}\left(-he^{-h}\right)\right] = \alpha$$

$$\lim_{h\to 0} \frac{1}{4h}\left[2he^{-2h} + 2he^{-h}\right] = \alpha$$

$$\lim_{h\to 0} \frac{2}{4}\left(e^{-2h} + e^{-h}\right) = \alpha$$

$\alpha = 1$

3. Let $f: R \to R$ be defined as

$$f(x) = \begin{cases} \dfrac{\lambda|x^2-5x+6|}{\mu(5x-x^2-6)}, & x < 2 \\ e^{\frac{\tan(x-2)}{x-[x]}} & , x > 2 \\ \mu & , x = 2 \end{cases}$$

where [x] is the greatest integer less than or equal to x. If f is continuous at $x = 2$, then $\lambda + \mu$:

[JEE(Main)July2021]

माना $f: R \to R$

$$f(x)=\begin{cases}\dfrac{\lambda|x^2-5x+6|}{\mu(5x-x^2-6)}, & x<2\\ e^{\frac{\tan(x-2)}{x-[x]}}, & x>2\\ \mu, & x=2\end{cases}$$

द्वारा परिभाषित है, जबकि [x] महत्तम पूर्णांक $\le x$ है। यदि $x = 2$ पर f संतत है, तो $\lambda + \mu$ बराबर है :

[JEE(Main)July2021]

(A) 1 (B) $2e - 1$ (C) $e(e - 1)$ (D) $e(-e + 1)$

Ans. D

Sol. $f(x)$ is continuous at $x = 2$

$f(2^-) = f(2) = f(2^+)$

$$\lim_{x\to2^-}\frac{\lambda|x^2-5x+6|}{-(x^2-5x+6)\mu}=\mu=\lim_{x\to2^+}e^{\frac{\tan(x-2)}{x-[x]}}$$

$$-\frac{\lambda}{\mu}=\mu=e$$

$\mu = e, \quad \lambda = -e^2$

$\lambda + \mu = e(-e + 1)$

4. Let $f: \left(-\dfrac{\pi}{4},\dfrac{\pi}{4}\right) \to R$ be defined as

$$f(x)=\begin{cases}(1+|\sin x|)^{\frac{3a}{|\sin x|}}, & -\dfrac{\pi}{4}<x<0\\ b, & x=0\\ e^{\cot 4x/\cot 2x}, & 0<x<\dfrac{\pi}{4}\end{cases}.$$

If f is continuous at $x = 0$, then the value of $6a + b^2$ is equal to : **[JEE(Main)July2021]**

माना $f: \left(-\frac{\pi}{4}, \frac{\pi}{4}\right) \to R$,

$$f(x) = \begin{cases} (1+|\sin x|)^{\frac{3a}{|\sin x|}} & , \quad -\frac{\pi}{4} < x < 0 \\ b & , \quad x = 0 \\ e^{\cot 4x / \cot 2x} & , \quad 0 < x < \frac{\pi}{4} \end{cases}$$

द्वारा परिभाषित है। यदि $x = 0$ पर f संतत है, $6a + b^2$ का मान बराबर है : **[JEE(Main)July2021]**

(A) $e - 1$ (B) $1 - e$ (C) e (D) $1 + e$

Ans. D

Sol. $f(0) = f(0^+) = f(0^-)$

$$b = \lim_{x \to 0^+} (1 + |\sin x|)^{\frac{3a}{|\sin x|}} = \lim_{x \to 0^-} e^{\frac{\cot 4x}{\cot 2x}}$$

$$b = e^{\lim_{x \to 0^+} \frac{3a}{|\sin x|}(1+|\sin x|-1)} = \lim_{x \to 0^-} e^{\frac{\tan 2x}{\tan 4x}}$$

$$b = e^{3a} = e^{1/2} \Rightarrow a = \frac{1}{6}, \; b = \sqrt{e}$$

$6a + b^2 = 1 + e$

5. Let $f: [0, \infty) \to [0, 3]$ be a function defined by $f(x) = \begin{cases} \max\{\sin t : 0 \le t \le x\}, & 0 \le x \le \pi \\ 2 + \cos x, & x > \pi \end{cases}$ Then which of the following is true : **[JEE(Main)July2021]**

(A) f is not continuous exactly at two points in $(0, \infty)$

(B) f is differentiable everywhere in $(0, \infty)$

(C) f is continuous everywhere but not differentiable exactly at one point in $(0, \infty)$

(D) f is continuous everywhere but not differentiable exactly at two points in $(0, \infty)$

माना $f: [0, \infty) \to [0, 3]$ $f(x) = \begin{cases} \max\{\sin t : 0 \le t \le x\}, & 0 \le x \le \pi \\ 2 + \cos x, & x > \pi \end{cases}$ द्वारा परिभाषित है। तो निम्न में से कौनसा सत्य है : **[JEE(Main)July2021]**

(A) $(0, \infty)$ के ठीक दो बिन्दु पर f संतत नहीं है

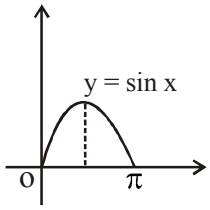
(B) $(0, \infty)$के प्रत्येक बिन्दु पर f अवकलनीय है

(C) $(0, \infty)$के प्रत्येक बिन्दु पर f संतत है परन्तु ठीक एक बिन्दु अवकलनीय नहीं है

(D) $(0, \infty)$के प्रत्येक बिन्दु पर f संतत है परन्तु ठीक दो बिन्दुओं पर अवकलनीय नहीं है

Ans. B

Sol.



$$f(x)=\begin{cases}\sin x & 0\le x<\dfrac{\pi}{2}\\ 1 & \dfrac{\pi}{2}\le x\le\pi\\ 2+\cos x & x>\pi\end{cases}$$

Doubtful points: $x=\dfrac{\pi}{2},\pi$

$\text{LHL}=\text{RHL}=f\left(\dfrac{\pi}{2}\right)=1 \quad \therefore$ hence Continuous at $x=\dfrac{\pi}{2}$

$\text{LHL}=\text{RHL}=f(\pi)=1 \quad \therefore$ hence Continuous at $x=\pi$

$$f'(x)=\begin{cases}\cos x & 0<x<\dfrac{\pi}{2}\\ 0 & \dfrac{\pi}{2}<x<\pi\\ -\sin x & x>\pi\end{cases}$$

$f'\left(\dfrac{\pi^-}{2}\right)=F'\left(\dfrac{\pi^+}{2}\right)=o \quad \therefore$ Derivable at $x=\dfrac{\pi}{2}$

$f'(\pi^-)=f'(\pi^+)=o \quad \therefore$ Derivable at $x=\pi$

$\therefore$ f is differentiable everywhere in $(0,\infty)$

6. Let a function $g:[0,4]\to R$ be defined as $g(x)=\begin{cases}\max\{t^3-6t^2+9t-3\}, 0\le x\le 3, 0\le t\le x\\ 4-x \qquad\qquad , 3<x\le 4\end{cases}$, then the number of points in the interval (0, 4) where g(x) is NOT differentiable, is _____: **[JEE(Main)July2021]**

माना एक फलन $g : [0, 4] \to R$, $g(x) = \begin{cases} \max\{t^3 - 6t^2 + 9t - 3\}, 0 \le x \le 3, 0 \le t \le x \\ 4 - x \qquad\qquad , 3 < x \le 4 \end{cases}$ द्वारा परिभाषित है, तो अन्तराल

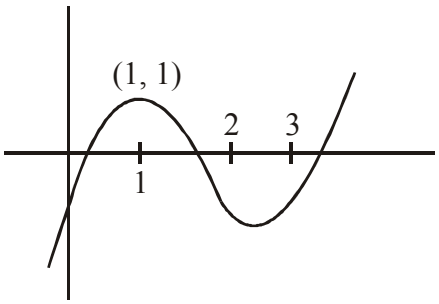
(0, 4) में जिन बिन्दुओं पर g(x) अवकलनीय नहीं है, उनकी संख्या है _____: **[JEE(Main)July2021]**

Ans. 1

Sol. $y = t^3 - 6t^2 + 9t - 3$

$$\frac{dy}{dx} = 3t^2 - 12t + 9 = 0$$

$t = 1 \qquad t = 3$



$g(x) \to x^3 - 6x^2 + 9x - 3 \qquad 0 \le n \le 1$

$\to 1 \qquad 1 < x \le 3$

$\to 4 - x \qquad 3 < x \le 4$

$g'(x) \to 3x^2 - 12x + 9 \qquad 0 \le x \le 1$

$\to 0 \qquad 1 < n \le 3$

$\to 4 \qquad 3 < x \le 4$

$g'(1^-) = g'(1^+)$ but $g'(3^-) \neq g'(3^+)$

nondifferentiable at one point

7. Let $f : R \to R$ be a function defined as $f(x) = \begin{cases} 3\left(1 - \frac{|x|}{2}\right) & \text{if } |x| \le 2 \\ 0 & \text{if } |x| > 2 \end{cases}$. Let $g : R \to R$ be given by

$g(x) = f(x + 2) - f(x - 2)$. If n and m denote the number of points in R where g is not continuous and not differentiable, respectively, then n + m is equal to _____. **[JEE(Main)July2021]**

माना फलन $f : R \to R$

$$f(x) = \begin{cases} 3\left(1-\frac{|x|}{2}\right) & \text{यदि } |x| \le 2 \text{ है} \\ 0 & \text{यदि } |x| > 2 \text{ है} \end{cases}$$

द्वारा परिभाषित है। माना $g : R \to R$ $g(x) = f(x+2) - f(x-2)$ द्वारा परिभाषित है। यदि R के उन बिन्दुओं की संख्या जहाँ g संतत नहीं है और जहाँ g अवकलनीय नहीं है, क्रमशः n और m हैं, तो $n + m$ बराबर है _____

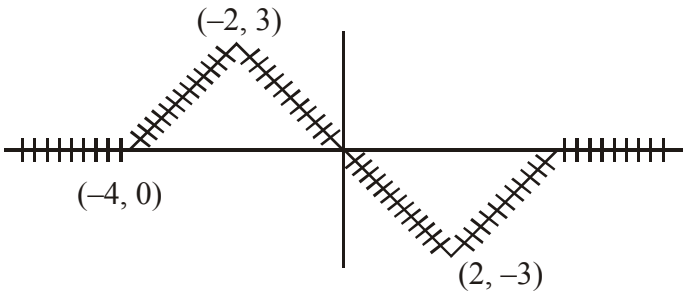
[JEE(Main)July2021]

Ans. 4

Sol. $$f(x-2) = \begin{cases} \frac{3x}{2} & : \quad -4 \le x \le -2 \\ \frac{-3x}{2} & : \quad -2 < x \le 0 \\ 0 & : \quad x \in (-\infty,-4) \cup (0,\infty) \end{cases}$$

$$f(x-2) = \begin{cases} \frac{3x}{2} & : \quad 0 \le x \le 2 \\ \frac{-3x}{2}+6 & : \quad 2 \le x \le 4 \\ 0 & : \quad x \in (-\infty,0) \cup (4,\infty) \end{cases}$$

$$g(x) = f(x+2) - f(x-2) = \begin{cases} \frac{3x}{2}+6 & : \quad -4 \le x \le -2 \\ \frac{-3x}{2} & : \quad -2 < x < 2 \\ \frac{3x}{2}-6 & : \quad 2 \le x \le 4 \\ 0 & : \quad x \in (-\infty,-4) \cup (4,\infty) \end{cases}$$



here $n = 0$

$m = 4$

hence $m + n = 4$

8. Consider the function $f(x) = \frac{P(x)}{\sin(x-2)}, \quad x \neq 2$

$= 7 \qquad , x = 2$

where P(x) is a polynomial such that P"(x) is always a constant and P(3) = 9. If f(x) is continuous at x = 2, then P(5) is equal to _____. **[JEE(Main)July2021]**

फलन $f(x) = \frac{P(x)}{\sin(x-2)}, \quad x \neq 2$

$= 7, \qquad x = 2$

का विचार कीजिए, जहाँ P(x) एक बहुपद है, जिसके लिए P"(x) हमेशा एक अचर है तथा P(3) = 9 है। यदि x = 2 पर f(x) संतत हैं, तो P(5) बराबर है _____ **[JEE(Main)July2021]**

Ans. 39

Sol. $f(x) = \begin{cases} \frac{P(x)}{\sin(x-2)} & x \neq 2 \\ 7 & x = 2 \end{cases}$

P" (x) = constant $\Rightarrow$ P(x) is 2 degree polynomial

$f(2^+) = f(2^-) = f(2)$

$\lim\limits_{x\to 2} \frac{P(x)}{\sin(x-2)} = 7$

$\lim\limits_{x\to 2} \frac{(x-2)P(x)}{(x-2)\sin(x-2)} = 7$

$\lim\limits_{x\to 2} \frac{P(x)}{(x-2)} = 7$

$P(x) = (x-2)(ax+b)$

$2a + b = 7$

$P(3) = 3a + b = 9$

$a = 2$

$4 + b = 7 \Rightarrow b = 3$

$P(x) = (x-2)(2x+3)$

$P(5) = 3(10 + 3) = 39$

9. Let f: [0,3] $\to$ R be defined by

$f(x) = \min \{x - [x], 1 + [x] - x\}$

where [x] is the greatest integer less than or equal to x.

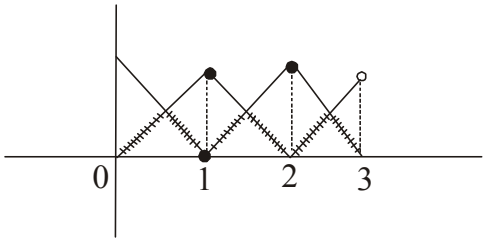
Let P denote the set containing all $x \in [0,3]$ where *f* is discontinuous, and Q denote the set containing all $x \in (0,3)$ where *f* is not differentiable. Then the sum of number of elements in P and Q is equal to_____: **[JEE(Main)July2021]**

माना f: [0,3] $\to$ R,

$f(x) = \min \{x - [x], 1 + [x] - x\}$, जहाँ [x] महत्तम पूर्णांक $\le x$ है, द्वारा परिभाषित है। माना सभी $x \in [0,3]$ जिनके लिए f असंतत है, का समुच्चय P है तथा सभी $x \in (0,3)$, जिनके लिए f अवकलनीय नहीं है का समुच्चय Q है। तो P तथा Q में अवयवों की संख्या का योगफल बराबर है _____ **[JEE(Main)July2021]**

Ans. 5

Sol. $f(x) = \min\left(\{x\}, 1 - \{x\}\right)$



$n(P) = 0$

$n(Q) = 5$

$n(P) + n(Q) = 5$

10. Let [t] denote the greatest integer less than or equal to t. Let $f(x) = x - [x]$, $g(x) = 1 - x + [x]$, and $h(x) = \min \{f(x), g(x)\}$. $x \in [-2, 2]$. Then h is : **[JEE(Main)Aug.2021]**

(A) Continuous in $[-2, 2]$ but not differentiable at exactly three points in $(-2, 2)$

(B) Not continuous at exactly four points in $[-1, 2]$

(C) Not continuous at exactly three points in $[-1, 2]$

(D) Continuous in $[-2, 2]$ but not differentiable at more than four points in $(-2, 2)$

माना [t] महत्तम पूर्णांक $\le$ t है। माना $f(x) = x - [x]$, $g(x) = 1 - x + [x]$ तथा $h(x) = \min \{f(x), g(x)\}$, $x \in [-2, 2]$ है, तो h : **[JEE(Main)Aug.2021]**

(A) $[-2, 2]$ में संतत है परन्तु $(-2, 2)$ में ठीक तीन बिन्दुओं से अधिक पर अवकलनीय नहीं है

(B) $[-1, 2]$ में ठीक तीन बिन्दुओं पर संतत नहीं है।

(C) $[-1, 2]$ में ठीक चार बिन्दुओं पर संतत नहीं है।

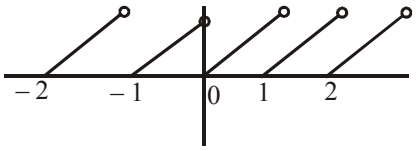
(D) $[-2, 2]$ में संतत है परन्तु $(-2, 2)$ में चार बिन्दुओं से अधिक पर अवकलनीय नहीं है।

Ans. D

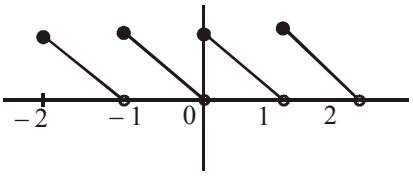
Sol. $f(x) = \{x\}$

$g(x) = 1 - \{x\}$

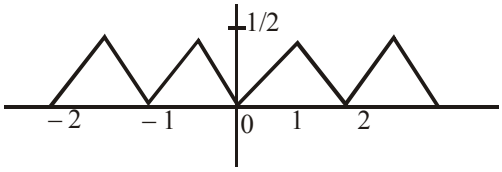
$y = \{x\}$



$g(x) = 1 - \{x\}$



$h(x) = \min\left(f(x), g(x)\right)$



non differentiable at 7 points

11. The function $f(x) = |x^2 - 2x - 3|.\ e^{|9x^2 - 12x + 4|}$ is not differentiable at exactly : **[JEE(Main)Aug.2021]**

(A) one point (B) three points (C) two points (C) four points

फलन $f(x) = |x^2 - 2x - 3|.\ e^{|9x^2 - 12x + 4|}$: **[JEE(Main)Aug.2021]**

(A) ठीक एक बिन्दु पर अवकलनीय नहीं है (2) ठीक तीन बिन्दुओं पर अवकलनीय नहीं है

(C) ठीक दो बिन्दुओं पर अवकलनीय नहीं है (D) ठीक चार बिन्दुओं पर अवकलनीय नहीं है

Ans. C

Sol. $f(x) = |x^2 - 2x - 3|\, e^{|(3x-2)^2|}$

$f(x) = |(x-3)(x+1)|\, e^{(3x-2)^2}$

Not differentiable at $x = -183$

12. If the function $f(x) = \begin{cases} \frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x < 0 \\ k & , \quad x = 0 \\ \frac{\cos^2 x - \sin^2 x - 1}{\sqrt{x^2+1}-1} & , \quad x > 0 \end{cases}$ is continuous at $x = 0$, then $\frac{1}{a} + \frac{1}{b} + \frac{4}{k}$ is equal to :

[JEE(Main)Aug.2021]

यदि फलन $f(x)=\begin{cases}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right) & , \quad x<0 \\ k & , \quad x=0 \\ \frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1} & , \quad x>0\end{cases}$

$x=0$ पर संतत हैं, तो $\frac{1}{a}+\frac{1}{b}+\frac{4}{k}$ बराबर है : **[JEE(Main)Aug.2021]**

(A) 5 (B) -5 (C) -4 (D) 4

Ans. B

Sol. $f(0)=k$

$$f(0^+)=\lim_{x\to 0^+}\frac{\cos^2 x-\sin^2 x-1}{\sqrt{x^2+1}-1}\times\frac{\sqrt{x^2+1}+1}{\sqrt{x^2+1}+1}$$

$$f(0^+)=\lim_{x\to 0^+}\frac{-2\sin^2 x}{x^2}(\sqrt{x^2+1}+1)$$

$$f(0^+)=-4$$

$$f(0^-)=\lim_{x\to 0^-}\frac{1}{x}\log_e\left(\frac{1+\frac{x}{a}}{1-\frac{x}{b}}\right)$$

$(9\sqrt{2},2)$

$$f(0^+)=\frac{1}{a}+\frac{1}{b}$$

$$f(0)=f(0^+)=f(0^-)$$

$$k=-4=\frac{1}{a}+\frac{1}{b}\Rightarrow\frac{1}{a}+\frac{1}{b}+\frac{4}{k}=-5$$

13. Let $a,b\in R, b\neq 0$. Define a function

$$f(x)=\begin{cases}a\sin\frac{\pi}{2}(x-1), & \text{for } x\le 0 \\ \frac{\tan 2x-\sin 2x}{bx^3}, & \text{for } x>0.\end{cases}$$

If f is continuous at $x=0$, then $10-ab$ is equal to _____. **[JEE(Main)Aug.2021]**

माना $a, b \in R, b \neq 0$ है। एक फलन

$$f(x) = \begin{cases} a\sin\frac{\pi}{2}(x-1), & \text{के लिए } x \leq 0 \\ \frac{\tan 2x - \sin 2x}{bx^3}, & \text{के लिए } x > 0. \end{cases}$$

द्वारा परिभाषित है। यदि x = 0 पर f संतत है, तो 10 – ab बराबर है_____ **[JEE(Main)Aug.2021]**

Ans. 14

Sol. $f(0) = -a = LHL$

$$RHL = \lim_{x\to 0}\frac{\tan 2x - \sin 2x}{bx^3} = \frac{4}{b}$$

$$-a = \frac{4}{b} \Rightarrow ab = -4$$

$$10 - ab = 10 + 4 = 14$$

14. Let [t] denote the greatest integer ≤ t. The number of points where the function $f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1], x \in (-2,2)$ is not continuous is _____

[JEE(Main)Aug.2021]

माना [t] महत्तम पूर्णांक ≤ t है। जहाँ फलन $f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1], x \in (-2,2)$ संतत नहीं है, ऐसे बिन्दुओं की संख्या है _____ **[JEE(Main)Aug.2021]**

Ans. 2

Sol. $f(x) = [x]|x^2 - 1| + \sin\left(\frac{\pi}{[x]+3}\right) - [x+1]$

$$x \in (-2,2)$$

$$f(x) = [x]\left(|x^2 - 1| - 1\right) + \sin\left(\frac{\pi}{[x]+3}\right) - 1$$

$$f(x) = (-2)\left((x^2 - 1) - 1\right) + \sin\left(\frac{\pi}{1}\right) - 1; \ (-2,-1]$$

$$= (-1)(-x^2 + 1 - 1) + \sin\left(\frac{\pi}{2}\right) - 1; \ (-1,0)$$

$$= 0 + \sin\frac{\pi}{3} - 1 \quad ; \ (0,1)$$

$= (1)((x^2-1)-1)+\sin\frac{\pi}{4}-1;\ (1,2)$

$$f(x)\begin{cases} -2(x^2-2)-1 & (-2,-1] \\ x^2 & (-1,0) \\ \sin\frac{\pi}{3}-1 & [0,1) \\ (x^2-3)+\sin\frac{\pi}{4} & [1,2) \end{cases}$$

discont. at $x = 0, 1$.

15. Let $f(x)=\begin{cases} \frac{\sin(x-[x])}{x-[x]} & , \quad x\in(-2,-1) \\ \max\{2x, 3[|x|]\} & , \quad |x|<1 \\ 1 & , \quad \text{otherwise} \end{cases}$

Where [t] denotes greatest integer $\le$ t. If m is the number of points where f is not continuous and n is the number of points where f is not differentiable, then the ordered pair (m, n) is : **[JEE(Main)June2022]**

माना $f(x)=\begin{cases} \frac{\sin(x-[x])}{x-[x]} & , \quad x\in(-2,-1) \\ \max\{2x, 3[|x|]\} & , \quad |x|<1 \\ 1 & , \quad \text{otherwise} \end{cases}$

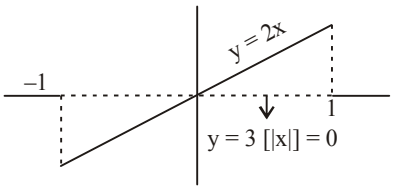
जहाँ [t] महत्तम पूर्णांक $\le$ t है। यदि m बिन्दुओं की संख्या है, जहाँ f संतत नहीं है, तथा n उन बिन्दुओं की संख्या है, जहाँ f अवकलनीय नहीं है, तो क्रमित युग्म (m, n) है : **[JEE(Main)June2022]**

(A) (3, 3) (B) (2, 4) (C) (2, 3) (D) (3, 4)

Ans. C

Sol. $f(x)=\begin{cases} \frac{\sin\{x\}}{\{x\}} & : \quad x\in(-2,-1) \\ \max(2x, 3[|x|]) & : \quad -1<x<1 \\ 1 & : \quad \text{otherwise} \end{cases}$

Let consider $y = \max(2x_1, 3[|x|]) : -1 , x < 1$



$$y = \max(2x, 0) = \begin{cases} 0 & : \ -1 < x < 0 \\ 2x & : \ 0 \le x < 1 \end{cases}$$

Now doubtful points for continuity and differentiability are $x = 0, 1, -1$

Now continuity at $x = 0$

RHL = 0

f(0) = 0

LHL = 0

Now RHD = 2

LHD = 0

Continuity at $x = 1$

LHL = 2

RHL = 1

f(1) = 1

Continuity at $x = -1$

RHL = 0

$$\text{LHL} = \frac{\sin(1-h)}{1-h} = \sin 1$$

f(–1) = 1

hence it is clear function is not continuous at x = 1, – 1 and not differentiable at x = 1, –1, 0

m = 2

n = 3

(2, 3)

16. Let f, g : R → R be two real valued functions defined as $f(x) = \begin{cases} -|x+3| & , \ x < 0 \\ e^x & , \ x \ge 0 \end{cases}$ and $g(x) = \begin{cases} x^2 + k_1 x & , \ x < 0 \\ 4x + k_2 & , \ x \ge 0 \end{cases}$, where k_1 and k_2 are real constants. If (gof) is differentiable at x = 0, then (gof) (–4) + (gof) (4) is equal to : **[JEE(Main)June2022]**

माना दो वास्तविक मान फलन $f, g : R \to R$

$$f(x) = \begin{cases} -|x+3| & , \ x < 0 \\ e^x & , \ x \ge 0 \end{cases} \text{ तथा } g(x) = \begin{cases} x^2 + k_1 x & , \ x < 0 \\ 4x + k_2 & , \ x \ge 0 \end{cases}$$

द्वारा परिभाषित हैं, जहाँ k_1 तथा k_2 वास्तविक अचर हैं। यदि x = 0 पर (gof) अवकलनीय है, तो (gof) (–4) + (gof) (4) बराबर है:

[JEE(Main)June2022]

(A) $4(e^4 + 1)$ (B) $2(2e^4 + 1)$ (C) $4e^4$ (D) $2(2e^4 - 1)$

Ans. D

Sol. $$f(x) = \begin{Bmatrix} x+3 & ; & x < -3 \\ -(x+3) & ; & -3 \le x < 0 \\ e^x & ; & x \ge 0 \end{Bmatrix}$$

$$g(x) = \begin{Bmatrix} x^2 + k_1 x & ; & x < 0 \\ 4x + k_2 & ; & x \ge 0 \end{Bmatrix}$$

$$g(f(x)) = \begin{Bmatrix} f(x)^2 + k_1 f(x) & ; & f(x) < 0 \\ 4f(x) + k_2 & ; & f(x) \ge 0 \end{Bmatrix}$$

$$g(f(x)) = \begin{Bmatrix} (x+3)^2 + k_1(x+3) & ; & x < -3 \\ (x+3)^2 - k_1(x+3) & ; & -3 \le x < 0 \\ 4e^x + k_2 & ; & x > 0 \end{Bmatrix}$$

check continuity at $x = 0$

$gof(0) = f(f(0^-)) = g(f(0^+))$

$4 + k_2 = 9 - 3k_1 = 4 + k_2$

$3k_1 + k_2 = 5$ ...(a)

differentiate

$$(g(f(x)))' = \begin{Bmatrix} 2(x+3) + k_1 & ; & x < -3 \\ 2(x+3) - k_1 & ; & -3 \le x < 0 \\ 4e^x & ; & x \ge 0 \end{Bmatrix}$$

$6 - k_1 = 4$

$k_1 = 2$ ...(b)

$\therefore k_1 = 2, k_2 = -1$

$$gof(x) = \begin{Bmatrix} (x+3)^2 + 2(x+3) & ; & x < -3 \\ (x+3)^2 - 2(x+3) & ; & -3 \le x < 0 \\ 4e^x - 1 & ; & x \ge 0 \end{Bmatrix}$$

$gof(-4) + gof(4) = 4e^4 - 2$

$\Rightarrow 2(2e^4 - 1)$

17. Let $f(x) = \min\{1, 1 + x \sin x\}$, $0 \le x \le 2\pi$. If m is the number of points, where f is not differentiable and n is the number of points, where f is not continuous, then the ordered pair (m, n) is equal to :

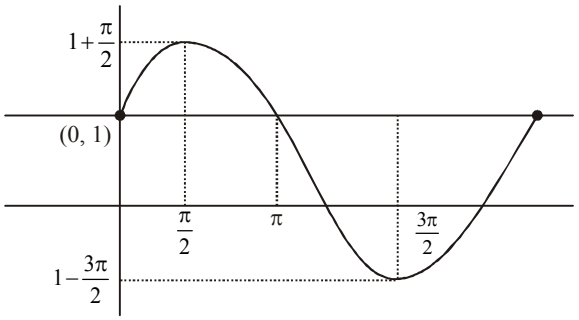
[JEE(Main)June2022]

माना $f(x) = \min\{1, 1 + x\sin x\}, 0 \le x \le 2\pi$ है। यदि उन बिन्दुओं की संख्या, जहाँ f अवकलनीय नहीं है, m है तथा उन बिन्दुओं की संख्या, जहाँ f संतत नहीं है, n है, तो क्रमित युग्म (m, n) बराबर है : **[JEE(Main)June2022]**

(A) (2, 0) (B) (1, 0) (C) (1, 1) (D) (2, 1)

Ans. B

Sol. $f(x) = \min\{1, 1 + x\sin x\}\ 0 \le x \le 2\pi$



$$f(x) = \begin{cases} 1 & 0 \le x < \pi \\ 1 + x\sin x & \pi \le x \le 2\pi \end{cases}$$

$f(0^+) = 1$ $f(0) = 1$

$f(\pi^-) = 1$ $f(\pi^+) = 1$ $f(\pi) = 1$

$f(2\pi^-) = 1$ $f(2\pi) = 1$

$n = 0$

$$f'(x) = \begin{cases} 0 & 0 < x < \pi \\ x\cos x + \sin x & \pi < x < 2\pi \end{cases}$$

$f'(\pi^-) = 0$ $f'(\pi^+) = \pi(-1) = -\pi$

$m = 1$

$(m, n) = (1, 0)$

18. Let $f : R \to R$ be defined as $f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \le x < 1 \\ b + [\sin(\pi x)], & 1 \le x < 2 \\ [e^{-x}] - c, & x \ge 2 \end{cases}$ where a, b, $c \in R$ and [t] denotes greatest integer less than or equal to t. Then, which of the following statements is true? **[JEE(Main)June2022]**

(A) There exists a, b, $c \in R$ such that f is continuous on R.

(B) If f is discontinuos at exactly one point, then $a + b + c = 1$

(C) If f is discontinuous at exactly one point, then $a + b + c \ne 1$

(D) f is discontinuous at atleast two points, for any values of a, b and c

माना $f: R \to R$ $f(x) = \begin{cases} [e^x], & x < 0 \\ ae^x + [x-1], & 0 \le x < 1 \\ b + [\sin(\pi x)], & 1 \le x < 2 \\ [e^{-x}] - c, & x \ge 2 \end{cases}$ द्वारा परिभाषित है, a, b, c ∈ R हैं तथा [t] महत्तम पूर्णांक t है। तो निम्न कथनों में से कौनसा सत्य है ? **[JEE(Main)June2022]**

(A) a, b, c ∈ R का अस्तित्व है जिनके लिए f, R पर संतत है

(B) यदि f मात्र एक बिन्दु पर असंतत है, तो a + b + c = 1

(C) यदि f मात्र एक बिन्दु पर असंतत है, तो a + b + c ≠ 1

(D) a, b, c के किन्हीं भी मानों के लिए f कम से कम बिन्दुओ पर असंतत है

Ans. C

Sol. f(x) is discontinuous at x = 1

For continuous at x = 0; a = 1

For continuous at x = 2; b + c = 1

a + b + c = 2

19. Let f, g : R → R be functions defined by $f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \ge 0 \end{cases}$ and $g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \ge 0 \end{cases}$ where [x] denote the greatest integer less than or equal to x. Then, the function fog is discontinuous at exactly : **[JEE(Main)June2022]**

(A) one point (B) two points (C) three points (D) four points

माना फलन f, g : R → R $f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \ge 0 \end{cases}$ तथा $g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \ge 0 \end{cases}$ द्वारा परिभाषित है, जहाँ [x] महत्तम पूर्णांक ≤ x है। तो फलन fog कितने बिन्दुओं पर असंतत है : **[JEE(Main)June2022]**

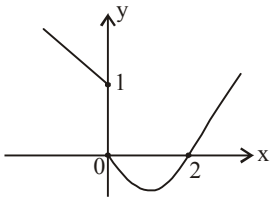
(A) एक बिन्दु (B) दो बिन्दु (C) तीन बिन्दु (D) चार बिन्दु

Ans. B

Sol. $f(x) = \begin{cases} [x], & x < 0 \\ |1-x|, & x \ge 0 \end{cases}$ and $g(x) = \begin{cases} e^x - x, & x < 0 \\ (x-1)^2 - 1, & x \ge 0 \end{cases}$

$$f(x)=\begin{cases}\left|1+x-e^{x}\right| & x<0\\ 1 & x=0\\ \left[(x-1)^{2}-1\right] & 0<x<2\\ \left|2-(x-1)^{2}\right| & x\geq 2\end{cases}$$

graph of fog(x)



So x = 0 and 2 are the 2 points where fog(x) is discontinuous.

20. The number of points where the function $f(x)=\begin{cases}|2x^{2}-3x-7| & \text{if } x\leq -1\\ [4x^{2}-1] & \text{if } -1<x<1\\ |x+1|+|x-2| & \text{if } x\geq 1,\end{cases}$

[t] denotes the greatest integer ≤ t, is discontinuous is ____. **[JEE(Main)June2022]**

उन बिन्दुओं, जहाँ फलन $f(x)=\begin{cases}|2x^{2}-3x-7| & \text{यदि } x\leq -1 \text{ है}\\ [4x^{2}-1] & \text{यदि } -1<x<1 \text{ है}\\ |x+1|+|x-2| & \text{यदि } x\geq 1 \text{ है,}\end{cases}$

[t] महत्तम पूर्णांक ≤ t है, असंतत है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. 7

Sol. $\because$ f(–1) = 2 and f(1) = 3

For $x\in(-1,1)$, $(4x^{2}-1)\in[-1,3)$

hence f(x) will be discontinuous at x = 1 and also

whenever $4x^{2}-1=0$, 1 or 2

$\Rightarrow x=\pm\frac{1}{2},\pm\frac{1}{\sqrt{2}}$ and $\pm\frac{\sqrt{3}}{2}$

So there are total 7 points of discontinuity

21. Let f(x) = [2x² + 1] and $g(x)=\begin{cases}2x-3, & x<0\\ 2x+3, & x\geq 0\end{cases}$, where [t] is the greatest integer ≤ t. Then, in the open interval (–1, 1), the number of points where fog is discontinuous is equal to _____. **[JEE(Main)June2022]**

माना $f(x)=[2x^2+1]$ तथा $g(x)=\begin{cases}2x-3, & x<0\\ 2x+3, & x\geq 0\end{cases}$ हैं, जहाँ [t] महत्तम पूर्णांक $\leq t$ है। तो विवृत अंतराल (–1, 1) में उन बिंदुओं, जहाँ fog असंतत है, की संख्या है _____ **[JEE(Main)June2022]**

Ans. 62

Sol. $f(x)=[2x^2]+1$

$$f(g(x))=\left[2g^2(x)\right]+1$$

$$f(g(x))=\begin{cases}\left[2(2x-3)^2\right]+1 & , \quad x\in(-1,0)\\ \left[2(2x+3)^2\right]+1 & , \quad x\in[0,1)\end{cases}$$

(i) $x \in (-1, 0)$

$2x \in (-2, 0)$

$2x - 3 \in (-5, -3)$

$(2x - 3)^2 \in (9, 25)$

$2(2x - 3)^2 \in (18, 50)$

f(g(x)) is DC when

$2(2x - 3)^2 \in \{19, 20, ...,49\}$

Total = 31

(ii) $x \in [0, 1)$

$2x \in [0, 2)$

$2x + 3 \in [3, 5)$

$(2x + 3)^2 \in [9, 25)$

$2(2x - 3)^2 \in [18, 50)$

f(g(x)) is DC when

$2(2x + 3)^2 \in \{19, 20, ...,49\}$

Total = 31

Total point of discontinuity = 31 + 31 = 62

22. If the function $f(x)=\begin{cases}\dfrac{\log_e\left(1-x+x^2\right)+\log_e\left(1+x+x^2\right)}{\sec x-\cos x} & , \quad x\in\left(\dfrac{-\pi}{2},\dfrac{\pi}{2}\right)-\{0\}\\ k & , \quad x=0\end{cases}$ is continuous at $x = 0$, then k is equal to : **[JEE(Main)July2022]**

यदि फलन $f(x)=\begin{cases}\dfrac{\log_e\left(1-x+x^2\right)+\log_e\left(1+x+x^2\right)}{\sec x-\cos x} & , \quad x\in\left(\dfrac{-\pi}{2},\dfrac{\pi}{2}\right)-\{0\}\\ k & , \quad x=0\end{cases}$

$x = 0$, पर संतत है, तो k बराबर है : **[JEE(Main)July2022]**

(A) 1 (B) –1 (C) e (D) 0

Ans. A

Sol. $\lim\limits_{x\to 0}\dfrac{\left(\ln\left(1+x^2+x^4\right)\right)\cos x}{1-\cos^2 x}$

$$\lim_{x\to 0}\frac{\left(\dfrac{\ln\left(1+x^2+x^4\right)}{x^2+x^4}\right)x^2\left(1+x^2\right)\cos x}{\left(\dfrac{\sin^2 x}{x^2}\right)x^2}=1$$

$\therefore$ k = 1

23. If $f(x)=\begin{cases}x+a & , \ x\le 0\\ |x-4| & , \ x>0\end{cases}$ and $g(x)=\begin{cases}x+1 & , \ x<0\\ (x-4)^2+b & , \ x\ge 0\end{cases}$ are continuous on R, then (gof) (2) + (fog) (–2) is equal to : **[JEE(Main)July2022]**

यदि $f(x)=\begin{cases}x+a & , \ x\le 0\\ |x-4| & , \ x>0\end{cases}$ तथा $g(x)=\begin{cases}x+1 & , \ x<0\\ (x-4)^2+b & , \ x\ge 0\end{cases}$ R पर संतत तो (gof) (2) + (fog) (–2) बराबर है: **[JEE(Main)July2022]**

(A) –10 (B) 10 (C) 8 (D) –8

Ans. D

Sol. $f(x)=\begin{cases}x+a, & x\le 0\\ |x-4|, & x>0\end{cases}$ and $g(x)=\begin{cases}x+1, & x<0\\ (x-4)^2+b, & x\ge 0\end{cases}$

$\because$ f(x) and g(x) are continuous on R

$\therefore$ a = 4 and b = 1 – 16 = –15

then (gof)(2) + (fog) (–2)

= g(2) + f(–1)

= –11 + 3 = – 8

24. If for $p\ne q\ne 0$, the function $f(x)=\dfrac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$ is continuous at x = 0, then : **[JEE(Main)July2022]**

यदि $p\ne q\ne 0$ के लिए, फलन $f(x)=\dfrac{\sqrt[7]{p(729+x)}-3}{\sqrt[3]{729+qx}-9}$, x = 0 पर संतत है, तो : **[JEE(Main)July2022]**

(A) 7pq f(0) – 1 = 0 (B) 63q f(0) – p^2 = 0

(C) 21q f(0) – p^2 = 0 (D) 7pq f(0) – 9 = 0

Ans. B

Sol. $f(0)=\lim_{x\to 0} f(x)$

Limit should be $\frac{0}{0}$ form

So, $\sqrt[7]{p.729}-3=0 \Rightarrow p.3^6=3^7 \Rightarrow p=3$

Now, $f(0)=\lim_{x\to 0}\frac{\sqrt[7]{3\left(3^6+x\right)}-3}{\sqrt[3]{3^6+qx}-9}$

$$=\lim_{x\to 0}\frac{3\left[\left(1+\frac{x}{3^6}\right)^{1/7}-1\right]}{9\left[\left(1+\frac{qx}{3^6}\right)^{1/3}-1\right]}=\frac{3}{9}\times\frac{\frac{1}{7.3^6}}{\frac{q}{3.3^6}}$$

$$\Rightarrow f(0)=\frac{1}{3}\times\frac{3}{7q}=\frac{1}{7q}$$

$\Rightarrow 7qf(0)-1=0$

$\Rightarrow 7.p^2.qf(0)-p^2=0$ (for option)

$\Rightarrow 63qf(0)-p^2=0$

25. The function $f: R \to R$ defined by $f(x)=\lim_{n\to\infty}\frac{\cos(2\pi x)-x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}}$ is continuous for all x in :

[JEE(Main)July2022]

$f(x)=\lim_{n\to\infty}\frac{\cos(2\pi x)-x^{2n}\sin(x-1)}{1+x^{2n+1}-x^{2n}}$ द्वारा परिभाषित फलन $f: R \to R$ किस समुच्चय के सभी बिन्दु x पर संतत है:

[JEE(Main)July2022]

(A) $R-\{-1\}$ (B) $R-\{-1, 1\}$ (C) $R-\{1\}$ (D) $R-\{0\}$

Ans. B

Sol. n should be given as a naturel number.

$$f(x=\begin{cases}\frac{-\sin(x-1)}{x-1} & x<-1\\ -(\sin 2+1) & x=-1\\ \cos 2\pi x & -1<x<1\\ \frac{-\sin(x-1)}{x-1} & x>1\end{cases}$$

f(x) is discontinuous at $x=-1$ and $x=1$

26. The number of points, where the function $f: R \to R$,

$f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)|x^2-5x+4|$, is NOT differentiable, is :

[JEE(Main)July2022]

उन बिन्दुओं की संख्या, जहाँ पर फलन $f: R \to R$,

$f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)|x^2-5x+4|$, अवकलनीय नहीं है, है : **[JEE(Main)July2022]**

(A) 1 (B) 2 (C) 3 (D) 4

Ans. B

Sol. $f(x) = |x-1| \cos|x-2| \sin|x-1| + (x-3)|x^2-5x+4|$

$= |x-1| \cos|x-2| \sin|x-1| + (x-3)|x-1||x-4|$

$= |x-1| [\cos|x-2| \sin|x-1| + (x-3)|x-4|]$

Non differentiable at $x = 1$ and $x = 4$.

27. Let the function $f(x) = \begin{cases} \dfrac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{if} \quad x \neq 0 \\ 10 & ; \text{if} \quad x = 0 \end{cases}$ be continuous at $x = 0$. Then α is equal to :

[JEE(Main)July2022]

माना फलन $f(x) = \begin{cases} \dfrac{\log_e(1+5x) - \log_e(1+\alpha x)}{x} & ; \text{if} \quad x \neq 0 \\ 10 & ; \text{if} \quad x = 0 \end{cases}$, $x = 0$ पर संतत है। तो α बराबर है :

[JEE(Main)July2022]

(A) 10 (B) –10 (C) 5 (D) –5

Ans. D

Sol. $f(x) = \begin{cases} \dfrac{\ln(1+5x) - \ln(1+\alpha x)}{x} & ; x \neq 0 \\ 10 & ; x = 0 \end{cases}$

$\lim\limits_{x \to 0} \dfrac{\ln(1+5x) - \ln(1+\alpha x)}{x} = 10$

Using expension

$\lim\limits_{x \to 0} \dfrac{(5x + \ldots\ldots) - (\alpha x + \ldots..)}{x} = 10$

$5 - \alpha = 10 \Rightarrow \alpha = -5$

28. $f(x)=\begin{cases}\left|4x^2-8x+5\right|, & \text{if} \quad 8x^2-6x+1\geq 0\\ \left[4x^2-8x+5\right] & \text{if} \quad 8x^2-6x+1<0\end{cases}$ where $[\alpha]$ denotes the greatest integer less than or equal to α.

Then the number of points in R where f is not differentiable is _____. **[JEE(Main)July2022]**

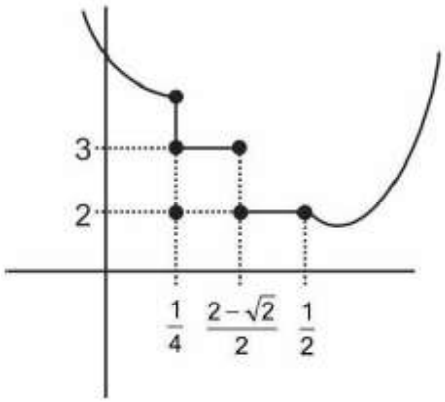
माना $f(x)=\begin{cases}\left|4x^2-8x+5\right|, & \text{if} \quad 8x^2-6x+1\geq 0\\ \left[4x^2-8x+5\right] & \text{if} \quad 8x^2-6x+1<0\end{cases}$ जहाँ $[\alpha]$ महत्तम पूर्णांक α है। तो R में उन बिन्दुओं की संख्या, जहाँ f अवकलनीय नहीं है, है _____ **[JEE(Main)July2022]**

Ans. 3

Sol. $f(x)=\begin{cases}\left|4x^2-8x+5\right|, & \text{if } 8x^2-6x+1\geq 0\\ \left[4x^2-8x+5\right], & \text{if } 8x^2-6x+1<0\end{cases}$

$$=\begin{cases}4x^2-8x+5, & \text{if } x\in\left[-\infty,\frac{1}{4}\right]\cup\left[\frac{1}{2},\infty\right)\\ \left[4x^2-8x+5\right] & \text{if } x\in\left(\frac{1}{4},\frac{1}{2}\right)\end{cases}$$

$$f(x)=\begin{cases}4x^2-8x+5 & \text{if } x\in\left(-\infty\frac{1}{4}\right]\cup\left[\frac{1}{2},\infty\right)\\ 3 & x\in\left(\frac{1}{4},\frac{2-\sqrt{2}}{2}\right)\\ 2 & x\in\left[\frac{2-\sqrt{2}}{2},\frac{1}{2}\right)\end{cases}$$



$\therefore$ Non–diff at $x=\frac{1}{4},\frac{2-\sqrt{2}}{2},\frac{1}{2}$

29. If [t] denotes the greatest integer ≤ t, then the number of points, at which the function $f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$ is not differentiable in the open interval (–20, 20), is _____.

[JEE(Main)July2022]

यदि [t] महत्तम पूर्णांक ≤ t है, तो उन बिन्दुओं, जिन पर फलन $f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$ विवृत अंतराल (–20, 20) में अवकलनीय नहीं है, कि संख्या_____ **[JEE(Main)July2022]**

Ans. 79

Sol. $f(x)=4|2x+3|+9\left[x+\frac{1}{2}\right]-12[x+20]$

$x\in(-20,20)$

f(x) is not Diff. at $x=I\in\{-19,-18,\ldots\ldots 19\}=39$

at $x=I+\frac{1}{2}$, $f(x)$ Not diff. at 39 points

Check at $x=\frac{-3}{2}$ Discount at $x=\frac{-3}{2}\therefore$ N.R(1)

No. of point of non-differentiabilty

= 39 + 39 + 1 = 79

Exercise J - A

1. For every integer n, let a_n and b_n be real numbers. Let function f: R → R be given by

$$f(x)=\begin{cases} a_n+\sin\pi x, & \text{for } x\in[2n,2n+1] \\ b_n+\cos\pi x, & \text{for } x\in(2n-1,2n)\end{cases}, \text{ for all integers n.}$$

If f is continuous, then which of the following hold(s) for all n : **[JEE-2012]**

प्रत्येक पूर्णांक n के लिये, माना a_n तथा b_n वास्तविक संख्याऐं है। फलन f : R → R इस प्रकार कि

$$f(x)=\begin{cases} a_n+\sin\pi x, & \text{for } x\in[2n,2n+1] \\ b_n+\cos\pi x, & \text{for } x\in(2n-1,2n)\end{cases}, \text{ सभी पूर्णांक n के लिये हो।}$$

यदि f सतत् है तो सभी n के लिये निम्न में से कौनसा /कौनसे कथन सत्य है : **[JEE-2012]**

(A) $a_{n-1}-b_{n-1}=0$ (B) $a_n-b_n=1$ (C) $a_n-b_{n+1}=1$ (D) $a_{n-1}-b_n=-1$

Ans. BD

Sol. For f to be continuous :

$f(2n^-)=f(2n^+)$.

$\Rightarrow \quad b_n+\cos 2n\pi=a_n+\sin 2n\pi$

$\Rightarrow \quad b_n+1=a_n$

$\Rightarrow \quad a_n-b_n=1$

($\therefore$ B is correct)

Also $$l(x)=\begin{cases} b_n+\cos\pi x & (2n-1,\ 2n) \\ a_n+\sin\pi x & [2n,\ 2n+1] \\ b_{n+1}+\cos\pi x & (2n+1,\ 2n+2) \\ a_n+\sin\pi x & [2n+2,\ 2n+3]\end{cases}$$

Again $f((2n+1)^-)=f((2n+1)^+)$

$\Rightarrow \quad a_n=b_{n+1}-1$

$\Rightarrow \quad a_n-b_{n+1}=-1$

$\Rightarrow \quad a_{n-1}-b_n=-1$ ($\therefore$ D is correct)

2. Let $f(x)=\begin{cases} x^2\left|\cos\dfrac{\pi}{x}\right|, & x\neq 0 \\ 0, & x=0\end{cases}$, $x\in R$, then f is : **[JEE-2012]**

(A) differentiable both at x = 0 and at x = 2

(B) differentiable at x = 0 but not differentiable at x = 2

(C) not differentiable at x = 0 but differentiable at x = 2

(D) differentiable neither at x = 2 nor at x = 0

माना $f(x)=\begin{cases} x^2\left|\cos\frac{\pi}{x}\right|, & x\neq 0 \\ 0, & x=0 \end{cases}$, $x\in R$,तो f होगा : **[JEE-2012]**

(A) x = 0 तथा x = 2 पर अवकलनीय है

(B) x = 0 पर अवकलनीय परन्तु x = 2 पर अवकलनीय नहीं है

(C) x = 2 पर अवकलनीय परन्तु x = 0 पर अवकलनीय नहीं है

(D) x = 0 तथा x = 2 दोनों पर अवकलनीय नहीं है

Ans. B

Sol. At $x=0$

$$R.H.D=\lim_{h\to 0}\frac{(0+h)-(0)}{h}=\lim_{h\to 0}\frac{h^2\left|\cos\frac{\pi}{h}\right|-0}{h}$$

$$=\lim_{h\to 0}h\left|\cos\frac{\pi}{h}\right|=0\times\cos(\infty)=0\times \text{finite}=0$$

$$LHD:=\lim_{h\to 0}\frac{l(0-h)-l(0)}{-h}$$

$$=\lim_{h\to 0}\frac{h^2\cos\left(\frac{\pi}{-h}\right)-0}{-h}=\lim_{h\to 0}-h\cos\left(\frac{\pi}{h}\right)$$

$$=0$$

$\because$ LHD = RHD at $x=0$

$\Rightarrow$ $f(x)$ is differentiable at $x=0$

At $x=2$

$$RHD=\lim_{h\to 0}\frac{l(2+h)-l(2)}{h}$$

$$=\lim_{h\to 0}\frac{(2+h)^2.\cos\left(\frac{\pi}{2+h}\right)-0}{h}$$

$$=4\lim_{h\to 0}\frac{\cos\left(\frac{\pi}{2+h}\right)}{h}$$

$$=-4\lim_{h\to 0}\frac{\sin\left(\frac{\pi}{2+h}\right).\left(-\frac{\pi}{(2+h)^2}\right)}{1}=\pi$$

$$\text{LHD}: \lim_{h\to 0}\frac{l(2-h)-l(2)}{-h}$$

$$=\lim_{h\to 0}\frac{(2-h)^2\left(-\cos\left(\frac{\pi}{2-h}\right)\right)}{-h}$$

$$=\lim_{h\to 0}\frac{4\left(\sin\frac{\pi}{(2-h)}\right)\left(\frac{\pi}{(2-h)^2}\right)}{-1}=-\pi.$$

LHD $\neq$ RHD at $x=2$

$\therefore$ Not differentiable at $x=2$.

3. Let $f_1: R\to R$, $f_2:[0,\infty)\to R$, $f_3: R\to R$ and $f_4: R\to[0,\infty)$ be defined by

$$f_1(x)=\begin{cases}|x| & \text{if } x<0,\\ e^x & \text{if } x\geq 0;\end{cases} \qquad f_2(x)=x^2; \qquad f_3(x)=\begin{cases}\sin x & \text{if } x<0,\\ x & \text{if } x\geq 0;\end{cases}$$

and $f_4(x)=\begin{cases}f_2(f_1(x)) & \text{if } x<0,\\ f_2(f_1(x))-1 & \text{if } x\geq 0.\end{cases}$: **[JEE(Adv.)2014]**

Column -I	**Column -II**
P. f_4 is	1. onto but not one-one
Q. f_3 is	2. neither continuous nor one-one
R. $f_2 O f_1$ is	3. differentiable but not one-one
S. f_2 is	4. continuous and one-one
(A) P-3; Q-1; R-4; S-2	(B) P-1; Q-3; R-4; S-2
(C) P-3; Q-1; R-2; S-4	(D) P-1; Q-3; R-2; S-4

माना $f_1: R\to R$, $f_2:[0,\infty)\to R$, $f_3: R\to R$ तथा $f_4: R\to[0,\infty)$ इस प्र कार परिभाषित है

$$f_1(x)=\begin{cases}|x| & \text{if } x<0,\\ e^x & \text{if } x\geq 0;\end{cases} \qquad f_2(x)=x^2; \qquad f_3(x)=\begin{cases}\sin x & \text{if } x<0,\\ x & \text{if } x\geq 0;\end{cases}$$

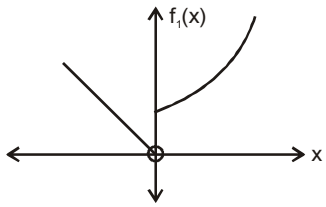
तथा $f_4(x)=\begin{cases}f_2(f_1(x)) & \text{if } x<0,\\ f_2(f_1(x))-1 & \text{if } x\geq 0.\end{cases}$: **[JEE(Adv.)2014]**

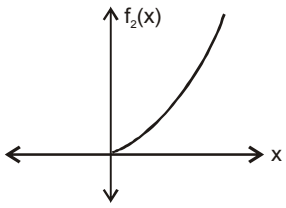
Column -I	**Column -II**
P. f_4 होगा	1. आच्छादक परन्तु एकैकी नहीं
Q. f_3 होगा	2. असतत् तथा बहुएकैकी
R. $f_2 O f_1$ होगा	3. अवकलनीय तथा बहुएकैकी
S. f_2 होगा	4. सतत् तथा एकैकी
(A) P-3; Q-1; R-4; S-2	(B) P-1; Q-3; R-4; S-2
(C) P-3; Q-1; R-2; S-4	(D) P-1; Q-3; R-2; S-4

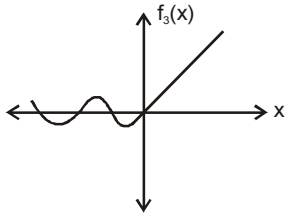
Ans. D

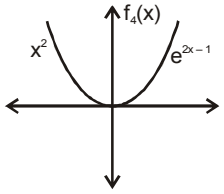
Sol. $f_2(f_1(x)) = (f_1(x))^2 - \begin{cases} x^2 & x < 0 \\ e^{2x} & x \geq 0 \end{cases}$

$f_4(x) \begin{cases} x^2 & x < 0 \\ e^{2x} - 1 & x \geq 0 \end{cases}$









$f_4(x)$ is many-one onto, continuous and non-derivable

$f_3(x)$ is many-one, into, continuous and derivable

$f_2(x)$ is one-one, into, differentiable

$f_4(x)$ बहुएकैकी आच्छादक, सतत् और अवकलनीय नही

$f_3(x)$ बहुएकैकी, अर्न्तक्षेपी, सतत्, अवकलनीय

$f_2(x)$ एकैकी, अर्न्तक्षेपी, अवकलनीय

Hence अतः $R \to 2$

so अतः (D)

$p \to 1, q \to 3, R \to 2, S \to 4$

4. Let $f : R \to R$ and $g : R \to R$ be respectively given by $f(x) = |x| + 1$ and $g(x) = x^2 + 1$. Define $h : R \to R$ by

$$h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0 \end{cases}$$

The number of points at which h(x) is not differentiable is : **[JEE(Adv.)2014]**

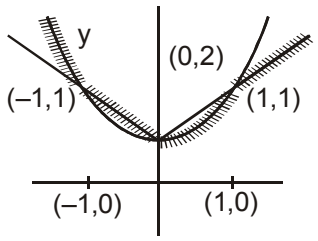
माना $f : R \to R$ तथा $g : R \to R$ क्रमशः दिये गये $f(x) = |x| + 1$ तथा $g(x) = x^2 + 1$. फलन $h : R \to R$ परिभाषित है

$$h(x) = \begin{cases} \max\{f(x), g(x)\} & \text{if } x \leq 0, \\ \min\{f(x), g(x)\} & \text{if } x > 0 \end{cases}$$ बिन्दुओं की संख्या होगी जहाँ h(x) अवकलनीय नहीं है। **[JEE Adv 2014]**

Ans. 3

Sol. $(x) = |x| + 1 = \begin{cases} x + 1 & x \geq 0 \\ -x + 1 & x < 0 \end{cases}$

$g(x) = x^2 = 1$



Number of Non-differential points 3.
अवलनीय नहीं होने वाले बिन्दुओं की संख्या 3

5. Let $g : R \to R$ be a differentiable function with $g(0) = 0$, $g'(0) = 0$ and $g'(1) \neq 0$.

Let $f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ and $h(x) = e^{|x|}$ for all $x \in R$. Let $(f \circ h)(x)$ denote $f(h(x))$ and

$(h \circ f)(x)$ denote $h(f(x))$. Then which of the following is (are) true ? : **[JEE(Adv.)2015]**

(A) f is differentiable at $x = 0$

(B) h is differentiable at $x = 0$

(C) $f \circ h$ is differentiable at $x = 0$

(D) $h \circ f$ is differentiable at $x = 0$

माना कि $g : R \to R$ एक अवकलनीय फलन है जहाँ कि $g(0) = 0$, $g'(0) = 0$ एवं $g'(1) \neq 0$ है। माना कि

$f(x) = \begin{cases} \frac{x}{|x|} g(x), & x \neq 0 \\ 0, & x = 0 \end{cases}$ और प्रत्येक $x \in R$ के लिए $h(x) = e^{|x|}$ है। माना कि $(f \circ h)(x)$, $f(h(x))$ को और $(h \circ f)(x)$, $h(f(x))$

को दर्शाते है। तब निम्नलिखित कथनों में से कौन कौन से सही है ? **[JEE(Adv.)2015]**

(A) $x = 0$ पर f अवकलनीय है

(B) $x = 0$ पर h अवकलनीय है

(C) $x = 0$ पर $f \circ h$ अवकलनीय है

(D) $x = 0$ पर $h \circ f$ अवकलनीय है

Ans. AD

Sol. $g(0) = 0, \quad g'(0) = 0 \quad g'(1) \neq 0$

$$f(x)\begin{cases} g(x) & ; \ x > 0 \\ -g(x) & ' \ x < 0 \\ 0 & ; \ 0 \end{cases} \quad h(x) = e^{|x|}$$

$$f\left(h\left(x\right)\right) = g\left(e^{|x|}\right) \ h\left(f\left(x\right)\right) = e^{|g(x)|}$$

$$R\left(f'\left(0\right)\right) = \lim_{x \to 10} \frac{g\left(x\right) - g\left(0\right)}{x - 0} = g'(0) = 0$$

$$L\left(f'(0)\right)=\lim_{x\to 10}\frac{-g(x)-g(0)}{x-0}=g'(0)=0$$

$R\left(h'(0)\right)=1$ & $L\left(h'(0)\right)=-1$ So h(x) is non derivable. at x = 0

अतः h(x), x = 0 पर अवकलनीय नहीं है।

Now $\lim_{x\to 0}\frac{f(h(x))-f(h(0))}{x}=\lim_{x\to 0}\frac{g\left(e^{|x|}\right)-g(1)}{x}$

$$R(f'(h(x)))=\lim_{x\to 0^+}\frac{g(e^x)-g(1)}{x}=\lim_{x\to 0^+}\frac{g(e^x)-g(1)}{e^x-1}\frac{e^x-1}{x}=g'(1)$$

$L\left(f'(h(x))\right)=-g'(1)$ Hence $f(h(x))$ is non derivable at x = 0

अतः $f(h(x))$, x = 0 पर अवकलनीय नहीं है।

Since x = 0 is repeated root of g(x) So $e^{|g|x||}$ is differetiable at x = 0

चूंकिx = 0, g(x) का पुनरावृत्ति मूल है। अतः $e^{|g|x||}$, x = 0 पर अवकलनीय है।

hence अतः (A), (D)

6. Let $a, b \in R$ and $f: R \to R$ be defined by f(x) = a cos ($|x^3 - x|$) + b |x|sin($|x^3 + x|$). Then f is :

[JEE(Adv.)2016]

(A) differentiable at x = 0 if a = 0 and b = 1

(B) differentiable at x = 1 if a = 1 and b = 0

(C) NOT differentaible at x=0 at a=1 and b=0

(D) NOT differentiable at x=1 if a=1 and b=1

माना कि $a, b \in R$ तथा $f: R \to R$, f(x) = a cos ($|x^3 - x|$) + b |x|sin($|x^3 + x|$) से परिभाषित है। तब f :

[JEE(Adv.)2016]

(A) x = 0 पर अवकलनीय है यदि a = 0 और b = 1

(B) x = 1 पर अवकलनीय है यदि a = 1 और b = 0

(C) x = 0 पर अवकलनीय नहीं है यदि a = 1 और b = 0

(D) x = 1 पर अवकलनीय नहीं है यदि a = 1 और b = 1

Ans. AB

Sol. $f(x) = a\cos(x^3 - x) + bx\sin(x^3 + x) \quad \forall\ x \in R$

Which is composition and sum of differentiable functions, therefore f(x) is always continuous and differentiable :

7. Let $f: \left[-\frac{1}{2}, 2\right] \to R$ and $g: \left[-\frac{1}{2}, 2\right] \to R$ be functions defined by $f(x) = [x^2 - 3]$ and $g(x) = |x|\, f(x) + |4x - 7|$

f(x), where [y] denotes the greatest integer less than or equal to y for $y \in R$. Then : **[JEE(Adv.)2016]**

(A) f is discontinuous exactly at three points in $\left[-\frac{1}{2}, 2\right]$

(B) f is discontinuous exactly at four points in $\left[-\frac{1}{2}, 2\right]$

(C) g is NOT differentiable exactly at four points in $\left(-\frac{1}{2}, 2\right)$

(D) g is NOT differentiable exactly at five point in $\left(-\frac{1}{2}, 2\right)$

माना कि फलन $f:\left[-\frac{1}{2}, 2\right] \to R$ और $g:\left[-\frac{1}{2}, 2\right] \to R$, $f(x) = [x^2 - 3]$ और $g(x) = |x|\, f(x) + |4x - 7|\, f(x)$ से परिभाषित हैं, जहाँ $y \in R$ के लिए y से कम या y के बराबर के महत्तम पूर्णांक (greatest integer less than or equal to y) को [y] द्वारा दर्शाया गया है। तब : **[JEE(Adv.)2016]**

(A) $\left[-\frac{1}{2}, 2\right]$ में f ठीक तीन (exactly three) बिन्दुओं पर असंतत (discontinuous) है

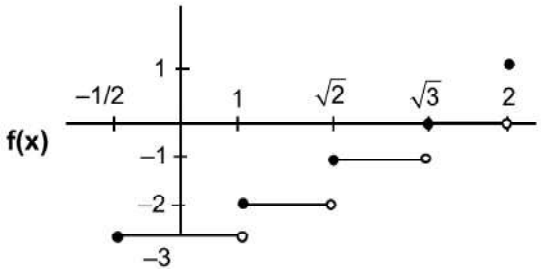
(B) $\left[-\frac{1}{2}, 2\right]$ में f ठीक चार (exactly four) बिन्दुओं पर असंतत है

(C) $\left(-\frac{1}{2}, 2\right)$ में g ठीक चार (exactly four) बिन्दुओं पर अवकलनीय (differentiable) नहीं है

(D) $\left(-\frac{1}{2}, 2\right)$ में g ठीक पाँच (exactly five) बिन्दुओं पर अवकलनीय (differentiable) नहीं है

Ans. BC

Sol.



Clearly from the graph f is discontinuous at four points.

$g(x) = f(x)(|x|+|4x-7|)$

f(x) is non-differentiable at $x = 1, \sqrt{2}, \sqrt{3}$

& $|x|+|4x-7|$ is non-differentiable at $x = 0, \frac{7}{4}$

But $f(x) = 0 \quad \forall x \in \left[\sqrt{3}, 2\right]$

Hence g(x) is non-differentiable at $x = 0, 1, \sqrt{2}, \sqrt{3}$

8. Let [x] be the greatest integer less than or equal to x. Then, at which of the following point(s) the function $f(x) = x \cos(\pi([x] + x))$ is discontinuous? **[JEE(Adv.)2017]**

माना कि x से छोटा या x के समान सबसे बढ़ा पूर्णांक [x] है। तब $f(x) = x \cos(\pi([x] + x))$, निम्न में से किन बिन्दु/बिन्दुओं पर असतत है? **[JEE(Adv.)2017]**

(A) $x = 2$ (B) $x = 1$ (C) $x = -1$ (D) $x = 0$

Ans. ABC

Sol. $f(x) = \cos(\pi x + \pi[x])$

$f(x) \begin{cases} x \cos \pi x & [x] = \text{even} \\ -\cos \pi x & [x] = \text{odd} \end{cases}$

Now check at integer

f(x) is continuous at x = 0

f(x) is disc. at x = 2, 1, –1

9. Let $f_1 : R \to R, f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R, f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \to R$ and $f_4 : R \to R$ defined by :

[JEE(Adv.)2018]

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$,

(ii) $f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & x = 0 \end{cases}$, where the inverse trigonometric function $\tan^{-1}x$ assumes values in $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$,

(iii) $f_3(x) = [\sin(\log_e(x+2))]$, where, for $t \in \mathbb{R}$ [t] denotes the greatest integer less than or equal to t,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Column -I	**Column - II**
P. The function f_1 is	1. NOT continuous at x = 0
Q. The function f_2 is	2. continuous at x = 0 and NOT differentiable at x = 0
R. The function f_3 is	3. differentiable at x = 0 and its derivative is NOT continuous at x = 0
S. The function f_4 is	4. differentiable at x = 0 and its derivative is continuous at x = 0

The correct option is: **[JEE(Adv.)2018]**

(A) P → 2; Q → 3; R → 1; S → 4 (B) P → 4; Q → 1; R → 2; S → 3

(C) P → 4; Q → 2; R → 1; S → 3 (D) P → 2; Q → 1; R → 4; S → 3

माना कि फलन $f_1 : R \to R, f_2 : \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \to R, f_3 : \left(-1, e^{\frac{\pi}{2}} - 2\right) \to R$ और $f_4 : R \to R$ इस प्रकार परिभाषित हैं कि

[JEE(Adv.)2018]

(i) $f_1(x) = \sin\left(\sqrt{1 - e^{-x^2}}\right)$,

(ii) $f_2(x) = \begin{cases} \frac{|\sin x|}{\tan^{-1} x} & \text{if } x \neq 0 \\ 1 & x = 0 \end{cases}$, जहाँ प्रतिलोम त्रिकोणमितीय फलन $\tan^{-1}x$, $\left(-\frac{\pi}{2}, \frac{\pi}{2}\right)$ में मान धारण करता है,

(iii) $f_3(x) = [\sin(\log_e(x+2))]$, जहाँ $t \in \mathbb{R}$ के लिये, [t], t से छोटा या t के बराबर महत्तम पूर्णांक को दर्शाता है ,

(iv) $f_4(x) = \begin{cases} x^2 \sin\left(\frac{1}{x}\right) & \text{if } x \neq 0 \\ 0 & \text{if } x = 0 \end{cases}$

Column -I	**Column - II**
P. फलन f_1	1. $x = 0$ पर संतत नहीं है
Q. फलन f_2	2. $x = 0$ पर संतत है और $x = 0$ पर अवकलनीय नहीं है
R. फलन f_3	3. $x = 0$ पर अवकलनीय है और $x = 0$ पर इसका अवकलज संतत नहीं है
S. फलन f_4	4. $x = 0$ पर अवकलनीय है और $x = 0$ पर इसका अवकलज संतत है

दिए हुए विकल्पों में से सही विकल्प है : **[JEE(Adv.)2018]**

(A) $P \to 2; Q \to 3; R \to 1; S \to 4$ (B) $P \to 4; Q \to 1; R \to 2; S \to 3$

(C) $P \to 4; Q \to 2; R \to 1; S \to 3$ (D) $P \to 2; Q \to 1; R \to 4; S \to 3$

Ans. D

Sol. (i) $f'_1(0) = \lim_{h\to 0} \frac{\sin\sqrt{1-e^{-h^2}} - 0}{h} = \lim_{h\to 0} \frac{\sin\sqrt{1-e^{-h^2}}}{\sqrt{1-e^{-h^2}}} \times \sqrt{\frac{1-e^{-h^2}}{h^2}} \times \frac{|h|}{h}$

$= 1 \times 1 \times \frac{|h|}{h} = 1 \times 1 \times \frac{|h|}{h}$

= limit does not exist.

$\Rightarrow$ for option (P), (2) is correct.

(ii) $\lim_{x\to 0} f_2(x) = \lim_{x\to 0} \frac{|\sin x|}{\tan^{-1} x}$

$= \lim_{x\to 0} \frac{|\sin x|}{|x|} \times \frac{x}{\tan^{-1} x} \times \frac{|x|}{x}$

$= \lim_{x\to 0} 1 \times 1 \times \frac{|x|}{x}$

= limit does not exist $\Rightarrow$ for option Q, (1) is correct.

(iii) $= \lim_{x\to 0} f_3(x) = \lim_{x\to 0}[\sin(\log_e(x+2) \text{ tends to } 2]$

now at x tends to zero (x + 2) tends to 2

$\Rightarrow$ $\log_e(x+2)$ tends to $ln2$

which is less than 1

$0 < \lim_{x\to 0} \sin(\log_e(x+2)) < \sin 1 \quad \Rightarrow \quad \lim_{x\to 0}[\sin(\log_e(x+2))] = 0$

$f_3(x) = \{0 \quad x \in [-1, e^{\pi/2} - 2)$

$\Rightarrow \quad f'_3(x) = 0 \quad \forall\, x \in (-1, e^{\pi/2} - 2)$

$\Rightarrow \quad f''_3(x) = 0 \quad \forall\, x \in (-1, e^{\pi/2} - 2)$

Hence for (R), (4) is correct.

(iv) $\lim_{x\to 0} f_4(x) = \lim_{x\to 0}\left(x^2 \sin\frac{1}{x}\right) = \lim_{x\to 0} x^2\left(\sin\frac{1}{x}\right) = 0$

$$f'_4(0) = \lim_{h\to 0}\frac{h^2 \sin\left(\frac{1}{x}\right) - 0}{x} = \lim_{h\to 0} h \sin\left(\frac{1}{x}\right) = 0$$

$$f'_4(x) = -\cos\frac{1}{x} + x\sin\frac{1}{x}, x \neq 0$$

$$f''_4(0) = \frac{-\cos\frac{1}{h} + h\sin\frac{1}{h} - 0}{h} \quad \Rightarrow \quad \text{does not exist}$$

hence for (S), (3) is correct.

10. Let $f: R \to R$ be a function. We say that f has

PROPERTY 1 if $\lim_{h\to 0}\frac{f(h) - f(0)}{\sqrt{|h|}}$ exists and is finite and

PROPERTY 2 if $\lim_{h\to 0}\frac{f(h) - f(0)}{h^2}$ exists and is finite and

Then which of the following options is/are correct ? **[JEE(Adv.)2019]**

(A) $f(x) = |x|$ has PROPERTY 1 (B) $f(x) = x|x|$ has PROPERTY 2

(C) $f(x) = \sin x$ has PROPERTY 2 (D) $f(x) = x^{2/3}$ has PROPERTY 1

माना कि $f: R \to R$ एक फलन है। हम कहते हैं कि f में

गुण 1 (PROPERTY 1) है यदि $\lim_{h\to 0}\frac{f(h)-f(0)}{\sqrt{|h|}}$ का अस्तित्व (exists) है और वह परिमित (finite) है,

गुण 2 (PROPERTY 2) है यदि $\lim_{h\to 0}\frac{f(h)-f(0)}{h^2}$ का अस्तित्व (exists) है और वह परिमित (finite) है।

तब निम्न में से कौन सा (से) विकल्प सही है (हैं) ? **[JEE(Adv.)2019]**

(A) $f(x) = |x|$ में गुण 1 है (B) $f(x) = x|x|$ में गुण 2 है

(C) $f(x) = \sin x$ में गुण 2 है (D) $f(x) = x^{2/3}$ में गुण 1 है

Ans. AD

Sol. Option (1) $f(x) = |x|$ $f(0) = 0$

Property -1

$$\lim_{h\to 0}\frac{|h|}{\sqrt{|h|}} = \lim_{h\to 0}|h|^{1/2} = 0 \text{ limit exists}$$

Option (2) $f(x) = x|x|$ $f(0) = 0$

Property -2

$$\lim_{h\to 0}\frac{h|h|-0}{h^2} = \lim_{h\to 0}|\frac{|h|}{h}$$

LHL = –1

RHL = 1

LHL ≠ RHL

⇒ Does not exist

Option (3) $f(x) = \sin x$ $f(0) = 0$

Property -2

$$\lim_{h\to 0}\frac{\sin h}{h^2} = \lim_{h\to 0}\left(\frac{\sin h}{h}\cdot\frac{1}{h}\right)$$

⇒ Does not exist

Option (4) $f(x) = x^{2/3}$

$= (x^2)^{1/3} = |x|^{2/3} f(0) = 0$

Property -1

$$\lim_{h\to 0}\frac{|h|^{2/3}-0}{\sqrt{|h|}} = \lim_{h\to 0}|h|^{2/3-1/2} = \lim_{h\to 0}|h|^{1/6} = 0 \text{ limit existsi}$$

11. Let the function $f: R \to R$ be defined by $f(x) = x^3 - x^2 + (x-1)\sin x$ and let $g: R \to R$ be an arbitrary function. Let $fg: R \to R$ be the product function defined by $(fg)(x) = f(x)\,g(x)$. Then which of the following statements is/are TRUE? **[JEE(Adv.)2020]**

(A) If g is continuous at x = 1, then fg is differentiable at x = 1

(B) If fg is differentiable at x = 1, then g is continuous at x = 1

(C) If g is differentiable at x =1, then f g is differentiable at x =1

(D) If f g is differentiable at x =1, then g is differentiable at x =1

माना कि फलन $f: R \to R$ को $f(x) = x^3 - x^2 + (x-1)\sin x$ द्वारा परिभाषित किया जाता है, एवं माना कि

$g: R \to R$ एक स्वेच्छ फलन है। माना कि $fg: R \to R$ गुणन फलन है जो कि $(fg)(x) = f(x)\,g(x)$ के द्वारा परिभाषित है। तब निम्न में से कौन सा (से) कथन सही है (हैं) ? **[JEE(Adv.)2020]**

(A) यदि x =1 पर g संतत है, तब x =1 पर f g अवकलनीय है

(B) यदि x = 1 पर f g अवकलनीय है, तब x =1 पर g संतत है

(C) यदि x =1 पर g अवकलनीय है, तब x =1 पर f g अवकलनीय है

(D) यदि x =1 पर f g अवकलनीय है, तब x =1 पर g अवकलनीय है

Ans. AC

Sol. $f(x) = x^3 - x^2 + (x-1)\sin(x)$

$f(1) = 0$

f is continous & differentiable

$f'(1) = 1 + \sin 1$

Let $h(x) = f(x)\,g(x)$, $h(1) = 0$

Differentiability at x = 1

$$h'(1^+) = \lim_{h\to\infty} \frac{f(1+h)\,g(1+h) - 0}{h}$$

$= f'(1)\,g(1^+)$

$$h'(1^-) = \lim_{h\to\infty} \frac{f(1-h)\,g(1-h) - 0}{-h}$$

$= f'(1)\,g(1^-)$

(A) If g is continuous at x = 1

$\Rightarrow g(1^+) = g(1) = g(1^-)$

$\Rightarrow h'(1^+) = h'(1^-)$

(B) If $h'(1^+) = h'(1^-)$

$\Rightarrow g(1^+) = g(1^-)$

but g(1) can be different

(C) If g is differentiable at x = 1

$\Rightarrow$ g is continuous at x = 1

from A we can say that C is true

(B) is false so (D) is also false

12. Let $f: R \to R$ and $g: R \to R$ be functions satisfying

$f(x+y) = f(x) + f(y) + f(x)f(y)$ and $f(x) = xg(x)$

for all $x, y \in R$. If $\lim_{x\to 0} g(x) = 1$, then which of the following statements is/are TRUE ? **[JEE(Adv.)2020]**

(A) f is differentiable at every $x \in R$ (B) If $g(0) = 1$, then g is differentiable at every $x \in R$

(C) The derivative $f'(1)$ is equal to 1 (D) The derivative $f'(0)$ is equal to 1

माना फलन $f: R \to R$ और $g: R \to R$ ऐसे फलन हैं जो कि सभी $x, y \in R$ के लिये

$f(x+y) = f(x) + f(y) + f(x)f(y)$ एवं $f(x) = xg(x)$

को सन्तुष्ट करते हैं। यदि $\lim_{x\to 0} g(x) = 1$ है, तब निम्न में से कौन सा (से) कथन सही है (हैं) ? **[JEE(Adv.)2020]**

(A) f प्रत्येक $x \in R$ पर अवकलनीय है

(B) यदि $g(0) = 1$ है, तब g प्रत्येक $x \in R$ पर अवकलनीय है

(C) अवकलज $f'(1)$ का मान 1 के बराबर है

(D) अवकलज $f'(0)$ का मान 1 के बराबर है

Ans. ABD

Sol. $f'(x) = \lim_{h\to 0} \frac{f(x+h) - f(x)}{h}$

$= \lim_{h\to 0} \frac{f(x+h) - f(x+0)}{h}$

$= \lim_{h\to 0} \frac{f(x) + f(h) + f(x)f(h) - f(x) - f(0) - f(x)f(0)}{h}$

$f'(x) = \lim_{h\to 0} \frac{f(h) - f(0) + f(x)\big(f(h) - f(0)\big)}{h}$

$f'(x) = \lim_{h\to 0} \left(\frac{f(h) - f(0)}{h}\right)\big(f(x) + 1\big)$

$f'(x) = \lim_{h\to 0} \left(\frac{hg(h) - (0)g(0)}{h}\right)\big(f(x) + 1\big)$

$f'(x) = \lim_{h\to 0} g(h).(f(x) + 1)$

$f'(x) = f(x) + 1$

$\int \frac{f'(x)}{f(x)+1} dx = \int 1 dx$

$\ln(f(x) + 1) = x + c \{ \because f(0) = 0\}$

$c = 0$

$f(x) = e^x - 1$

f is differentiable at every $x \in R$

$g(x) = \frac{f(x)}{x} = \frac{e^x - 1}{x}$

Check at $x = 0$

$$g'(0) = \lim_{h\to 0} \frac{g(0+h) - g(0)}{h}$$

$$\lim_{h\to 0} \frac{\frac{e^h - 1}{h} - 1}{h}$$

$$\lim_{h\to 0} \frac{1 + h + \frac{h^2}{2!} + \ldots\ldots\ldots\ldots - 1 - h}{h}$$

$$g'(0) = \frac{1}{2}$$

If $g(0) = 1$ then g is differentiable at every $x \in R$

$f'(1) = f(1) + 1 = e - 1 + 1 = e$

$f'(0) = f(0) + 1 = 1$

13. Let the functions $f: (-1, 1) \to R$ and $g: (-1, 1) \to (-1, 1)$ be defined by

$f(x) = |2x - 1| + |2x + 1|$ and $g(x) = x - [x]$,

where $[x]$ denotes the greatest integer less than or equal to x. Let $f \circ g: (-1, 1) \to R$ be the composite function defined by $(f \circ g)(x) = f(g(x))$. Suppose c is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is NOT continuous, and suppose d is the number of points in the interval $(-1, 1)$ at which $f \circ g$ is NOT differentiable. Then the value of $c + d$ is ———. **[JEE(Adv.)2020]**

माना कि फलनों $f: (-1, 1) \to R$ एवं $g: (-1, 1) \to (-1, 1)$ को $f(x) = |2x - 1| + |2x + 1|$ एवं $g(x) = x - [x]$ से परिभाषित किया जाता है, जहाँ $[x]$ उस महत्तम पूर्णांक को दर्शाता है जो x से कम या x के बराबर है। माना कि $f \circ g: (-1, 1) \to R$ संयुक्त फलन है जो कि $(f \circ g)(x) = f(g(x))$ द्वारा परिभाषित है। मान लीजिये कि c, अन्तराल $(-1, 1)$ में उन बिंदुओं की संख्या है, जिन पर $f \circ g$ संतत नहीं है, एवं d, अन्तराल $(-1, 1)$ में उन बिंदुओं की संख्या है, जिन पर $f \circ g$ अवकलनीय नहीं है। तब $c + d$ का मान है ——— **[JEE(Adv.)2020]**

Ans. 4

Sol. $f(x) = |2x - 1| + |2x + 1|$ $g(x) = x - [x]$

$g(x) = [x] + \{x\} - [x]$

$g(x) = \{x\}$

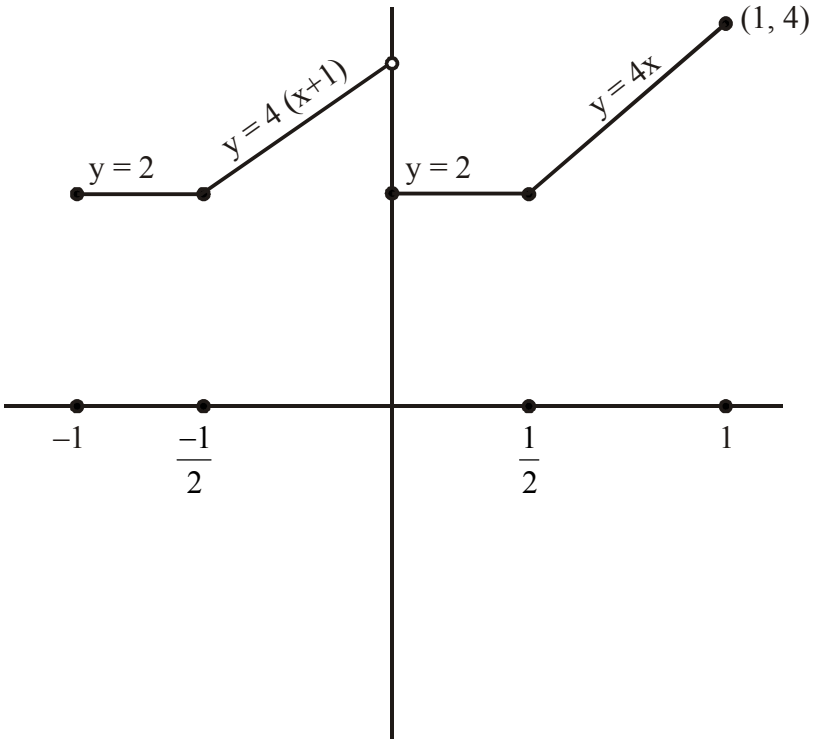
$$f(x) = \begin{cases} 4x & : x \geq \frac{1}{2} \\ 2 & : \frac{-1}{2} < x < \frac{1}{2} \\ -4x & : x \leq \frac{-1}{2} \end{cases}$$

$$f\left(g(x)\right)=\begin{cases}4\{x\} & :\{x\}\geq\dfrac{1}{2}\\ 2 & :\dfrac{-1}{2}<\{x\}<\dfrac{1}{2}\\ -4\{x\} & :\{x\}\leq\dfrac{-1}{2}\end{cases}$$

$$f\left(g(x)\right)=\begin{cases}4\{x\} & :\{x\}\geq\dfrac{1}{2}\\ 2 & :0\leq\{x\}<\dfrac{1}{2}\end{cases}$$

$$f\left(g(x)\right)=\begin{cases}4\{x\} & :x\in\left[\dfrac{1}{2},1\right)\cup\left[\dfrac{-1}{2},0\right)\\ 2 & :\left[0,\dfrac{1}{2}\right)\cup\left(\dfrac{-1}{2},-1\right)\end{cases}$$

$$f\left(g(x)\right)=\begin{cases}4x & :x\in\left[\dfrac{1}{2},1\right)\\ 4\left(x+1\right) & :x\in\left[\dfrac{-1}{2},0\right)\\ 2 & :x\in\left[0,\dfrac{1}{2}\right)\cup\left(\dfrac{-1}{2},-1\right)\end{cases}$$



Not continuous at x = 0

Not differetiable at x = $\pm\frac{1}{2}, 0$

c = 1

d = 3

c + d = 1 + 3 = 4

14. Let $S = (0,1) \cup (1,2) \cup (3,4)$ and $T = \{0,1,2,3\}$. Then which of the following statements is(are) true?

[JEE(Adv.)2023]

(A) There are infinitely many functions from S to T

(B) There are infinitely many strictly increasing functions from S to T

(C) The number of continuous functions from S to T is at most 120

(D) Every continuous function from S to T is differentiable

माना कि $S = (0,1) \cup (1,2) \cup (3,4)$ एवं $T = \{0,1,2,3\}$ है। तब निम्न में से कौनसा(से) कथन सत्स है(हैं)?

[JEE(Adv.)2023]

(A) S से T तक अनन्ततः अनेक (Infinitely many) फलन हैं

(B) S से T तक अनंततः अनेक (infinitely many) निरंतर वर्धमान (strictly increasing) फलन हैं

(C) S से T तक संतत (continuous) फलनों की संख्या 120 या उससे कम है

(D) S से T तक प्रत्येक संतत फलन अवकलनीय (differentiable) है

Ans. ACD

Sol. $S=(0,1)\cup(1,2)\cup(3,4)$

$T=\{0,1,2,3\}$

Number of functions

Each element of S have 4 choice

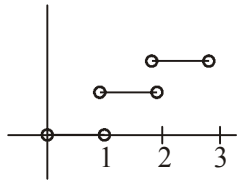
Let n be the number of element in set S.

Number of function $=4^n$

Here $n\to\infty$

$\Rightarrow$ Option (A) is correct.

Option (B) is incorrect (obvious)

(C) 

For continuous function

Each interval will have 4 choices

$\Rightarrow$ Number of continuous functions

$=4\times4\times4=64$

$\Rightarrow$ Option (C) is correct.

(D) Every continuous function is piecewise constant functions

$\Rightarrow$ Differentiable.

Option (D) is correct.

15. Let $f:(0,1)\to R$ be the function defined as $f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right)$, where [x] denotes the greatest integer less than or equal to x. Then which of the following statements is (are) true ? **[JEE(Adv.)2023]**

(A) The function f is discontinuous exactly at one point in (0, 1)

(B) There is exactly one point in (0, 1) at which the function f is continuous but **NOT** differentiable

(C) The function f is **NOT** differentiable at more than three points in (0, 1)

(D) The minimum value of the function f is $-\frac{1}{512}$

माना कि फलन $f:(0,1)\to R$ इस प्रकार परिभाषित है कि $f(x)=[4x]\left(x-\frac{1}{4}\right)^2\left(x-\frac{1}{2}\right)$, जहाँ x से कम या x के बराबर महत्तम पूर्णांक (greatest integer) को निरूपित करता है। तब निम्न में से कौनसा (से) कथन सत्य है (हैं) ?

[JEE(Adv.)2023]

(A) फलन f अन्तराल (0, 1) में केवल एक बिन्दु पर असंतत (discontinuous) है

(B) अन्तराल (0, 1) में केवल एक बिन्दु है जिस पर फलन f संतत है किन्तु अवकलनीय नहीं (continuous but not differentiable) है

(C) अन्तराल (0, 1) के तीन से अधिक बिन्दुओं पर फलन f अवकलनीय नहीं (not differentiable) है

(D) फलन f का न्यूनतम मान (minimum value) $-\frac{1}{512}$ है

Ans. AB

Sol. $f: (0, 1) \to R$

$f(x) = [4x]\left(x - \frac{1}{4}\right)^2\left(x - \frac{1}{2}\right)$ $\Rightarrow$ Critical point $= \frac{1}{4}, \frac{1}{2}, \frac{3}{4}$

Discontinuity at $x = \frac{3}{4}$

Continuous and differentiable at $x = \frac{1}{4}$

Continuous but non-differentiable at $x = \frac{1}{2}$

LHD $\left(\text{at } x = \frac{1}{4}\right)$ RHD $\left(\text{at } x = \frac{1}{4}\right)$

$\lim\limits_{h \to 0^+} \frac{0 - 0}{-h} = 0$ $\lim\limits_{h \to 0^+} \frac{h^2\left(-\frac{1}{2} + h\right)}{h} = 0$

LHD $\left(\text{at } x = \frac{1}{2}\right)$ RHD $\left(\text{at } x = \frac{1}{2}\right)$

$\lim\limits_{h \to 0^+} \frac{\left(\frac{1}{4} - h\right)^2(-h) - 0}{-h} - \frac{1}{16}$ $\lim\limits_{h \to 0^+} \frac{2\left(\frac{1}{4} + h\right)^2 h - 0}{h} - \frac{1}{8}$

Minimum -ve value will exist between $\frac{1}{4}$ and $\frac{1}{2}$

$f(x) = \left(x - \frac{1}{4}\right)^2\left(x - \frac{1}{2}\right)$ $\frac{1}{4} \le x \le \frac{1}{2}$

$f'(x) = \left(x - \frac{1}{4}\right)\left(3x - \frac{5}{4}\right)$ $\Rightarrow$ minima at $x = \frac{5}{12}$

$f\left(\frac{5}{12}\right) = \frac{1}{36} \times \frac{-1}{12} = \frac{-1}{432}$

16. Let f : R $\to$ R and g : R $\to$ R be functions defined by $f(x)=\begin{cases} x|x|\sin\left(\frac{1}{x}\right) & ,x\neq 0 \\ 0, & x=0, \end{cases}$ and

$g(x)=\begin{cases} 1-2x, & 0\le x\le\frac{1}{2}, \\ 0, & \text{otherwise,} \end{cases}$. Let $a,b,c,d \in R$. Define the function $h : R \to R$ by

$$h(x)=a\,f(x)+b\left(g(x)+g\left(\frac{1}{2}-x\right)\right)+c\left(x-g(x)\right)+dg(x), x\in R .$$

Match each entry in **List-I** to the correct entery in **List -II**.

List -I	**List -II**
(P) If a = 0, b = 1, c = 0, and d = 0, then	(1) h is one-one
(Q) If a = 1, b = 0, c = 0, and d = 0, then	(2) h is onto
(R) If a = 0, b = 0, c = 1, and d = 0, then	(3) h is differentiable on R
(S) If a = 0, b = 0, c = 0, and d = 1, then	(4) the range of h is [0, 1]
	(5) the range of h is {0, 1}

The correct option is : **[JEE(Adv.)2024]**

माना कि फलन $f : R \to R$ और $g : R \to R$, $f(x)=\begin{cases} x|x|\sin\left(\frac{1}{x}\right) & ,x\neq 0 \\ 0, & x=0, \end{cases}$ और $g(x)=\begin{cases} 1-2x, & 0\le x\le\frac{1}{2}, \\ 0, & \text{otherwise,} \end{cases}$ द्वारा परिभाषित हैं। माना कि $a,b,c,d \in R$ हैं। फलन $h : R \to R$ को

$h(x)=a\,f(x)+b\left(g(x)+g\left(\frac{1}{2}-x\right)\right)+c\left(x-g(x)\right)+dg(x), x\in R$ द्वारा परिभाषित कीजिये।

सूची-I की प्रत्येक प्रविष्टि का **सूची-II** की सही प्रविष्टि से मिलान कीजिए।

सूची-I	**सूची-II**
(P) यदि a = 0, b = 1, c = 0, और d = 0 है, तब	(1) h एकैकी है।
(Q) यदि a = 1, b = 0, c = 0, और d = 0 है, तब	(2) h आच्छादी है।
(R) यदि a = 0, b = 0, c = 1, और d = 0 है, तब	(3) h, R पर अवकलनीय है।
(S) यदि a = 0, b = 0, c = 0, और d = 1 है, तब	(4) h का परिसर [0, 1] है।
	(5) h का परिसर {0, 1} है।

सही विकल्प है : **[JEE(Adv.)2024]**

(A) (P) $\to$ (4); (Q) $\to$ (3); (R) $\to$ (1); (S) $\to$ (2)
(B) (P) $\to$ (5); (Q) $\to$ (2); (R) $\to$ (4); (S) $\to$ (3)
(C) (P) $\to$ (5); (Q) $\to$ (3); (R) $\to$ (2); (S) $\to$ (4)
(D) (P) $\to$ (4); (Q) $\to$ (2); (R) $\to$ (1); (S) $\to$ (3)

Ans. C

Sol. (P) $h(x)=g(x)+g\left(\frac{1}{2}-x\right)$

$$g(x)=\begin{cases}1-2x & ; \quad 0\le x\le\frac{1}{2}\\ 0 & ; \quad \text{otherwise}\end{cases}$$

$$g\left(\frac{1}{2}-x\right)=\begin{cases}2x & : \quad 0\le x<\frac{1}{2}\\ 0 & : \quad \text{otherwise}\end{cases}$$

$$h(x)=\begin{cases}1 & : \quad 0\le x<\frac{1}{2}\\ 0 & : \quad \text{otherwise}\end{cases}$$

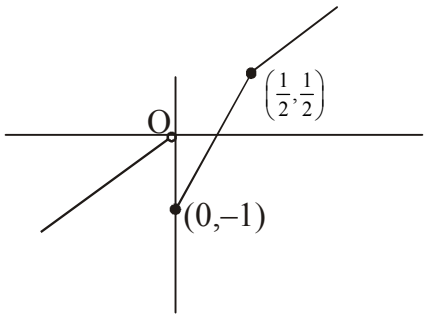
hence, correct option is (5).

(Q) $h(x)=\begin{cases}x|x|\sin\frac{1}{x}; & x\neq 0\\ 0; & \text{otherwise}\end{cases}$

$$h(x)=\begin{cases}x^2\sin\frac{1}{x} & : \quad x>0\\ -x^2\sin\frac{1}{x} & : \quad x<0\\ 0 & : \quad x=0\end{cases}$$

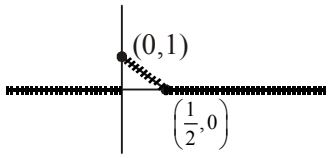
Clearly, function is differentiable everywhere. Hence correct option is (3).

(R) $h(x)=\begin{cases}3x-1 & : \quad 0\le x\le\frac{1}{2}\\ x & : \quad \text{otherwise}\end{cases}$



Clearly onto. Hence, correct option is (2).

(S) $g(x)=\begin{cases}1-2x & : \quad 0 \le x \le \frac{1}{2} \\ 0 & : \quad \text{otherwise}\end{cases}$



Clearly, range is [0,1].

Hence, correct option is (4).

Hence, overall answer is (C).

17. Let R denote the set of all real numbers. Define the function $f: R \to R$ by :

$$f(x)=\begin{cases}2-2x^2-x^2\sin\frac{1}{x} & \text{if} \quad x \neq 0, \\ 2 & \text{if} \quad x = 0.\end{cases}$$

Then which of the following statements is TRUE ? **[JEE(Adv.)2025]**

(A) The function f is **NOT** differentiable at $x = 0$

(B) There is a positive real number δ, such that f is a decreasing function on the interval $(0, \delta)$

(C) For any positive real number δ, the function f is **NOT** an increasing function on the interval $(-\delta, 0)$

(D) $x = 0$ is a point of local minima of f

मान लीजिए कि R सभी वास्तविक संख्याओं (real numbers) के समुच्चय (set) को दर्शाता है।

फलन (function) $f: R \to R$ को :

$$f(x)=\begin{cases}2-2x^2-x^2\sin\frac{1}{x} & \text{यदि} \quad x \neq 0, \\ 2 & \text{यदि} \quad x = 0.\end{cases}$$

द्वारा परिभाषित कीजिये। तब निम्नलिखित कथनों में से कौन सा एक कथन सत्य है ? **[JEE(Adv.)2025]**

(A) $x = 0$ पर, फलन f अवकलनीय **नहीं** है

(B) एक धनात्मक वास्तविक संख्या (positive real number) δ इस प्रकार है कि अंतराल $(0, \delta)$ में फलन f एक ह्रासमान फलन है

(C) किसी भी धनात्मक वास्तविक संख्या δ के लिए अंतराल $(-\delta, 0)$ में फलन f एक वर्धमान फलन **नहीं** है

(D) $x = 0$, फलन f का एक स्थानीय निम्नतम का बिन्दु है

Ans. C

Ans. (By Martix B)

Sol. Calculate LHD & RHD at $x = 0$

$$f'\left(0^+\right) = \lim_{h\to 0}\frac{f\left(0+h\right)-f\left(0\right)}{h}$$

$$= \lim_{h\to 0}\frac{-2h^2 - h^2\sin\left(\frac{1}{h}\right)}{h}$$

$$= \lim_{h\to 0}\left(-h\left(2+\sin\left(\frac{1}{h}\right)\right)\right)$$

$f'(0^+) = 0$

$$f'\left(0^-\right) = \lim_{h\to 0}\frac{f\left(0-h\right)-f\left(0\right)}{-h}$$

$$= \frac{-2h^2 + h^2\sin\left(\frac{1}{h}\right)}{-h}$$

$$= \lim_{h\to 0} h\left(2-\sin\left(\frac{1}{h}\right)\right) = 0$$

In nbd of $x = 0$

$f'(0^+) < 0 \qquad f'(0^-) > 0$

$\Rightarrow x = 0$ is point of local maxima

f is increasing in left nbd of $x = 0$

f is decreasing in right nbd of $x = 0$

Exercise C - I

1. Which statement is false :

(A) $f(x) = \underset{n\to\infty}{\text{Limit}} \frac{1}{1+n\sin^2\pi x}$ is continuous at $x = 1$.

(B) The function $f(x) = 2^{-2^{1/(1-x)}}$ if $x \neq 1$ & $f(1) = 1$ is not continuous at $x = 1$.

(C) There exists a continuous function f: [0, 1] onto [0, 10], but there exists no continuous function g : [0, 1] onto (0, 10).

(D) If $f(x)$ is continuous in $[0, 1]$ & $f(x) = 1$ for all rational numbers in $[0, 1]$ then $f\left(1/\sqrt{2}\right)$ equal to 1.

(E) If $f(x) = \begin{cases} \frac{\cos\pi x + \sin(\pi x/2)}{(x-1)(3x^2-2x-1)} & \text{if } x \neq 1 \\ k & \text{if } x = 1 \end{cases}$ is continuous at $x = 1$, then the value of k is $\frac{3\pi^2}{32}$.

निम्नलिखित में से असत्य कथन है :

(A) $f(x) = \underset{n\to\infty}{\text{Limit}} \frac{1}{1+n\sin^2\pi x}$, $x = 1$ पर सतत् है।

(B) फलन $f(x) = 2^{-2^{1/(1-x)}}$, $x \neq 1$ और $f(1) = 1$, $x = 1$ पर सतत् नहीं है।

(C) एक सतत् आच्छादक फलन f: $[0, 1] \to [0, 10]$ में हो सकता है लेकिन $g : [0, 1] \to (0, 10)$ सतत् आच्छादक नही हो सकता।

(D) फलन $f(x)$, $[0, 1]$ के अन्तराल में सतत् है तथा इसी अन्तराल में प्रत्येक परिमेय संख्या x के लिये $f(x) = 1$ तो $f\left(1/\sqrt{2}\right) = 1$.

(E) फलन $f(x) = \begin{cases} \frac{\cos\pi x + \sin(\pi x/2)}{(x-1)(3x^2-2x-1)} & \text{if } x \neq 1 \\ k & \text{if } x = 1 \end{cases}$, $x = 1$ पर सतत है तो k का मान $\frac{3\pi^2}{32}$ होगा।

Ans. A

Sol. (B) $f(x) = \begin{cases} 2^{-2^{\frac{1}{1-x}}} & x \neq 1 \\ 1 & x = 1 \end{cases}$

at $x = 1$

$$RHL = \lim_{h\to 0} 2^{-2^{-\frac{1}{h}}} = 1$$

$$LHL = \lim_{h\to 0} 2^{-2^{\frac{1}{h}}} = 0$$

$\therefore \lim_{x\to 1} f(x)$ does not exists.

$\Rightarrow f(x)$ is not continuous at $x = 1$.

(C) $f:[0,1]\xrightarrow[\text{onto}]{\text{continuous}}[0,10] \Rightarrow$ let $f(x) = 10x^2$

$g:[0,1]\xrightarrow[\text{onto}]{}(0,10)$

$\Rightarrow g:[0,1]\xrightarrow[\text{onto}]{\text{continuous}}(0,10)$

(D) $f(x)=\begin{cases} 1 & x\in Q\cap[0,1] \\ b(\text{let}) & x\in Q^C\cap[0,1] \end{cases}$

Given f(x) is continuous $\forall x\in[0,1]$

$\Rightarrow b = 1$

$\Rightarrow f(x) = 1 \qquad \forall x\in[0,1]$

$\Rightarrow f\left(\frac{1}{\sqrt{2}}\right)=1$

(E) $f(x)=\begin{cases} \dfrac{\cos\pi x+\sin\left(\dfrac{\pi x}{2}\right)}{(x-1)\left(3x^2-2x-1\right)} & \text{if} \quad x\neq 1 \\ k & \text{if} \quad x=1 \end{cases}$

$$\Rightarrow \lim_{x\to 1}\frac{\cos\pi x+\sin\frac{\pi x}{2}}{(x-1)\left(3x^2-2x-1\right)}$$

$$=\lim_{x\to 1}\frac{\sin\frac{\pi}{2}x+\sin\left(\frac{\pi}{2}-\pi x\right)}{(x-1)^2(3x+1)}$$

$$=\frac{1}{4}\lim_{x\to 1}\frac{2\sin\left(\dfrac{\frac{\pi x}{2}+\frac{\pi}{2}-\pi x}{2}\right)\cos\left(\dfrac{\frac{\pi x}{2}-\frac{\pi}{2}+\pi x}{2}\right)}{(x-1)^2}$$

$$=\frac{1}{2}\lim_{x\to 1}\frac{\sin\left(\frac{\pi}{4}-\frac{\pi x}{4}\right)\cos\left(\frac{3\pi x}{4}-\frac{\pi}{4}\right)}{(x-1)^2}$$

$$= \frac{1}{2}\left(-\frac{\pi}{4}\right)\lim_{x\to 1}\frac{\sin\left(\frac{\pi}{2}-\frac{3\pi x}{4}+\frac{\pi}{4}\right)}{(x-1)}$$

$$= +\frac{\pi}{8}\lim_{x\to 1}\frac{3\pi}{4}\frac{\sin\left(\frac{3\pi}{4}(x-1)\right)}{\frac{3\pi}{4}(x-1)} = \frac{3\pi^2}{32}$$

(A) $f(x) = \lim_{n\to\infty}\frac{1}{1+n\sin^2\pi x}$

$$= \begin{cases} 0 & 0 < x < 1 \\ 1 & x = 1 \\ 0 & 2 > x > 1 \end{cases}$$

$\therefore$ f(x) is not continuous at x = 1.

2. Given $f(x) = \frac{[\{|x|\}]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)\text{sgn}(\sin x)}$ for $x \neq 0$

$= 0$ for $x = 0$

where {x} is the fractional part function; [x] is the step up function and sgn(x) is the signum function of x then, f(x)

(A) is continuous at x = 0 (B) is discontinuous at x = 0

(C) has a removable discontinuity at x = 0 (D) has an irremovable discontinuity at x = 0

दिया है कि $f(x) = \frac{[\{|x|\}]e^{x^2}\{[x+\{x\}]\}}{\left(e^{1/x^2}-1\right)\text{sgn}(\sin x)}$ for $x \neq 0$

$= 0$ for $x = 0$

जहाँ {x} एक अपूर्णांश फलन [x] एक महत्तम पूर्णांक फलन है तथा sgn(x) सिग्नम फलन हो, तो फलन f(x)

(A) x = 0 पर सतत् होगा (B) x = 0 पर असतत्

(C) x = 0 पर विस्थापनीय असतता (D) x = 0 पर अविस्थापनीय असतत्ता

Ans. A

Sol. $$f(x) = \begin{cases} \frac{\left[\{|x|\}\right]e^{x^2}\{[x+\{x\}]\}}{\left(e^{\frac{1}{x^2}}-1\right)\text{sgn}(\sin x)} & x \neq 0 \\ 0 & x = 0 \end{cases}$$

at x = 0

$$\text{LHL} = \lim_{h\to 0} f(0-h)$$

$$= \lim_{h\to 0} \frac{\left[\{h\}\right] e^{h^2} \left\{\left[-h+\{-h\}\right]\right\}}{\operatorname{sgn}(-\sinh)\left(e^{\frac{1}{h^2}}-1\right)}$$

$$= \frac{0\cdot e^{h^2}\cdot(1-2h)}{-1\cdot\left(e^{\frac{1}{h^2}}-1\right)} = 0$$

$$\text{RHL} = \lim_{h\to 0} \frac{\left[\{h\}\right] e^{h^2} \left\{\left[h+\{h\}\right]\right\}}{\operatorname{sgn}(\sinh)\left(e^{\frac{1}{h^2}}-1\right)} = 0$$

f(0) = 0

$\therefore$ f(x) is continuous at x = 0.

3. Consider $f(x) = \begin{cases} x[x]^2 \log_{(1+x)} 2 & \text{for} -1 < x < 0 \\ \dfrac{ln\left(e^{x^2} + 2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & \text{for} 0 < x < 1 \end{cases}$

where [ ] & {} are the greatest integer function & fractional part function respectively, then

(A) f(0) = ln2 $\Rightarrow$ f is continuous at x = 0 (B) f(0) = 2 $\Rightarrow$ f is continuous at x = 0

(C) f(0) = e^2 $\Rightarrow$ f is continuous at x = 0 (D) f has an irremovable discontinuity at x = 0

दिया है $f(x) = \begin{cases} x[x]^2 \log_{(1+x)} 2 & \text{for} -1 < x < 0 \\ \dfrac{ln\left(e^{x^2} + 2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & \text{for} 0 < x < 1 \end{cases}$

जहाँ [ ] महत्तम पूर्णांक फलन तथा {} भिन्नांश फलन हो, तो

(A) f(0) = ln2 $\Rightarrow$ x = 0 पर f सतत है। (B) f(0) = 2 $\Rightarrow$ x = 0 पर f सतत है

(C) f(0) = e^2 $\Rightarrow$ x = 0 पर f सतत् है। (D) x = 0 पर f अविस्थापनीय असतत् है

Ans. D

Sol.
$$f(x)=\begin{cases} x[x]^2\log_{1+x}2 & -1<x<0 \\ \dfrac{\ln\left(e^{x^2}+2\sqrt{\{x\}}\right)}{\tan\sqrt{x}} & 0<x<1 \end{cases}$$

$$f(x)=\begin{cases} -x\log_{(1+x)}2 & -1<x<0 \\ \dfrac{\ln\left(e^{x^2}+2\sqrt{x}\right)}{\tan\sqrt{x}} & 0<x<1 \end{cases}$$

at $x=0$

RHL

$$=\lim_{h\to 0}\frac{\ln\left(e^{h^2}+2\sqrt{h}\right)}{\sqrt{h}}$$

4. Consider $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}}$, $x\neq 0$;

$$g(x)=\cos 2x,\ -\frac{\pi}{4}<x<0,$$

$$h(x)-\begin{cases} \dfrac{1}{\sqrt{2}}f(g(x)) & \text{for } x<0 \\ 1 & \text{for } x=0 \\ f(x) & \text{for } x>0 \end{cases}$$

then, which of the following holds good.

where $\{x\}$ denotes fractional part function.

(A) 'h' is continuous at $x=0$ (B) 'h' is discontinuous at $x=0$

(C) f(g(x)) is an even function (D) f(x) is an even function

दिया है $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}}$, $x\neq 0$;

$$g(x)=\cos 2x,\ -\frac{\pi}{4}<x<0,$$

$$h(x)-\begin{cases} \dfrac{1}{\sqrt{2}}f(g(x)) & \text{for } x<0 \\ 1 & \text{for } x=0 \\ f(x) & \text{for } x>0 \end{cases}$$

तो निम्न में से कौन सा सत्य होगा

यहाँ $\{x\}$ भिन्नांश फलन है।

(A) x = 0 पर 'h' सतत् है (B) x = 0 पर 'h' असतत् है (C) f(g(x)) समफलन है (D) f(x) एक समफलन है

Ans. A

Sol. $f(x)=\dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}};x\neq 0$

$$f(-x)=\frac{\sqrt{1-x}-\sqrt{1+x}}{\{x\}}=\frac{-\left(\sqrt{1+x}-\sqrt{1-x}\right)}{-x-[-x]}$$

$\neq f(x)$

$\therefore$ f(x) is not an even function.

$$h(x)=\begin{cases}\dfrac{1}{\sqrt{2}}f(g(x)) & x<0\\ 1 & x=0\\ f(x) & x>0\end{cases}$$

$$=\begin{cases}\dfrac{1}{\sqrt{2}}f(\cos 2x) & x<0\\ 1 & x=0\\ \dfrac{\sqrt{1+x}-\sqrt{1-x}}{\{x\}} & x>0\end{cases}$$

at x = 0

$$\text{LHL}=\lim_{h\to 0}\frac{1}{\sqrt{2}}f(\cos 2h)=\frac{1}{\sqrt{2}}\left(\frac{\sqrt{1+\cos 2h}-\sqrt{1-\cos 2h}}{\{\cos 2h\}}\right)$$

$$=\frac{1}{\sqrt{2}}\sqrt{2}\lim_{h\to 0}\frac{(\cosh-\sinh)}{\cos 2h}=1$$

$$\text{RHL}=\lim_{x\to 0^+}\frac{2x}{\sqrt{1+x}+\sqrt{1-x}}\cdot\frac{1}{\{x\}}=\lim_{h\to 0}\frac{2h}{(h)\left(\sqrt{1-h}+\sqrt{1+h}\right)}=1$$

$\Rightarrow$ h(x) is continuous at x = 0.

$$-\frac{\pi}{4}<x<0$$

$$\Rightarrow f(g(x))=f(\cos 2x)$$

$$=\frac{\sqrt{1+\cos 2x}-\sqrt{1-\cos 2x}}{\cos 2x-[\cos 2x]}$$

$$= \frac{\sqrt{2}\left(|\cos 2x| - |\sin x|\right)}{\cos 2x - [\cos 2x]}$$

$$\because x \in \left(-\frac{\pi}{4}, 0\right) \Rightarrow 2x \in \left(-\frac{\pi}{2}, 0\right) \Rightarrow [\cos 2x] = 0$$

$$\Rightarrow f(g(x)) = \frac{\sqrt{2}(\cos x + \sin x)}{\cos 2x}$$

$\because$ f(g(x)) is not an even function.

5. Consider the function $f(x) = \lim\limits_{n\to\infty} \dfrac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$, where $n \in N$

Statement-1: $f(x)$ is discontinuous at $x = 1$. **because**

Statement-2: $f(1) = 0$.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true

दिया है कि $f(x) = \lim\limits_{n\to\infty} \dfrac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$, जहाँ $n \in N$

कथन-1: $f(x)$, $x = 1$ पर असत्त है।

क्योंकि

कथन-2: $f(1) = 0$.

(A) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण है।

(B) कथन–1 सत्य है, कथन–2 सत्य है, कथन–2 कथन–1 का सही स्पष्टीकरण नही है।

(C) कथन–1 सत्य है, कथन–2 असत्य है।

(D) कथन–1 असत्य है, कथन– 2 सत्य है।

Ans. B

Sol. $$f(x) = \lim_{n\to\infty} \frac{\sin \pi x - x^{2n} \sin(x-1)}{1 + x^{2n+1} - x^{2n}}$$

$$= \lim_{n\to\infty} \frac{\frac{\sin \pi x}{x^{2n}} - \sin(x-1)}{\frac{1}{x^{2n}} + (x-1)}$$

$$= \begin{cases} \sin \pi x & x < 1 \\ 0 & x = 1 \\ -\dfrac{\sin(x-1)}{x-1} & x > 1 \end{cases}$$

$\Rightarrow$ LHL = 0

RHL = –1

$\therefore$ f(x) is not continuous at x = 1.

6. Given $f(x) = \begin{cases} \log_a\left(a^{|[x]+[-x]|}\right)^x \left(\dfrac{a^{\frac{2}{\left(\frac{[x]+[-x]}{|x|}\right)}-5}}{3+a^{\frac{1}{|x|}}}\right) & \text{for } |x| \neq 0 \ ; \ a > 1 \\ 0 & \text{for } x = 0 \end{cases}$ where [ ] represents the integral part function, then :

(A) f is continuous but not differentiable at x = 0

(B) f is cont. & diff. at x = 0

(C) the differentiability of 'f' at x = 0 depends on the value of a

(D) f is cont. & diff. at x = 0 and for a = e only

दिया है कि $f(x) = \begin{cases} \log_a\left(a^{|[x]+[-x]|}\right)^x \left(\dfrac{a^{\frac{2}{\left(\frac{[x]+[-x]}{|x|}\right)}-5}}{3+a^{\frac{1}{|x|}}}\right) & \text{for } |x| \neq 0 \ ; \ a > 1 \\ 0 & \text{for } x = 0 \end{cases}$ जहाँ [ ] महत्तम पूर्णांक फलन है, तो

(A) x = 0 पर f सतत् होगा लेकिन अवकलनीय नहीं होगा

(B) x = 0 पर f सतत् तथा अवकलनीय दोनों होगा

(C) x = 0 पर 'f' की अवकलनीयता a के मान पर निर्भर करेगी

(D) केवल a = e होने पर ही f, x = 0 पर सतत् तथा अवकलनीय होगा।

Ans. B

Sol. $f(x) = \begin{cases} x.\dfrac{a^{2|x|}-5}{3+a^{\frac{1}{|x|}}} & x \neq 0 \\ 0 & x = 0 \end{cases}$ $x \neq 0$

$$f(0^-) = \lim_{h\to 0} -h.\frac{a^{2h-5}}{3+a^{\frac{1}{h}}} = 0$$

$$f\left(0^+\right)=\lim_{h\to 0}-h.\frac{a^{2h-5}}{3+a^{\frac{1}{h}}}=0$$

f(a) = 0 f is cannot at x = 0

$$\text{L.H.D.} = f'\left(0^-\right)=\lim_{h\to 0}-h\,\frac{a^{2h-5}}{\frac{3+a^{\frac{1}{h}}}{-h}}=\frac{a^{-5}}{3}$$

$$\text{R.H.D.} = f'\left(0^+\right)=\lim_{h\to 0}h\,\frac{a^{2h-5}}{\frac{3+a^{\frac{1}{h}}}{h}}=\frac{a^{-5}}{3}$$

L.H.D. = R.H.D. $\Rightarrow$ f is diff. at x = 0

7. Let $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ and p is a prime number. The number of points where $f(x)$ is not differentiable is

(A) p – 1 (B) p + 1 (C) 2p + 1 (D) 2p – 1.

Here [x] denotes greatest integer function.

माना कि $f(x) = [n + p \sin x]$, $x \in (0, \pi)$, $n \in I$ तथा p एक अभाज्य संख्या हो, तो ऐसे कुल कितने बिन्दु होगें जहाँ $f(x)$ अवकलनीय नहीं होगा।

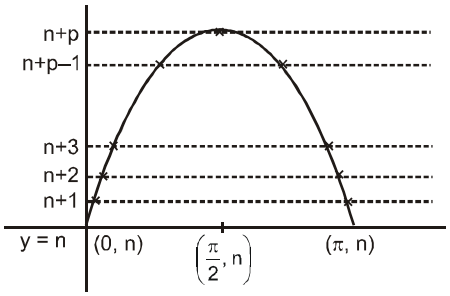
(A) p – 1 (B) p + 1 (C) 2p + 1 (D) 2p – 1.

यहाँ [.] महत्तमपूर्णांक फलन है।

Ans. D

Sol. f(x) = [n + psinx] , $x \in (0, \pi)$

graph of y = n + p sinx



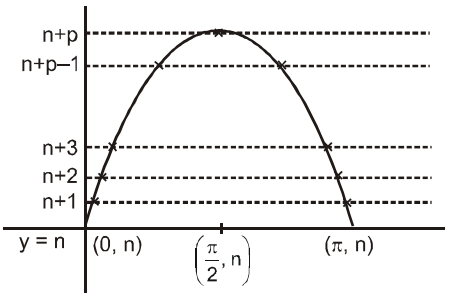
obviously

f(x) = [n +psinx] is discontinous at points mark in above curve

$\Rightarrow$ number of such points $\Rightarrow$ (p –1) + 1 + p –1 = 2p –1

Hindi. f(x) = [n + psinx] , $x \in (0, \pi)$

y = n + p sinx का आलेख



स्पष्ट रूप से

f(x) = [n +psinx] असतत् है उन बिन्दुओं पर जो आलेख से चिन्हित है।

$\Rightarrow$ ऐसे बिन्दुओं की संख्या $\Rightarrow$ (p –1) + 1 + p –1 = 2p –1

8. Consider $f(x) = \left[\dfrac{2\left(\sin x - \sin^3 x\right) + \left|\sin x - \sin^3 x\right|}{2\left(\sin x - \sin^3 x\right) - \left|\sin x - \sin^3 x\right|}\right]$, $x \neq \dfrac{\pi}{2}$ for $x \in (0, \pi)$

$f(\pi/2) = 3$ where [ ] denotes the greatest integer function then,

(A) f is continuous & differentiable at $x = \pi/2$

(B) f is continuous but not differentiable at $x = \pi/2$

(C) f is neither continuous nor differentiable at $x = \pi/2$

(D) none of these

दिया है कि $f(x) = \left[\dfrac{2\left(\sin x - \sin^3 x\right) + \left|\sin x - \sin^3 x\right|}{2\left(\sin x - \sin^3 x\right) - \left|\sin x - \sin^3 x\right|}\right]$, $x \neq \dfrac{\pi}{2}$ $x \in (0, \pi)$

$f(\pi/2) = 3$ जहाँ [ ] महत्तम पूर्णांक फलन हो, तो

(A) $x = \pi/2$ पर f सतत् तथा अवकलनीय दोनों होगें।

(B) $x = \pi/2$ पर f सतत् तो होगा लेकिन अवकलनीय नहीं है।

(C) $x = \pi/2$ पर f न तो सतत् हो और न ही अवकलनीय है।

(D) इनमें से कोई नहीं

Ans. A

Sol. In the immediate neighborhood of $x = \pi/2$, $\sin x > \sin^3 x \Rightarrow |\sin x - \sin^3 x| = \sin x - \sin^3 x$

Hence for $x \neq \pi/2$, $f(x) = \left[\dfrac{2(\sin x - \sin^3 x) + \sin x - \sin^3 x}{2(\sin x - \sin^3 x) - \sin x + \sin^3 x}\right] = \dfrac{3\sin x - 3\sin^3 x}{\sin x - \sin^3 x} = 3$

Hence f is continuous and diff. at $x = \pi/2$]

9. Let $f: [a, b] \to R$ be any function which is such that $f(x)$ is rational for irrational x and that $f(x)$ is irrational for rational x, then in [a, b]

(A) *f* is discontinuous everywhere in [a, b]

(B) *f* is continuous only at x = 0 and discontinuous everywhere

(C) f is continuous for all irrational x and discontinuous for rational x

(D) f is continuous for rational x and discontinuous for irrational x

f : [a, b] $\to$ R एक ऐसा फलन है कि यदि $x \notin Q$ तो $f(x)$ परिमेय है तथा $x \in Q$ तो $f(x)$ अपरिमेय हो, तो अन्तराल [a, b] में

(A) [a, b] सभी जगह f असतत् होगा

(B) फलन f सिर्फ $x = 0$ पर सतत् होगा बाकी बिन्दुओं पर असतत् होगा।

(C) यदि x अपरिमेय हो, तो f सतत् होगा तथा यदि x परिमेय हो, तो f असतत् होगा।

(D) यदि x परिमेय हो तो f सतत् होगा तथा यदि x अपरिमेय हो, तो f असतत् होगा।

Ans. A

Sol. $f(x) = \begin{cases} \text{rational} & \text{if } x \notin Q \text{ in } [a,b] \\ \text{irrational} & \text{if } x \in Q \text{ in } [a,b] \end{cases}$

Let $c \in Q$

$$\left.\begin{array}{l} f(c) = \text{irrational} \\ f(c+h) = \text{rational} \\ \lim\limits_{h\to 0} \text{ through irrational} \end{array}\right] \Rightarrow f \text{ is discontinuous for } x \in Q \text{ in } [a, b]$$

now let $r \notin Q$

$$\left.\begin{array}{l} f(r) = \text{rational} \\ f(r+h) = \text{irrational} \\ \lim\limits_{h\to 0} \text{ through rational} \end{array}\right] \Rightarrow f \text{ is discontinuous for } x \notin Q \text{ in } [a, b]$$

10. The graph of function f contains the point P (1, 2) and Q(s, r). The equation of the secant line through P and Q is $y = \left(\frac{s^2+2s-3}{s-1}\right)x - 1 - s$. The value of $f'(1)$, is

(A) 2 (B) 3 (C) 4 (D) non existent

फलन $f(x)$ का आरेख बिन्दु P (1, 2) तथा Q(s, r) से गुजरता है यदि P और Q से गुजरने वाली रेखा खण्ड का समीकरण $y = \left(\frac{s^2+2s-3}{s-1}\right)x - 1 - s$ हो, तो $f'(1)$ का मान बताओं।

(A) 2 (B) 3 (C) 4 (D) अस्तित्व विहीन

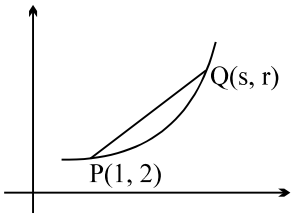
Ans. C

Sol. **I** By definition $f'(1)$ is the limit of the slope of the secant line when $s \to 1$.

$$\text{Thus } f'(1) = \lim_{s\to 1}\frac{s^2+2s-3}{s-1}$$

$$= \lim_{s\to 1}\frac{(s-1)(s+3)}{s-1}$$

$$= \lim_{s\to 1}(s+3) = 4 \qquad \Rightarrow \quad \text{(D)}$$



II By substituting x = s into the equation of the secant line, and cancelling by s – 1 again, we get $y = s^2 + 2s - 1$. This is f(s), and its derivative is $f'(s) = 2s + 2$, so $f'(1) = 4$.]

11. If $f(x + y) = f(x) + f(y) + |x|y + xy^2$, $\forall$ x, y $\in$ R and $f'(0) = 0$, then

(A) f need not be differentiable at every non zero x (B) f is differentiable for all x $\in$ R

(C) f is twice differentiable at x = 0 (D) none

यदि $f(x + y) = f(x) + f(y) + |x|y + xy^2$, $\forall$ x, y $\in$ R और $f'(0) = 0$, तो

(A) कोई जरूरी नहीं की प्रत्येक अशून्य (zero) के लिये f अवकलनीय हो

(B) x के प्रत्येक मान के लिये f एक अवकलनीय फलन होगा

(C) x = 0 पर f तथा f' दोनो अवकलनीय होगा

(D) इनमें से कोई नहीं

Ans. B

Sol. $f'(x) = \lim_{h\to 0}\frac{f(x+h)-f(x)}{h} = \lim_{h\to 0}\frac{f(h)+|x|h+xh^2}{h}$ where x = h and y = x

$\because$ f(0) = 0; hence $f'(x) = \lim_{h\to 0}\left(\frac{f(h)-f(0)}{h}+|x|+xh\right)$

$f'(x) = f'(0) + |x| = |x|$

12. For x > 0, let $h(x) = \begin{cases}\frac{1}{q} & \text{if } x=\frac{p}{q} \\ 0 & \text{if x is irrational}\end{cases}$ *where p & q > 0 are relatively prime integers*

then which one does not hold good?

(A) h(x) is discontinuous for all x in $(0, \infty)$

(B) h(x) is continuous for each irrational in $(0, \infty)$

(C) h(x) is discontinuous for each rational in $(0, \infty)$

(D) h(x) is not derivable for all x in $(0, \infty)$

x > 0 के लिये $h(x) = \begin{cases}\frac{1}{q} & \text{यदि } x = \frac{p}{q} \\ 0 & \text{यदि x अपरिमेय है}\end{cases}$

जहाँ *p, q > 0* तथा p और q तुलनात्मक अविभाज्य संख्याऐं हो, तो इनमें से कौन सा विकल्प असत्य होगा?

(A) $(0, \infty)$ के अन्तराल में सभी x के लिए h(x) असतत् फलन होगा

(B) $(0, \infty)$ के अन्तराल में प्रत्येक अपरिमेय के लिये h(x) सतत् फलन है।

(C) $(0, \infty)$ के अन्तराल में प्रत्येक परिमेय के लिये $h(x)$ असतत् फलन है।

(D) $(0, \infty)$ के अन्तराल में x के किसी भी मान के लिये $h(x)$ अवकलनीय् फलन नहीं होगा

Ans. A

Sol. Let $x = \frac{2}{3}$ which is rational

$\Rightarrow \quad h\left(\frac{2}{3}\right) = \frac{1}{3}$

$\lim\limits_{t \to 0} h\left(\frac{2}{3} + t\right) = 0 \quad \Rightarrow$ discontinuous at $x \in Q$

Let $x = \sqrt{2} \notin Q$

$h\left(\sqrt{2}\right) = 0$ consider $\sqrt{2} = 1.41401235839$

$h\left(\sqrt{2}\right) = h\left(\frac{1.414023583}{10^{10}}\right) = \frac{1}{10^{10}} \to 0$

Hence h is continuous for all irrational $\Rightarrow$ A

13. Let $f(x) = \max. \left\{\left|x^2 - 2|x|\right|, |x|\right\}$ and $g(x) = \min. \left\{\left|x^2 - 2|x|\right|, |x|\right\}$ then

(A) both f(x) and g(x) are non differentiable at 5 points.

(B) f(x) is not differentiable at 5 points whether g(x) is non differentiable at 7 points.

(C) number of points of non differentibility for f(x) and g(x) are 7 and 5 respectively.

(D) both f(x) and g(x) are non differentiable at 3 and 5 points respectively

माना $f(x) = \max. \left\{\left|x^2 - 2|x|\right|, |x|\right\}$ तथा $g(x) = \min. \left\{\left|x^2 - 2|x|\right|, |x|\right\}$ है तो –

(A) f(x) तथा g(x) दोनों 5 बिन्दुओं पर अवकलनीय नहीं है

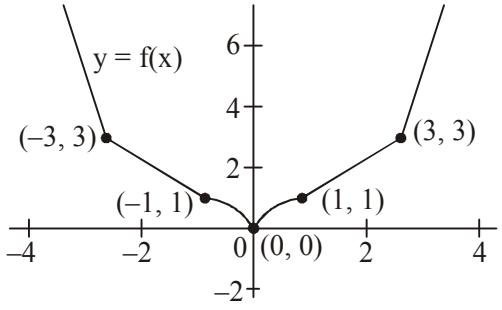
(B) f(x), 5 बिन्दुओं पर अवकलनीय नहीं है जबकि g(x), 7 बिन्दुओं पर अवकलनीय नहीं है

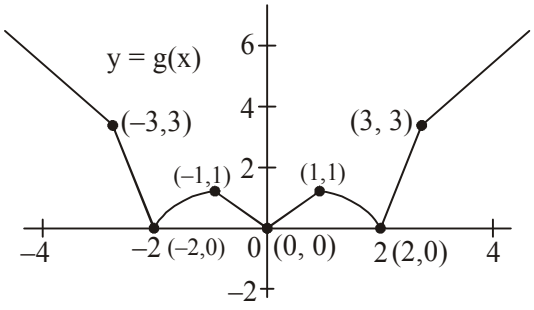
(C) f(x) तथा g(x) क्रमशः 7 तथा 5 बिन्दुओं पर अवकलनीय नहीं है

(D) f(x) तथा g(x) क्रमशः 3 तथा 5 बिन्दुओं पर अवकलनीय नहीं है

Ans. B

Sol. 0





ONE OR MORE THAN ONE CORRECT TYPE

14. If f is defined on an interval [a, b]. Which of the following statement(s) is/are INCORRECT?

(A) If $f(a)$ and $f(b)$, have opposite sign, then there must be a point $c \in (a, b)$ such that $f(c) = 0$.

(B) If f is continuous on [a, b], $f(a) < 0$ and $f(b) > 0$, then there must be a point $c \in (a, b)$ such that $f(c) = 0$

(C) If f is continuous on [a, b] and there is a point c in (a, b) such that $f(c) = 0$, then $f(a)$ and $f(b)$ have opposite sign.

(D) If f has no zeroes on [a, b], then $f(a)$ and $f(b)$ have the same sign

फलन f अन्तराल [a, b] में परिभाषित हो, तो गलत विकल्पों को चुनिये।

(A) यदि $f(a)$ और $f(b)$ का चिन्ह विपरीत हो, तो एक ऐसा $c \in (a, b)$ होगा जब $f(c) = 0$.

(B) यदि f अन्तराल [a, b] में सतत् है तथा $f(a) < 0$ और $f(b) > 0$ हो, तो एक ऐसा $c \in (a, b)$ होगा जब $f(c) = 0$

(C) यदि f अन्तराल [a, b] में सतत् है तथा $c \in (a, b)$ में एक ऐसा बिन्दु है जब $f(c) = 0$ तो $f(a)$ और $f(b)$ का चिन्ह विपरीत होगा।

(D) यदि [a, b] में f का कोई मूल नही है तो $f(a)$ व $f(b)$ का चिन्ह समान होगा

Ans. ACD

Sol. According to intermediate value theorem if $f(a)\, f(b) < 0$ then there exist at least one 'c' $\in (a, b)$ for which $f(x) = 0$.

15. If $f(x) = \text{Min}(\tan x, \cot x)$ then:

(A) f(x) is discontinuous at $x = 0, \frac{\pi}{4}$ & $\frac{5\pi}{4}$ (B) f(x) is continuous at $x = \frac{\pi}{2}$ & $\frac{3\pi}{2}$

(C) $\int_0^{\pi/2} f(x)\, dx = 2\,\ell n\sqrt{2}$ (D) f(x) is periodic with period π

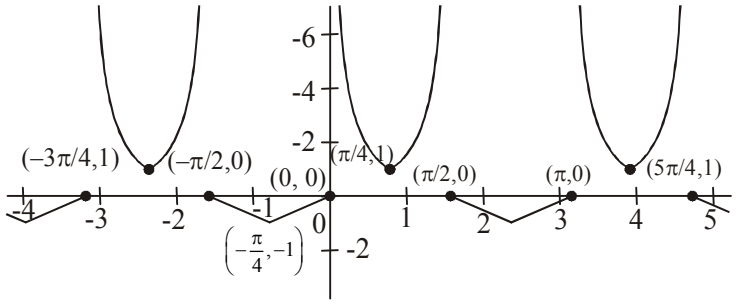
यदि $f(x) = \text{Min}(\tan x, \cot x)$ तो :

(A) f(x) $x = 0, \frac{\pi}{4}$ तथा $\frac{5\pi}{4}$ पर असत्त है (B) फलन f(x) $x = \frac{\pi}{2}$ तथा $\frac{3\pi}{2}$ पर सत्त है

(C) $\int_0^{\pi/2} f(x)\, dx = 2\,\ell n\sqrt{2}$ (D) f(x) का आवर्तकाल π है

Ans. CD

Sol.



(C) $\int_0^{\frac{\pi}{2}} f(x)\, dx = \int_0^{\frac{\pi}{4}} \cot x\, dx + \int_{\frac{\pi}{4}}^{\frac{\pi}{2}} \tan x\, dx$

$= 2\ln\sqrt{2}$

(D) Time period $= \pi$

16. If $f(x) = |2x + 1| + |2x - \pi| + \tan x$ then $f(x)$ is

(A) Continuous at all points except $x = n\pi + \frac{\pi}{2}$ (where $n \in I$)

(B) Continuous at $x = -\frac{1}{2}$, but not differentiable at $x = \pi/2$

(C) Not differentiable at $x = n\pi + \frac{\pi}{2}$ (where $n \in I$)

(D) Limit at $x = \frac{\pi}{2}$ does not exist

यदि $f(x) = |2x + 1| + |2x - \pi| + \tan x$ तो फलन $f(x)$ है –

(A) $x = n\pi + \frac{\pi}{2}$ (जहाँ $n \in I$) के अलावा सभी बिन्दुओं पर संत्त

(B) $x = -\frac{1}{2}$ पर संत्त परन्तु $x = \pi/2$ पर अवकलनीय नहीं

(C) $x = n\pi + \frac{\pi}{2}$ (जहाँ $n \in I$) अवकलनीय नहीं है

(D) $x = \frac{\pi}{2}$ पर सीमा विद्यमान नहीं है

Ans. ABCD

Sol. $$f(x) = \begin{cases} -4x + \pi - 1 + \tan x & x < -\frac{1}{2} \\ \pi + 1 + \tan x & -\frac{1}{2} \le x < \frac{\pi}{2} \\ 4x - \pi + 1 + \tan x & x > \frac{\pi}{2} \end{cases}$$

$x = \frac{\pi}{2}$

$f\left(\frac{\pi^-}{2}\right) \to \infty$

$f\left(\frac{\pi^+}{2}\right) \to -\infty$

$x = -\frac{1}{2}$

$$f\left(-\frac{1}{2}^{-}\right) = -4\left(-\frac{1}{2}\right) + \pi - 1 + \tan\left(-\frac{1}{2}\right) = \pi + 1 - \tan\frac{1}{2}$$

$$f\left(-\frac{1}{2}^{+}\right) = \pi + 1 + \tan\left(-\frac{1}{2}\right) = \pi + 1 - \tan\frac{1}{2}$$

$$LHL = RHL = f\left(\frac{1}{2}\right)$$

17. The function defined as $f(x) = \lim_{n\to\infty} \begin{cases} \cos^{2n} x & \text{if } x < 0 \\ \sqrt[n]{1+x^n} & \text{if } 0 \le x \le 1 \\ \dfrac{1}{1+x^n} & \text{if } x > 1 \end{cases}$. Which of the following does not hold good?

(A) continuous at x = 0 but discontinuous at x = 1.

(B) continuous at x = 1 but discontinuous at x = 0.

(C) continuous both at x = 1 and x = 0.

(D) discontinuous both at x = 1 and x = 0

दिया है $f(x) = \lim_{n\to\infty} \begin{cases} \cos^{2n} x & \text{if } x < 0 \\ \sqrt[n]{1+x^n} & \text{if } 0 \le x \le 1 \\ \dfrac{1}{1+x^n} & \text{if } x > 1 \end{cases}$. तो इनमें से कौन सा विकल्प फलन f(x) के लिये सत्य नहीं होगा?

(A) x = 0 पर सतत् तथा x = 1 पर असतत् (B) x = 1 पर सतत् तथा x = 0 पर असतत्

(C) x = 0 , x = 1 पर सतत् (D) x = 0, x = 1 पर असतत्

Ans. ABC

Sol. $f(0^-) = 0$; $f(0^+) = 1$; $f(0) = 1$ $\Rightarrow$ discontinuous at x = 0

$f(1^-) = 1$; $f(1) = 1$; $f(1^+) = 0$ $\Rightarrow$ discontinuous at x = 1

18. f is a continous function in [a, b]; g is a continuous functin in [b, c]

A function h (x) is defined as

$h(x) = f(x)$ for $x \in [a, b)$

$= g(x)$ for $x \in (b, c]$

if $f(b) = g(b)$, then

(A) h(x) has a removable discontinuity at x=b.

(B) h(x) may or may not be continuous in [a, c]

(C) $h(b^-) = g(b^+)$ and $h(b^+) = f(b^-)$

(D) $h(b^+) = g(b^-)$ and $h(b^-) = f(b^+)$

फलन f [a, b] में सतत् है, फलन g [b, c] में सतत् है। एक फलन h (x) इस प्रकार से परिभाषित किया जाता है कि

$h(x) = f(x)$ for $x \in [a, b)$

$= g(x)$ for $x \in (b, c]$

यदि $f(b) = g(b)$, तो

(A) h(x), x = b पर विस्थापनीय असतत्ा (B) [a, c] में h(x) सतत हो भी सकता है, और नहीं भी

(C) $h(b^-) = g(b^+)$ & $h(b^+) = f(b^-)$ (D) $h(b^+) = g(b^-)$ & $h(b^-) = f(b^+)$

Ans. AC

Sol. Given f is continuous in [a, b] ....(1)

g is continuous in [b, c] ....(2)

$f(b) = g(b)$ ....(3)

$h(x) = f(x)$ for $x \in [a, b)$

$= f(b) = g(b)$ for $x = b$ } ....(4)

$= g(x)$ for $x \in (b, c]$

h (x) is continuous in $[a, b) \cup (b, c]$ [using (1), (2)]

also $f(b^-) = f(b)$; $g(b^+) = g(b)$ ....(5) [using (1), (2)]

$\therefore$ $h(b^-) = f(b^-) = f(b) = g(b) = g(b^+) = h(b^+)$ [using (4), (5)

now, verify each alternative. Of course! $g(b^-)$ and $f(b^+)$ are undefined.

$h(b^-) = f(b^-) = f(b) = g(b) = g(b^+)$

and $h(b^+) = g(b^+) = g(b) = f(b) = f(b^-)$

hence $h(b^-) = h(b^+) = f(b) = g(b)$

and h (b) is not defined $\Rightarrow$ (A)]

19. Let f be a continuous function on the closed interval [a, b]. Then which of the following statement(s) need not be true?

(A) There is a number c in the open interval (a, b) such that $f(a) < f(c) < f(b)$.

(B) There is a number c in the closed interval [a, b] such that $f(c) \geq f(x)$ for all x in [a, b].

(C) There is a number c in the open interval (a, b) such that $f'(c) = 0$

(D) There is a number c in the open interval (a, b) such that $f'(c) = \dfrac{f(b) - f(a)}{b - a}$

माना f एक सतत् फलन है अन्तराल [a, b] मे, तो निम्न में से कौनसा कथन सही होना आवश्यक नही है ?

(A) अन्तराल (a, b) में एक संख्या c है जिसके लिये $f(a) < f(c) < f(b)$ है

(B) अन्तराल [a, b] में एक संख्या c है जिसके लिये $f(c) \geq f(x)$ सभी $x \in [a, b]$ के लिये

(C) अन्तराल (a, b) में एक संख्या c है जिसके लिये $f'(c) = 0$ है

(D) अन्तराल (a, b) में एक संख्या c है जिसके लिये $f'(c) = \dfrac{f(b) - f(a)}{b - a}$ है

Ans. ACD

Sol. (A) Can be true only if function is strictly increasing in (a, b).

(B) Correct because by extreme value theorem for continuous function.

(C) False. Why $f'(c) = 0$?

(D) Can be true if differentiable in (a, b)].

20. Let $f(x) = \begin{cases} x^2 \cos\frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin\frac{1}{x}, & x > 0 \end{cases}$,

then which of the following is(are) correct?

(A) f(x) is continuous but not differentiable at $x = 0$.

(B) f(x) is continuous and differentiable at $x = 0$.

(C) $f'(x)$ is continuous but not differentiable at $x = 0$.

(D) $f'(x)$ is discontinuous at $x = 0$

माना $f(x) = \begin{cases} x^2 \cos\frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ x^2 \sin\frac{1}{x}, & x > 0 \end{cases}$, तो निम्न में से कौनसा सही है ?

(A) f(x), $x = 0$ सतत् है परन्तु अवकलनीय नही है

(B) f(x) $x = 0$ सतत् है तथा अवकलनीय है

(C) $f'(x)$ $x = 0$ सतत् है परन्तु अवकलनीय नही है

(D) $f'(x)$, $x = 0$ असतत् है

Ans. BD

Sol. Clearly f(x) is continuous and differentiable at $x = 0$ such that $f'(0) = 0$

$$f'(x) = \begin{cases} \sin\frac{1}{x} + 2x\cos\frac{1}{x}, & x < 0 \\ 0, & x = 0 \\ 2x\sin\frac{1}{x} - \cos\frac{1}{x}, & x > 0 \end{cases}$$

$\therefore$ Clearly $f'(x)$ is discontinuous at $x = 0$

21. If $f(x) = \min.(1, x^2, x^3)$, then

(A) f(x) is continuous $\forall\ x \in R$

(B) $f'(x) > 0$, $\forall\ x > 1$

(C) f(x) is not differentiable but continuous $\forall\ x \in R$

(D) f(x) is not differentiable for two values of x

फलन $f(x) = \min.(1, x^2, x^3)$, के लिये

(A) फलन $f(x)$, $\forall\ x \in R$ के लिये सतत् होगा

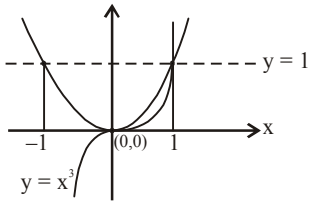
(B) $f'(x) > 0$, $\forall\ x > 1$

(C) f(x), $\forall\ x \in R$ के लिए सतत् तो होगा लेकिन अवकलनीय नही होगा।

(D) f(x), x के दो मान के लिये अवकलनीय नहीं होगा।

Ans. AC

Sol. $f(x) = \min(1, x^2, x^3)$



$$f(x) = \begin{cases} x^3 & x < 0 \\ x^3 & 0 \le x < 1 \\ 1 & x \ge 1 \end{cases}$$

$$f(x) = \begin{cases} x^3 & x < 1 \\ 1 & \ge 1 \end{cases}$$

$\therefore$ f(x) is continuous $\forall\ x \in R$

f(x) is not differentiable at x = 1.

22. $$f(x) = \begin{cases} 2(x+1) & ; \quad x \le -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1 \\ |||x|-1|-1| & ; \quad x \ge 1 \end{cases}$$, then

(A) f(x) is non differentiable at exactly three points

(B) f(x) is continuous in $(-\infty, 1]$

(C) f(x) is differentiable in $(-\infty, -1)$

(D) f(x) is finite type of discontinuity at x = 1, but continuous at x = –1

$$f(x) = \begin{cases} 2(x+1) & ; \quad x \le -1 \\ \sqrt{1-x^2} & ; \quad -1 < x < 1 \\ |||x|-1|-1| & ; \quad x \ge 1 \end{cases}$$ है तो –

(A) f(x), ठीक तीन बिन्दुओं पर अवकलनीय नहीं है

(B) $(-\infty, 1]$ में f(x) सतत है

(C) $(-\infty, -1)$ में f(x) अवकलनीय है

(D) x = 1 पर f(x) की विस्थापनीय असतता है लेकिन x = –1 पर सतत है

Ans. ACD

Sol.
$$f(x)=\begin{cases}2(x+1) & x\le -1\\ \sqrt{1-x^2} & -1<x<1\\ \big||x-1|-1\big| & x\ge 1\end{cases}$$

$$f(x)=\begin{cases}2(x+1) & x\le 1\\ \sqrt{1-x^2} & -1<x<1\\ -(x-2) & 1\le x<2\\ x-2 & 2\le x<3\end{cases}$$

If $x \in (-\infty, 1)$

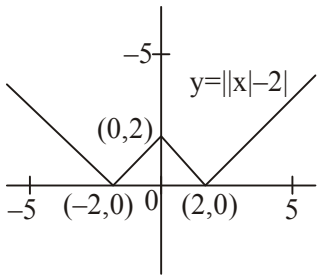
$$f'(x)=\begin{cases}2 & x<-1\\ \dfrac{-2x}{2\sqrt{1-x^2}} & -1<x<1\end{cases}$$

$\Rightarrow$ f(x) is not differentiable at $x=-1$.

$\therefore$ f(x) is differentiable $\forall\ x \in (-\infty,-1)$.

23. If $f(x) = |||x| - 2| + p|$ have more than 3 points of non-derivability then the value of p can be
दि $f(x) = |||x| - 2| + p|$, 3 से अधिक बिन्दुओं पर अवकलनीय नहीं है तो p का मान हो सकता है
(A) 0 (B) –1 (C) –2 (D) 2

Ans. BC

Sol.


Clearly, for $p = 0$ $f(x) = |||x| - 2| + p|$ will be same as the above graph, and it is non differentiable for exactly 3 points.

Now, for more than 3 points of non diffferentiability, we need to shift the graph down and take the modulus.

$\therefore$ We need to shift the graph of $||x| - 2|$ down. So, p has to be negative. i.e. $p < 0$

Exercise C - II

1. Let $y_n(x) = x^2 + \frac{x^2}{1+x^2} + \frac{x^2}{(1+x^2)^2} + \ldots\ldots\ldots + \frac{x^2}{(1+x^2)^{n-1}}$ and $y(x) = \lim_{n\to\infty} y_n(x)$

Discuss the continuity of $y_n(x)$ $(n \in N)$ and $y(x)$ at $x = 0$.

माना कि $y_n(x) = x^2 + \frac{x^2}{1+x^2} + \frac{x^2}{(1+x^2)^2} + \ldots\ldots\ldots + \frac{x^2}{(1+x^2)^{n-1}}$ तथा $y(x) = \lim_{n\to\infty} y_n(x)$

तो $y_n(x)$ $(n \in N)$ और $y(x)$, $x = 0$ के सतता की जाँच करिये।

Ans. $y_n(x)$ is continuous at $x = 0$ for all n and $y(x)$ is dicontinuous at $x = 0$

Sol. $y_n(x)$ = Sum of G.P. of n terms with first term 'x^2' and common ratio $\frac{1}{1+x^2}$

$$y_n(x) = x^2 \cdot \left[\frac{1-\left(\frac{1}{1+x^2}\right)^n}{1-\frac{1}{1+x^2}}\right]$$

$$y_n(x) = \left(1+x^2\right)\left[1-\frac{1}{\left(1+x^2\right)^n}\right], x \neq 0, n \in N \qquad \text{_____(1)}$$

$$y(x) = \begin{cases} \lim\limits_{n\to\infty} y_n(x) & x \neq 0 \\ 0 & x = 0 \end{cases}$$

Now,

$$y(x) = \begin{cases} 1+x^2 & x \neq 0 \\ 0 & x = 0 \end{cases}$$

$y_n(0) = y_n(0+0) = y_n(0-0) = 0$

$y_n(x)$ is continuous at $x = 0$

also,

$y(0) = 0$, $y(0-0) = y(0+0) = 1$

$y(x)$ discontinuous at $x = 0$.

2. Let [x] denote the greatest integer function and f(x) be defined in a neighbourhood of 2 by

$$f(x) = \begin{cases} \frac{\left(\exp\{(x+2)\ell n4\}\right)^{\frac{[x+1]}{4}} - 16}{4^x - 16} & , x<2 \\ A\frac{1-\cos(x-2)}{(x-2)\tan(x-2)} & , x>2 \end{cases}.$$

Find the values of A & f(2) in order that f(x) may be continuous at $x = 2$.

यदि [x] महत्तम पूर्णांक फलन है तथा f(x) x = 2 के पडोस में इस प्रकार से परिभाषित है कि

$$f(x)=\begin{cases}\dfrac{\left(\exp\{(x+2)\ell n4\}\right)^{\frac{[x+1]}{4}}-16}{4^x-16} & ,x<2\\ A\dfrac{1-\cos(x-2)}{(x-2)\tan(x-2)} & ,x>2\end{cases}.$$

तो A व f(2) का मान ज्ञात करो जबकि फलन f(x), x = 2 पर सतत् है।

Ans. A = 1 ; f(2) = 1/2

Sol. Given f is continuous at x = 2.

then f(2) = f(2 + 0) = f(2 – 0)

Now

LHL = f(2 – 0)

$$=\lim_{h\to0} f(2-h)$$

$$=\lim_{h\to0}\frac{\left\{e^{((4-h)\ell n4)}\right\}^{\frac{[3-h]}{4}}-16}{4^{2-h}-16}$$

$$\text{LHL}=\lim_{h\to0}\frac{\left(4^{(4-h)}\right)^{\frac{2}{4}}-4^2}{4^2 4^{-h}-4^2}=\lim_{h\to0}\frac{4^2 4^{-\frac{h}{2}}-4^2}{4^2\left(4^{-h}-1\right)}$$

$$=\lim_{h\to0}\frac{4^2\left[4^{-\frac{4}{2}}-1\right]}{4^2\left(h^{-h}-1\right)}$$

$$=\lim_{h\to0}\frac{4^{-\frac{h}{2}}-1}{-\frac{4}{2}}\cdot\frac{-h}{4^{-h}-1}\cdot\frac{1}{2}$$

$$=\frac{\ell n4}{\ell n4}\cdot\frac{1}{2}=\frac{1}{2}$$

$$\lim_{x\to0}\frac{a^x-1}{x}=\ell na$$

$$\because f(2)=f(2-0)=\frac{1}{2}$$

Now, f(2) = f(2 + 0)

$$\Rightarrow\frac{1}{2}=\lim_{h\to0}A\cdot\frac{1-\cosh}{h\cdot\tanh}$$

$$\frac{1}{2} = A \lim_{h\to 0} \frac{1-\cosh}{h^2} \cdot \frac{h}{\tanh}$$

$$\frac{1}{2} = A \cdot \frac{1}{2} \cdot 1 \Rightarrow A = 1$$

3. Let $g(x) = \lim\limits_{n\to\infty} \dfrac{x^n f(x) + h(x) + 1}{2x^n + 3x + 3}$, $x \neq 1$ and $g(1) = \lim\limits_{x\to 1} \dfrac{\sin^2(\pi \cdot 2^x)}{\ell n\left(\sec(\pi \cdot 2^x)\right)}$ be a continuous function at $x = 1$, find the value of $4\,g(1) + 2\,f(1) - h(1)$. Assume that $f(x)$ and $h(x)$ are continuous at $x = 1$.

माना कि $g(x) = \lim\limits_{n\to\infty} \dfrac{x^n f(x) + h(x) + 1}{2x^n + 3x + 3}$, $x \neq 1$ तथा $g(1) = \lim\limits_{x\to 1} \dfrac{\sin^2(\pi \cdot 2^x)}{\ell n\left(\sec(\pi \cdot 2^x)\right)}$, $x = 1$ पर सतत् फलन हो, तो $4g(1) + 2f(1) - h(1)$ का मान ज्ञात करो। यहाँ पर $f(x)$ तथा $h(x)$, $x = 1$ पर सतत् फलन है।

Ans. 5

Sol. $$g(1) = \lim_{x\to 1} \frac{\sin^2\left[2\pi - \pi 2^x\right]}{\left(2\pi - \pi 2^x\right)^2} \frac{4\pi^2\left\{2^{x-1} - 1\right\}^2}{\left\{-\ell n \cos \pi 2^x\right\}}$$

$$= -\lim_{x\to 1} \frac{4\pi^2\left\{2^{x-1} - 1\right\}^2}{\dfrac{\ell n\left\{1 - \left(1 - \cos\left(2\pi - \pi 2^x\right)\right)\right\}}{1 - \cos\left(2\pi - \pi 2^x\right)} \cdot \left[1 - \cos\left(2\pi - \pi 2^x\right)\right]}$$

$$= -1 \cdot \frac{1}{-1} \cdot 2 = 2$$

$$g(x) = \lim_{n\to\infty} \frac{x^n \cdot f(x) + h(x) + 1}{2x^n + 3x + 3}, x \neq 1$$

for

$x = 1 - h \cdot x^n = 0$ $\quad\quad$ $\because$ h(x) continuous at x = 1

$\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad\quad$ $h(1) = h(1 - 0) = h(1 + 0)$

$$g(x) = \frac{h(x) + 1}{3x + 3}$$

at $x = 1$ $\quad\quad$ $\text{L.H.L.} = \dfrac{h(1-0) + 1}{3(1-0) + 3} = \dfrac{h(1) + 1}{6}$

$$g(1) = \text{L.H.L.} \Rightarrow 2 = \frac{h(1) + 1}{6}$$

$\Rightarrow h(1) = 11$

for

$x = 1 + h \qquad x^n = \infty$

$$g(x) = \lim_{n\to\infty} \frac{f(x) + \frac{h(x)+1}{x^n}}{2 + \frac{3x+3}{x^n}}$$

$$g(x) = \frac{f(x)}{2} \qquad \because \text{ f continuous at } x = 1.$$

$$\text{at } x = 1, \text{ R.H.L.} = \frac{f(1+0)}{2} \qquad f(+0) = f(1-0) = f(1)$$

$$g(1+0) = \frac{f(1)}{2}$$

$$g(1) = \frac{f(1)}{2} \Rightarrow f(1) = 4$$

Now $4g(1) + 2f(1) - h(1)$

$= 4(2) + 2(4) - 11 = 5$

4. Let $f(x) = \underset{n\to\infty}{\text{Lim}} \frac{x^{2n-1} + ax^3 + bx^2}{x^{2n} + 1}$. If f(x) is continuous for all $x \in R$, find the bisector of angle between the lines $2x + y - 6 = 0$ and $2x - 4y + 7 = 0$ which contains the point (a, b).

माना $f(x) = \underset{n\to\infty}{\text{Lim}} \frac{x^{2n-1} + ax^3 + bx^2}{x^{2n} + 1}$ यदि f(x) सभी $x \in R$ के लिये सतत् हो तो रेखाओं $2x + y - 6 = 0$ तथा $2x - 4y + 7 = 0$ के कोणार्द्धक ज्ञात कीजिये जो (a, b) को अन्तर्विष्ट करते है।

Ans. $6x - 2y - 5 = 0$

Sol. $$f(x) = \lim_{n\to\infty} \frac{x^{2n-1} + ax^3 + bx^2}{x^{2n} + 1}$$

$$f(x) = \begin{cases} ax^3 + bx^2 & x \in (-1,1) \\ \frac{1}{x} & x \in (-\infty,-1) \cup (1,\infty) \\ \frac{\frac{1}{x} + ax^3 + bx^2}{2} & x = \pm 1 \end{cases}$$

f is continuous $\forall x \in R$ it mean it continuous also at x = 1, –1.

at x = 1 $\quad f(1) = f(1-0)$

$$\frac{a+b+1}{2} = a+b \Rightarrow a+b=1 \qquad \text{_____(1)}$$

at x = –1 $\quad f(-1+0) = f(-1-0)$

$\Rightarrow -a+b=-1 \Rightarrow a-b=1$ _____(2)

Solving (1) & (2)

$a = 1, b = 0 \Rightarrow (a, b) = (1,0)$

equation of angle bisector $\quad L_1 = 2x + y - 6, L_2 = 2x - 4y + 7$

$$\frac{2x+y-6}{\sqrt{5}} = \pm\frac{2x-4y+7}{2\sqrt{5}}$$

$\Rightarrow 4x + 2y - 12 \pm (2x - 4y + 7)$

+ve $\quad 2x + 6y - 19 = 0$ $\qquad$ –ve $\quad 6x - 2y - 5 = 0$

Here $L_1(1,0) = -4$ $\qquad L_2(1,0) = 9$

–ve $\qquad$ +ve

Product = –ve

angle nisector which obtain by –ve sign that angle bisector of angle between lines contain (1,0)

Hence, $6x - 2y - 5 = 0$

5. $$f(x) = \begin{cases} \dfrac{a^{\sin x} - a^{\tan x}}{\tan x - \sin x} & \text{for } x > 0 \\ \dfrac{\ln(1+x+x^2) + \ln(1-x+x^2)}{\sec x - \cos x} & \text{for } x < 0 \end{cases}$$, if f is continuous at x = 0, find 'a'

now if $g(x) = \ln\left(2 - \frac{x}{a}\right)\cot(x-a)$ for $x \neq a, a \neq 0, a > 0$. If g is continuous at x=a then show that $g(e^{-1}) = -e$.

दिये है $f(x) = \begin{cases} \dfrac{a^{\sin x} - a^{\tan x}}{\tan x - \sin x} & \text{for } x > 0 \\ \dfrac{\ln(1+x+x^2) + \ln(1-x+x^2)}{\sec x - \cos x} & \text{for } x < 0 \end{cases}$ यदि फलन f, x = 0 पर सतत् है, तो 'a' का मान बताओ।

यदि फलन $g(x) = \ln\left(2 - \frac{x}{a}\right)\cot(x-a)$, $x \neq a, a \neq 0, a > 0$ तथा यदि g(x), x = a पर सतत् हो, तो सिद्ध करो कि $g(e^{-1}) = -e$.

Ans. $a = \dfrac{1}{e}$

Sol. $\text{R.H.L.} = \lim\limits_{x\to 0^+} \dfrac{a^{\sin x-\tan x}-1}{\sin x-\tan x}\cdot\left(\dfrac{a^{\tan x}}{-1}\right)$

$\text{R.H.L.} = -\ell na$

$$\text{L.H.L.}: \lim_{x\to 0^-} \frac{\ell n\left(1+x\left(x+1\right)\right)+\ell n\left\{1-x\left(1-x\right)\right\}\cos x}{1-\cos^2 x}$$

$$= \lim_{x\to 0^-} \frac{\ell n\left\{1+\left(x+x^2\right)\right\}+\ell n\left\{1+x^2-x\right\}}{x^2}\cdot\frac{x^2}{\left(\sin x\right)^2}\cdot\cos x$$

$$= \lim_{x\to 0} \frac{\left\{\left(x+x^2\right)-\dfrac{\left(x+x^2\right)^2}{2}+.....\right\}+\left\{x^2-x-\dfrac{\left(x^2-x\right)^2}{2}+.....\right\}}{x^2}$$

$$=1\frac{(1+0)^2}{2}+1\frac{(0-1)^2}{2}$$

$= 2 - 1 = 1$

f is continuous at $x = 0 \Rightarrow$ L.H.L. = R.H.L. = f(0)

$\Rightarrow 1 = -\ell na = f(0)$

$$\Rightarrow \frac{1}{a} = e \Rightarrow a = \frac{1}{e}$$

Now, $g\left(x\right) = \ell n\left(2-\dfrac{x}{a}\right)\cot\left(x-a\right), x \neq a$ is continuous at $x = a = \dfrac{1}{e}$

$$g\left(a\right) = \lim_{x\to a} g\left(x\right)$$

$$= \lim_{x\to a} \frac{\ell n\left[1+\left(1-\dfrac{x}{a}\right)\right]}{\left(1-\dfrac{x}{a}\right)}\cdot\frac{a-x}{a}\cdot\frac{\cos\left(x-a\right)}{\sin\left(x-a\right)}$$

$$g\left(a\right) = \frac{1}{a}\cdot(-1)(1) = -\frac{1}{a}$$

$a = \dfrac{1}{e}$ $\qquad g(e^{-1}) = -e$ $\qquad$ H.P.

6. Let $f(x) = \begin{cases} (\sin x + \cos x)^{\text{cosec } x} ; & \frac{-\pi}{2} < x < 0 \\ a ; & x = 0 \\ \frac{e^{\frac{1}{x}} + e^{\frac{2}{x}} + e^{\frac{3}{|x|}}}{a e^{\frac{2}{x}} + b e^{\frac{3}{|x|}}} ; & 0 < x < \frac{\pi}{2} \end{cases}$

If $f(x)$ is continuous at $x = 0$, find the value of $\left(a^2 + b^2\right)$.

माना $f(x) = \begin{cases} (\sin x + \cos x)^{\text{cosec } x} ; & \frac{-\pi}{2} < x < 0 \\ a ; & x = 0 \\ \frac{e^{\frac{1}{x}} + e^{\frac{2}{x}} + e^{\frac{3}{|x|}}}{a e^{\frac{2}{x}} + b e^{\frac{3}{|x|}}} ; & 0 < x < \frac{\pi}{2} \end{cases}$

यदि $f(x), x = 0$ पर सतत् है $\left(a^2 + b^2\right)$ का मान ज्ञात कीजिये।

Ans. $e^2 + e^{-2}$

Sol. Given f is continuous at $x = 0$

$f(0) = f(0-0) = f(0+0)$

$$f(0-0) = \lim_{x \to 0^-} (\sin x + \cos x)^{\text{cosec}x} \qquad (1^\infty)$$

$$= e^{\lim_{x \to 0^-} \frac{\sin x + \cos x - 1}{\sin x}}$$

$$= e^{\lim_{x \to 0} 1 - \frac{1 - \cos x}{x^2} \cdot \frac{x}{\sin x} \cdot x}$$

$$= e^1$$

$f(0) = f(0-0) \Rightarrow a = e$

$$f(0+0) = \lim_{x \to 0^+} \frac{e^{\frac{1}{x}} + e^{\frac{2}{x}} + e^{\frac{3}{x}}}{a e^{\frac{2}{x}} + b e^{\frac{3}{x}}}$$

put $x = \frac{1}{y}$

$$= \lim_{y\to\infty} \frac{e^{y} + e^{2y} + e^{3y}}{ae^{2y} + e^{3y}b}$$

$$f(0+0) = \lim_{y\to\infty} \frac{e^{-2y} + e^{-y} + 1}{ae^{-y} + b} = \frac{1}{b}$$

$$f(0+0) = f(0) \Rightarrow a = \frac{1}{b} \Rightarrow b = \frac{1}{e}$$

$$a^2 + b^2 = e^2 + \frac{1}{e^2}$$

7. Given $f(x) = \sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right)$; $r, n \in N$

$$g(x) = \underset{n\to\infty}{\text{Limit}} \frac{\ell n\left(f(x) + \tan\frac{x}{2^n}\right) - \left(f(x) + \tan\frac{x}{2^n}\right)^n \cdot \left[\sin\left(\tan\frac{x}{2}\right)\right]}{1 + \left(f(x) + \tan\frac{x}{2^n}\right)^n}$$

$= k$ for $x = \frac{\pi}{4}$ and the domain of $g(x)$ is $(0, \pi/2)$.

where [ ] denotes the greatest integer function.

Find the value of k, if possible, so that $g(x)$ is continuous at $x = \pi/4$. Also state the points of discontinuity of $g(x)$ in $(0, \pi/4)$, if any.

दिया गया है कि $f(x) = \sum_{r=1}^{n} \tan\left(\frac{x}{2^r}\right) \sec\left(\frac{x}{2^{r-1}}\right)$; $r, n \in N$

$$g(x) = \underset{n\to\infty}{\text{Limit}} \frac{\ell n\left(f(x) + \tan\frac{x}{2^n}\right) - \left(f(x) + \tan\frac{x}{2^n}\right)^n \cdot \left[\sin\left(\tan\frac{x}{2}\right)\right]}{1 + \left(f(x) + \tan\frac{x}{2^n}\right)^n}$$

$= k$, $x = \frac{\pi}{4}$ तथा $g(x)$ का प्रान्त $(0, \pi/2)$ है।

जहाँ [ ] महत्तम पूर्णांक फलन हो, तो k का मान ज्ञात करो। जबकि $g(x)$, $x = \pi/4$ पर सतत् हो तथा अन्तराल $(0, \pi/4)$ में वो बिन्दु ज्ञात करें जहाँ $g(x)$ असतत् है।

Ans. $k = 0$; $g(x) = \begin{cases} \ell n\ (\tan\ x) & \text{if} \quad 0 < x < \frac{\pi}{4} \\ 0 & \text{if} \quad \frac{\pi}{4} \le x < \frac{\pi}{2} \end{cases}$. Hence $g(x)$ is continuous everywhere.

Sol. $\because \tan A \cdot \sec 2A = \frac{\tan A}{\cos 2A} = \frac{\sin(2A - A)}{\cos A \cos 2A}$

$= \tan 2A - \tan A$

$A = \frac{x}{2^r}$

$$\Rightarrow \tan\left(\frac{2x}{2^r}\right)-\tan\left(\frac{x}{2^r}\right)\Rightarrow \tan\left(\frac{x}{2^r}\right)\sec\left(\frac{x}{2^{r-1}}\right)=\tan\left(\frac{x}{2^{r-1}}\right)-\tan\left(\frac{x}{2^r}\right)$$

$$=\tan\left(\frac{x}{2^{r-1}}\right)-\tan\left(\frac{x}{2^r}\right)$$

Now, $f(x)=\sum_{r=1}^{n}\left[\tan\left(\frac{x}{2^{r-1}}\right)-\tan\left(\frac{x}{2^r}\right)\right]$

$$f(x)=\tan x-\tan\left(\frac{x}{2^n}\right)$$

$$f(x)+\tan\left(\frac{x}{2^n}\right)=\tan x$$

then $g(x)=\lim_{n\to\infty}\dfrac{\ell n(\tan x)-(\tan x)^n\left[\sin\tan\frac{x}{2}\right]}{1+(\tan x)^n}$

$$g(x)=\begin{cases}\ell n(\tan x) & x<\frac{\pi}{4}\\ -\left[\sin\left(\tan\frac{x}{2}\right)\right] & x>\frac{\pi}{4}\end{cases}$$

$$g\left(\frac{\pi}{4}-h\right)=\lim_{h\to 0}\ell n\left(\tan\left(\frac{\pi}{4}-h\right)\right)=0$$

$$g\left(\frac{\pi}{4}+h\right)=\lim_{h\to 0}-\left[\sin\left(\tan\left(\frac{\pi}{8}-\frac{h}{2}\right)\right)\right]$$

$=0$

g(x) is continuous at $x=\frac{\pi}{4}\Rightarrow g\left(\frac{\pi}{4}\right)=k=0$

g(x) is continuous in $\left(0,\frac{\pi}{2}\right)$

8. Consider the function $g(x)=\begin{cases}\dfrac{1-a^x+xa^x\,\ell na}{a^x x^2} & \text{for } x<0\\ \dfrac{2^x a^x-x\ell n2-x\ell na-1}{x^2} & \text{for } x>0\end{cases}$ where $a>0$.

find the value of 'a' & 'g(0)' so that the function g(x) is continuous at $x=0$.

दिया हुआ फलन $g(x)=\begin{cases}\dfrac{1-a^x+xa^x\,\ell na}{a^x x^2} & \text{for } x<0\\ \dfrac{2^x a^x-x\ell n2-x\ell na-1}{x^2} & \text{for } x>0\end{cases}$. जहाँ $a>0$.

तो 'a' व 'g(0)' का मान ज्ञात करो जिससे g(x), $x=0$ पर सतत् हो जाये।

Ans. $a=\dfrac{1}{\sqrt{2}}$, $g(0)=\dfrac{(\ell n2)^2}{8}$

Sol. $g\left(0^-\right)=\lim\limits_{h\to0}\dfrac{1-a^{-h}+(-h)a^{-h}\ell na}{a^{-h}(-h)^2}$

$=\lim\limits_{h\to0}\dfrac{a^h-1-h\ell na}{h^2}$

$=\lim\limits_{h\to0}\dfrac{\left\{1+h\ell na+\dfrac{h^2(\ell na)^2}{2!}+....\right\}-h\ell na-1}{h^2}$

$=\dfrac{(\ell na)^2}{2!}$

$g\left(0^+\right)=\lim\limits_{h\to0}\dfrac{2^h a^h-h\ell n2-h\ell na-1}{h^2}$

$=\dfrac{1+h\ell n(2a)+\dfrac{h^2}{2!}(\ell n2a)^2+....-h\ell n2-h\ell na-1}{h^2}$

$g\left(0^+\right)=\dfrac{\ell n(2a)^2}{2!}$

$g\left(0^-\right)=g\left(0^+\right)\Rightarrow\dfrac{\ell n(2a)^2}{2!}=\dfrac{(\ell na)^2}{2!}$

$\Rightarrow\ell na=-\dfrac{1}{2}\ell n2\Rightarrow a=\dfrac{1}{\sqrt{2}}$

$$g(0)=g\left(0^{-}\right)=\frac{\left(\log\left(\frac{1}{\sqrt{2}}\right)\right)^{2}}{2}=\frac{1}{8}(\ell n2)^{2}$$

9. Let $f(0)=0$ and $f'(0)=1$. For a positive integer k, show that

$$\lim_{x\to0}\frac{1}{x}\left(f(x)+f\left(\frac{x}{2}\right)+.....f\left(\frac{x}{k}\right)\right)=1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k}.$$

यदि $f(0)=0$ तथा $f'(0)=1$ तो किसी धनात्मक पूर्णांक k के लिये सिद्ध करो कि

$$\lim_{x\to0}\frac{1}{x}\left(f(x)+f\left(\frac{x}{2}\right)+.....f\left(\frac{x}{k}\right)\right)=1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k}$$

Ans.

Sol. $\lim_{x\to0}\dfrac{f(x)+f\left(\frac{x}{2}\right)+.....f\left(\frac{x}{k}\right)}{x}$ $\left[\frac{0}{0}\right]$ $[f(0)=0]$

By L-Hospital Rule

$$=\lim_{x\to0}f'(0)+f'\left(\frac{x}{2}\right)\frac{1}{2}+.....+f'\left(\frac{x}{k}\right)\cdot\frac{1}{k}$$

$$=1+\frac{1}{2}+\frac{1}{3}+.....+\frac{1}{k} \qquad [f'(0)=1]$$

H.P.

10. For any real number x, let [x] denote the largest integer less than or equal to x. Let f be a real valued function defined on the interval $[-3,3]$ by $f(x)=\begin{cases}-x-[-x] & \text{if } [x] \text{ is even}\\ x-[x] & \text{if } [x] \text{ is odd}\end{cases}$.

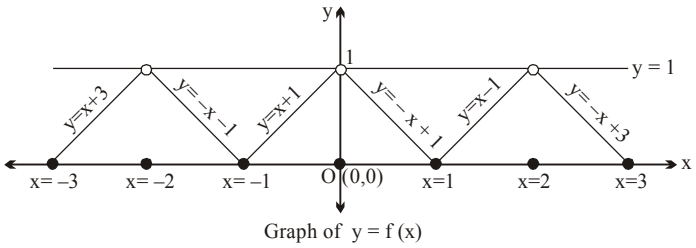
If L denotes the number of points of discontinuity and M denotes the number of points of non-derivability of f (x), then find (L + M).

किसी भी वास्तविक x के लिये माना [x] , x के महत्तम पूर्णांक फलन को दर्शाता है। यदि f एक वास्तविक मान का फलन जो अन्तराल $[-3,3]$ में इस प्रकार परिभाषित है $f(x)=\begin{cases}-x-[-x] & \text{यदि } [x] \text{ सम हो}\\ x-[x] & \text{यदि } [x] \text{ विषम हो}\end{cases}$

यदि L असतत् बिन्दुओं की संख्या तथा M उन बिन्दुओं की संख्या जहाँ $f(x)$ अवकलनीय नही है को दर्शाता है तो (L + M) का मान ज्ञात कीजिये।

Ans. 8

Sol.



Graph of y = f (x)

$$f(x)=\begin{cases}\{-x\} & \text{if } [x] \text{ is even}\\ \{x\} & \text{if } [x] \text{ is odd}\end{cases}$$

Cleary graph of $f(x)$ is discontinuous at $x=-2, 0, 2$ and non-derivable at $x=-2,-1, 0, 1, 2$

$\therefore$ $L = 3$ and $M= 5$

Hence $(L + M) = 3 + 5 = 8$

Note : f(x) is periodic with period = 2

11. $f(x)=\begin{cases}1-x & , \ (0\le x\le 1)\\ x+2 & , \ (1<x<2)\\ 4-x & , \ (2\le x\le 4)\end{cases}$ Discuss the continuity & differentiability of $y = f[f(x)]$ for $0 \le x \le 4$.

यदि $f(x)=\begin{cases}1-x & , \ (0\le x\le 1)\\ x+2 & , \ (1<x<2)\\ 4-x & , \ (2\le x\le 4)\end{cases}$ तो फलन $y = f[f(x)]$ की अन्तराल $0 \le x \le 4$ में सतत्ता तथा अवकलनीयता की जाँच करो।

Ans. f is discontinuous and not differentiable at x = 1, discontinuous and not differentiable at x = 2 & . cont.& diff.at all other points

Sol. $f(x)=\begin{cases}1-x & 0\le x\le 1\\ x+2 & 1<x<2\\ 4-x & 2\le x\le 4\end{cases}$

$y = f[f(x)]$ $\forall\ x \in [0,4]$

$f(1) = 0$ $f(1^+) = 3$ discontinuous, not differentiable at $x = 1$

$f(2) = 2$ $f(2^-) = 4$ discontinuous, not differentiable at $x = 2$

12. Consider the function, $f(x)=\begin{cases}x^2\left|\cos\dfrac{\pi}{2x}\right| & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$

(a) Show that $f'(0)$ exists and find its value

(b) Show that $f'(1/3)$ does not exist.

(c) For what values of x, $f'(x)$ fails to exist

दिया हुआ है कि $f(x)=\begin{cases}x^2\left|\cos\dfrac{\pi}{2x}\right| & \text{if } x\neq 0\\ 0 & \text{if } x=0\end{cases}$

(a) सिद्ध करों कि $f'(0)$ अस्तित्व में है और इसका मान भी बताओं

(b) सिद्ध करो कि $f'(1/3)$ अस्तित्व में नही है।

(c) x के किस मान के लिये $f'(x)$ अस्तित्व में नहीं होगा।

Ans. (a) $f'(0) = 0$, (b) $f'\left(\frac{1^-}{3}\right) = -\frac{\pi}{2}$ and $f'\left(\frac{1^+}{3}\right) = \frac{\pi}{2}$, (c) $x = \frac{1}{2n+1}$ $n \in I$

Sol. (a) L.H.D. $= \lim_{h\to 0} \frac{f(0-h)-f(0)}{-h}$

$$= \lim_{h\to 0} \frac{h^2 \cos\left(\frac{2\pi}{h}\right)}{h} = 0$$

$$\text{R.H.D.} = \lim_{h\to 0} \frac{f(0+h)-f(0)}{h} = \lim_{h\to 0} \frac{h^2 \cos\left(\frac{2\pi}{h}\right)}{h} = 0$$

hence, f is differentiable at $x = 0$ and $f'(0) = 0$

$$\text{(b) L.H.D.} = \lim_{h\to 0} \frac{f\left(\frac{1}{3}-h\right)-f\left(\frac{1}{3}\right)}{-h} = f'\left(\frac{1^-}{3}\right)$$

$$= \lim_{h\to 0} \frac{\left(\frac{1}{3}-h\right)^2 \left|\cos\frac{\pi}{2\left(\frac{1}{3}-h\right)}\right|}{-h}$$

$$= \lim_{h\to 0} \frac{(1-3h)^2}{9} \frac{\cos\left(\frac{3\pi}{2(1-3h)}\right)}{h}$$

$$= \lim_{h\to 0} \frac{-\sin\left(\frac{3\pi}{2}-\frac{3\pi}{2}\frac{1}{(1-3h)}\right)}{9h}$$

$$= \lim_{h\to 0} \frac{-\sin\left[\frac{3\pi}{2}\left\{\frac{+3h}{3h-1}\right\}\right]}{\frac{3\pi}{2}\left(\frac{3h}{3h-1}\right)} \cdot \frac{\frac{3\pi}{2}\left(\frac{3h}{3h-1}\right)}{9h}$$

$$= -\frac{\pi}{2}$$

$$f\left(\frac{1}{3}^{+}\right)=\lim_{h\to 0}\frac{f\left(\frac{1}{3}+h\right)-f\left(\frac{1}{3}\right)}{h}$$

$$=\lim_{h\to 0}\frac{\left(\frac{3h+1}{3}\right)^2\left|\cos\frac{3\pi}{2(3h+1)}\right|}{h}$$

$$=\lim_{h\to 0}\frac{\sin\left[\frac{3\pi}{2}-\frac{3\pi}{2(3h+1)}\right]}{\frac{3\pi}{2}\frac{3h}{3h+1}}\frac{3\pi}{2}\frac{3h}{3h+1}.\frac{1}{9h}$$

$$=\frac{\pi}{2}$$

$$f'\left(\frac{1}{3}\right)\to \text{not exist}\to f'\left(\frac{1}{3}^{-}\right)=-\frac{\pi}{2} \qquad f'\left(\frac{1}{3}^{+}\right)=\frac{\pi}{2}$$

(c) $\cos\frac{\pi}{2x}=0$ if $\frac{1}{x}=2n+1$

$$x=\frac{1}{2n+1},n\in I$$

$$\left|\cos\frac{\pi}{2x}\right|\begin{cases}+\text{ve at } x=\frac{1}{(2n+1)}^{+},\ x=\left(\frac{1}{2n+1}\right)^{-1} \text{ like } 2n+1=5\\ -\text{ve at } x=\frac{1}{2n+1}^{+}\ x=\left(\frac{1}{2n+1}\right)^{-} \text{ like } 2n+1=3\end{cases}$$

hence f is not differentiable at $x=\frac{1}{2n+1},n\in I$

13. Let f (x) be a differentiable function such that f ' (0) = 1, and the sequence $\{a_n\}$ is defined as $a_1=2$ and $a_n=\underset{x\to\infty}{\text{Lim}}\ x^2\left(f\left(\frac{a_{n-1}}{x}\right)-f(0)\right)^2$, $n\in N$. If the value of $\prod_{i=1}^{10}a_i=2^k$, find k.

माना कि f (x) एक अवकलनीय फलन है जहाँ f ' (0) = 1 तथा अनुक्रम $\{a_n\}$ इस प्रकार से परिभाषित है कि $a_1=2$ और $a_n=\underset{x\to\infty}{\text{Lim}}\ x^2\left(f\left(\frac{a_{n-1}}{x}\right)-f(0)\right)^2$, $n\in N$. यदि $\prod_{i=1}^{10}a_i=2^k$, तो k का मान बताईये।

Ans. 1023

Sol. $a_n = \lim_{x\to\infty} x^2\left(f\left(\frac{a_{n-1}}{x}\right)-f(0)\right)^2\Bigg]; n \in N$

put $t = \frac{1}{x}$

$= \lim_{t\to 0^+}\left[\frac{f(a_{n-1}t)-f(0)}{a_{n-1}t}\right]^2 \cdot (a_{n-1})^2$

$a_n = f'(0)\cdot(a_{n-1})^2 = (a_{n-1})^2$

$a_2 = a_1^2 = 2^2$

$a_3 = 2^{2^2}$ $\qquad\qquad$ $a_n = 2^{2^{n-1}}$

Now $2^k = \prod_{i=1}^{10} a_i = a_1\cdot a_2\cdot \ldots\ldots\ldots\cdot a_{10}$

$= 2^{2^0}2^{2^1}\ldots\ldots 2^{2^9}$

$= 2^{1\cdot(2^{10}-1)} = 2^{1023}$

$k = 1023$

14. Let $f(x) = \begin{bmatrix} 1+x & , \ 0\le x\le 2 \\ 3-x & , \ 2<x\le 3 \end{bmatrix}$. Determine the form of $g(x) = f[f(x)]$ & hence find the point of discontinuity of g, if any.

माना कि $f(x) = \begin{bmatrix} 1+x & , \ 0\le x\le 2 \\ 3-x & , \ 2<x\le 3 \end{bmatrix}$, तो $g(x) = f[f(x)]$ परिभाषित करे और यदि $g(x)$ कही पर असतत् हो, तो वह बिन्दु ज्ञात करें।

Ans. $g(x) = \begin{bmatrix} x+2 & 0\le x\le 1 \\ 2-x & 1<x\le 2 \\ 4-x & 2<x\le 3 \end{bmatrix}$, g is discontinuous at x = 1 & x=2

Sol. $f(x) = \begin{bmatrix} 1+x & , \ 0\le x\le 2 \\ 3-x & , \ 2<x\le 3 \end{bmatrix}$

$g(x) = fof(x) = \begin{bmatrix} 1+f(x) & \text{when} \quad 0\le f(x)\le 2 \\ 3-f(x) & \text{when} \quad 2<f(x)\le 3 \end{bmatrix} = \begin{cases} 1+(1+x) & \text{when } 0\le x\le 1 \\ 1+(3-x) & \text{when } 2\le x\le 3 \\ 3-(1+x) & \text{when } 1\le x\le 2 \end{cases}$

$g(1^-) = g(1) = 3 \neq g(1^+) = 1$ $\qquad\qquad$ $g(2^-) = g(2) = 0 \neq g(2^+) = 2$

Paragraph for Question nos. 15 to 17

For $0 < x < \frac{\pi}{2}$, let $f_1(x) = \sum_{r=1}^{n} \sec\left(x + \frac{r\pi}{6}\right) \sec\left(x + (r-1)\frac{\pi}{6}\right)$ and $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$

Also $f_2(x) + f_3(x) = 0$ and $f_4(x) = \begin{cases} (\text{exp.})\dfrac{e^{(e^{x+f_2(x)+\tan x}-1)}-1}{2\cdot(e^x-1)}; & x<0 \\ k_1 \quad ; & x=0 \\ \left(1+|f_2(x)|\right)^{\frac{k_2}{f_3(x)}} ; & x>0 \end{cases}$

$0 < x < \frac{\pi}{2}$ के लिये $f_1(x) = \sum_{r=1}^{n} \sec\left(x + \frac{r\pi}{6}\right) \sec\left(x + (r-1)\frac{\pi}{6}\right)$ और $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$

तथा $f_2(x) + f_3(x) = 0$ and $f_4(x) = \begin{cases} (\text{exp.})\dfrac{e^{(e^{x+f_2(x)+\tan x}-1)}-1}{2\cdot(e^x-1)}; & x<0 \\ k_1 \quad ; & x=0 \\ \left(1+|f_2(x)|\right)^{\frac{k_2}{f_3(x)}} ; & x>0 \end{cases}$

15. The value of k_1 and k_2, if $f_4(x)$ is continuous at $x = 0$ is
यदि $f_4(x)$, $x = 0$ पर सतत् हो, तो k_1 व k_2 का मान होगा

(A) $\frac{1}{\sqrt{e}}, 2$ (B) $e, 1$ (C) $\frac{e}{2}, 2$ (D) $\sqrt{e}, \frac{1}{2}$

Ans. D

Sol. $0 < x < \frac{\pi}{2}$

$$f_1(x) = \sum_{r=1}^{n} \frac{1}{\sin\frac{\pi}{6}} \frac{\sin\left[x + \frac{r\pi}{6} - \left(x + (r-1)\frac{\pi}{6}\right)\right]}{\cos\left(x + \frac{r\pi}{6}\right)\cos\left(x + (r-1)\frac{\pi}{6}\right)}$$

$$= 2\sum_{r=1}^{n} \tan\left(x + \frac{r\pi}{6}\right) - \tan\left(x + (r-1)\frac{\pi}{6}\right)$$

$$f_1(x) = 2\left[\tan\left(x + \frac{n\pi}{6}\right) - \tan x\right] \qquad \text{_____(1)}$$

Given $f_1(x) - 2f_2(x) = 2\tan\left(x + \frac{n\pi}{6}\right)$

$\Rightarrow$ from (1) $\Rightarrow f_2(x) = -\tan x$

Also given $f_3(x) = -f_2(x) = \tan x$

$$f_4(x) = \begin{cases} e^{\frac{e^{\left[e^x + f_2(x) + \tan x - 1\right]} - 1}{2\left(e^x - 1\right)}} & x < 0 \\ k_1 & x = 0 \\ \left(1 + \left|f_2(x)\right|\right)^{\frac{k_2}{f_3(x)}} & x > 0 \end{cases}$$

$$f_4(x) = \begin{cases} e^{\frac{e^{\left(e^x - 1\right)} - 1}{2\left(e^x - 1\right)}} & x < 0 \\ k_1 & x = 0 \\ \left(1 + \left|\tan x\right|\right)^{\frac{k_3}{\tan x}} & x > 0 \end{cases}$$

$f_4(x)$ continuous at $x = 0$

$f(0-0) = f(0+0) = f(0)$

Now

$$\Rightarrow f(0-0) = \lim_{x \to 0^-} e^{\frac{e^{e^x - 1} - 1}{2\left(e^x - 1\right)}} = e^{\frac{1}{2}}$$

$$f(0) = f(0-0) \Rightarrow e^{\frac{1}{2}} = k_1$$

$$\Rightarrow f(0+0) = \lim_{x \to 0^+} \left(1 + \left|\tan x\right|\right)^{\frac{k_2}{\tan x}} \qquad (1^\infty)$$

$$= e^{\lim_{x \to 0^+} \left|\tan x\right| \frac{k_2}{\tan x}} = e^{k_2}$$

$$f(0) = f(0+0) \Rightarrow e^{k_2} = e^{\frac{1}{2}} \Rightarrow k_2 = \frac{1}{2}$$

16. $y = f_3(x)$ is

(A) discontinuous and non derivable at $x = \frac{\pi}{4}$ and $x = \frac{\pi}{3}$

(B) neither continuous nor derivable at $x = \frac{2\pi}{5}$

(C) continuous and derivable in $\left(0, \frac{\pi}{2}\right)$

(D) continuous but not derivable at $x = \frac{2\pi}{5}$

फलन $y = f_3(x)$

(A) $x = \frac{\pi}{4}, x = \frac{\pi}{3}$ पर सतत् तथा अवकलनीय दोनों ही नही होगा

(B) $x = \frac{2\pi}{5}$ पर न तो सतत् न ही अवकलनीय होगा

(C) अन्तराल $\left(0, \frac{\pi}{2}\right)$ में सतत् तथा अवकलनीय दोनो ही होगा

(D) $x = \frac{2\pi}{5}$ पर सतत् होगा लेकिन अवकलनीय नहीं होगा।

Ans. C

Sol. $f_3(x) = \tan x$

continuous and derivable in $\left(0, \frac{\pi}{2}\right)$

17. For $n = 3$, the solution of equation $f_1(x) + 4 = 0$, in $\left(0, \frac{\pi}{2}\right)$ is

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) non existent

$n = 3$ के लिये समीकरण $f_1(x) + 4 = 0$ का $\left(0, \frac{\pi}{2}\right)$ के अन्तराल में हल होगा

(A) $\frac{\pi}{6}$ (B) $\frac{\pi}{4}$ (C) $\frac{\pi}{3}$ (D) अस्तित्व में नहीं है

Ans. B

Sol. For $n = 3$, $f_1(x) + 4 = 0$

$$2\left[\tan\left(x + \frac{3\pi}{6}\right) - \tan x\right] + 4 = 0$$

$-\cot x - \tan x = -2$

$\Rightarrow \tan x + \cot x = 2 \Rightarrow \tan x = 1$

$x = \frac{\pi}{4}$

Paragraph for Question nos. 18 to 20

Consider a function, $f(x) = \frac{2}{3} ln^{3/2}(\sin x + \cos x)$ for $x \in \left(0, \frac{\pi}{2}\right)$

फलन $f(x) = \frac{2}{3} ln^{3/2}(\sin x + \cos x)$ अन्तराल $x \in \left(0, \frac{\pi}{2}\right)$ में दिया गया है।

18. The value of $\lim\limits_{x \to \frac{\pi}{2}} \frac{f(x)}{\left(\frac{\pi}{2} - x\right)^{3/2}}$ is

(A) is $\frac{2}{3}$ (B) is 1 (C) is $\frac{2}{3}(2)^{3/4}$ (D) non existent

$\lim\limits_{x \to \frac{\pi}{2}} \frac{f(x)}{\left(\frac{\pi}{2} - x\right)^{3/2}}$ का मान होगा

(A) $\frac{2}{3}$ (B) 1 (C) $\frac{2}{3}(2)^{3/4}$ (D) अस्तित्व में नहीं है।

Ans. A

Sol. $L = \lim\limits_{x \to \frac{\pi}{2}} \frac{f(x)}{\left(\frac{\pi}{2} - x\right)^{\frac{3}{2}}}$

Put $\frac{\pi}{2} - x = t$

$$L = \lim_{t \to 0} \frac{2}{3} \frac{\left[\ell n\left(\sin t + \cos t\right)\right]^{\frac{3}{2}}}{t^{\frac{3}{2}}}$$

$$= \frac{2}{3} \lim_{t \to 0} \left[\frac{\ell n\left(1 + \tan t\right) - \ell n \cos t}{t}\right]^{\frac{3}{2}}$$

$$= \frac{2}{3} \lim_{t \to 0} \left[\frac{\ell n\left(1 + \tan t\right)}{\tan t} \cdot \frac{\tan t}{t} - \frac{\ell n \cos t}{t}\right]^{\frac{3}{2}}$$

$$= \frac{2}{3}\lim_{t\to 0}\left[1-\frac{\ell n\left[1-(1-\cos t)\right]}{(1-\cos t)}\frac{1-\cos t}{t^2}\cdot t\right]$$

$$= \frac{2}{3}[1-0] = \frac{2}{3}$$

19. The function g (x) is defined as $g(x) = \begin{cases} \left(\frac{3}{2}f(x)\right)^{2/3}; & x \in \left(0, \frac{\pi}{2}\right) \\ \frac{e^{\left(\frac{3}{2}f\left(x-\frac{\pi}{2}\right)\right)^{2/3}}}{\sqrt{2}}; & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$. Then g (x)

(A) is continuous at $x = \frac{\pi}{2}$ for $g\left(\frac{\pi}{2}\right) = 0$

(B) has a removable discontinuity at $x = \frac{\pi}{2}$

(C) g (x) is discontinuous at $x = \frac{\pi}{2}$ and jump of discontinuity is equals to $\sqrt{2}$

(D) has a non-removable discontinuity at $x = \frac{\pi}{2}$

दिया है कि $g(x) = \begin{cases} \left(\frac{3}{2}f(x)\right)^{2/3}; & x \in \left(0, \frac{\pi}{2}\right) \\ \frac{e^{\left(\frac{3}{2}f\left(x-\frac{\pi}{2}\right)\right)^{2/3}}}{\sqrt{2}}; & x \in \left(\frac{\pi}{2}, \pi\right) \end{cases}$ तो फलन g (x) के बारे में कौन सा विकल्प सही है।

(A) यदि $g\left(\frac{\pi}{2}\right) = 0$ तो फलन g(x), $x = \frac{\pi}{2}$ पर सतत् है।

(B) g(x) का $x = \frac{\pi}{2}$ पर विस्थापनीय असतत्ता है।

(C) g (x), $x = \frac{\pi}{2}$ पर असतत् है तथा इसके असतत्ता का उछाल (jump) $\sqrt{2}$ है।

(D) फलन की $x = \frac{\pi}{2}$ पर अविस्थापनीय असतत्ता है।

Ans. D

Sol.
$$g(x)=\begin{cases}\left(\frac{3}{2}f(x)\right)^{\frac{2}{3}} & x\in\left(0,\frac{\pi}{2}\right)\\ \frac{e^{\left(\frac{3}{2}f\left(x-\frac{\pi}{2}\right)\right)^{\frac{2}{3}}}}{\sqrt{2}} & x\in\left(\frac{\pi}{2},\pi\right)\end{cases}$$

$$=\begin{cases}\ell n|\sin x+\cos x| & x\in\left(0,\frac{\pi}{2}\right)\\ \frac{e^{\ell n|\sin x+\cos x|}}{\sqrt{2}} & x\in\left(\frac{\pi}{2},\pi\right)\end{cases}$$

$$=\begin{cases}\ell n|\sin x+\cos x| & \text{as } x\in\left(0,\frac{\pi}{2}\right)\\ \frac{|\sin x+\cos x|}{\sqrt{2}} & \text{as } \frac{\pi}{2}-x\in\left(0,\frac{\pi}{2}\right)\end{cases}$$

$$g\left(\frac{\pi}{2}-0\right)=0\neq g\left(\frac{\pi}{2}+0\right)=\frac{1}{\sqrt{2}}$$

g discontinuous and has non removable discontinuity.

20. The range of g (x) is :
g (x) का परिसर होगा :

(A) (0, 1] (B) $\left(0,\frac{1}{2}\ell n 2\right]$

(C) $\left(\frac{1}{2},1\right]$ (D) $\left(0,\frac{1}{2}\ell n 2\right]\cup\left(\frac{1}{\sqrt{2}},1\right]$

Ans. D

Sol. $\because \sin x+\cos x=\sqrt{2}\sin\left(\frac{\pi}{4}+x\right)$

$$x\in\left(0,\frac{\pi}{2}\right)\Rightarrow x+\frac{\pi}{4}\in\left(\frac{\pi}{4},\frac{3\pi}{4}\right)$$

$$\Rightarrow \sin\left(x+\frac{\pi}{4}\right)\in\left(\frac{1}{\sqrt{2}},1\right]$$

$\sin x + \cos x = \sqrt{2}\sin\left(\frac{\pi}{4} + x\right) \in \left(1, \sqrt{2}\right]$

$\ell n\left(\sin x + \cos x\right) \in \left(0, \ell n\sqrt{2}\right)$

$x \in \left(\frac{\pi}{2}, \pi\right) \Rightarrow x + \frac{\pi}{4} \in \left(\frac{3\pi}{4}, \frac{5\pi}{4}\right)$

$\left|\sin\left(x + \frac{\pi}{4}\right)\right| \in \left(\frac{1}{\sqrt{2}}, 1\right]$

$\left|\sin x + \cos x\right| = \left|\sqrt{2}\sin\left(x + \frac{\pi}{4}\right)\right| \in \left(1, \sqrt{2}\right]$

$\Rightarrow \frac{\left|\sin x + \cos x\right|}{\sqrt{2}} \in \left(\frac{1}{\sqrt{2}}, 1\right]$

Hence, range of $g(x) \equiv \left(0, \frac{1}{2}\ell n2\right] \cup \left(\frac{1}{\sqrt{2}}, 1\right]$

MATCH THE COLUMN

21. In the following, [ ] and { } are greatest integer function and fractional part function respectively. Match the functions in column-I with the properties in column-II.

Column-I	**Column-II**
(A) $f_1(x) = \left[\frac{4x}{\pi}\right]$	(P) Discontinuous at more than 3 points but less than 6 points in [–2, 2]
(B) $f_2(x) = \cos^{-1}\left(\text{sgn}\left(\cos\frac{2x-1}{2}\pi\right)\right)$	(Q) Non derivable at more than 2 points but atmost 5 points in [–2, 2]
(C) $f_3(x) = \max.(\{x+1\}, \{5-x\})$	(R) Range contains atleast one integer but not more than seven and no irrational value in [–2, 2]
(D) $f_4(x) = \sqrt{x^2} + [x]^2$	(S) many one but not even function in [–2, 2]
	(T) neither odd nor periodic in [–2, 2]

यदि [ ] महत्तम पूर्णांक फलन तथा { } भिन्नांक फलन हो, तो

स्तम्भ-I	**स्तम्भ-II**
(A) $f_1(x)=\left[\frac{4x}{\pi}\right]$	(P) [– 2, 2] के अन्तराल में 3 से ज्यादा तथा 6 से कम बिन्दु पर असतत् होगा।
(B) $f_2(x)=\cos^{-1}\left(\text{sgn}\left(\cos\frac{2x-1}{2}\pi\right)\right)$	(Q) [– 2, 2] के अन्तराल में 2 से अधिक परन्तु अधिकतम 5 बिन्दुओं पर अवकलनीय नही होगा।
(C) $f_3(x)=$ अधिकतम$(\{x+1\},\{5-x\})$	(R) परिसर में [– 2, 2] अन्तराल में कम से कम एक व अधिक से अधिक 7 पूर्णांक होगे तथा इस अन्तराल की कोई भी अपरिमेय संख्या नहीं होगी
(D) $f_4(x)=\sqrt{x^2}+[x]^2$	(S) अन्तराल [–2, 2] में बहुएकैकी लेकिन समफलन नही होगा (T) अन्तराल [–2, 2] में न तो विषम तथा न ही आवर्ती होगा।

Ans. $A\to P,Q,R,S,T;\ B\to P,Q,S,T;\ C\to P,T;\ D\to P,Q,S,T$

Sol. (A) $f_1(x)=\left[\frac{4x}{\pi}\right]$ $\quad x\in[-2,2]$

$$\frac{4x}{\pi}\in\left[-\frac{8}{\pi},\frac{8}{\pi}\right]$$

$f_1(t)=[t]$ $\quad t\in[-2.54,2.54]$

discontinuous $-2,-1,0,1,2$

Nondis $-2,-1,0,1,2$

$A\to P,\ Q,\ R,\ S,\ T$

(B) $f_2(x)=\cos^{-1}\left(\text{sgn}\left(\frac{\cos 2x-1}{2}\pi\right)\right)$

$x\in[-2,2]$ $\quad 2x\in[-4,4]$

$2x-1\in[-5,3]$

$$\frac{2x-1}{2}\pi\in\left[-\frac{5\pi}{2},\frac{3\pi}{2}\right]$$

$$t\in\left[-\frac{5\pi}{2},\frac{3\pi}{2}\right]$$

$\cos t\in[-1,1]$

$\text{sgn}\cos t=-1,0,1$

$$\cos^{-1}\text{sgn}\cos t=\left\{\pi,\frac{\pi}{2},0\right\}$$

$\left(-\frac{5\pi}{2},\frac{3\pi}{2}\right)\to \text{sgn}(\cos t)\to 1$

$-\frac{5\pi}{2},\frac{3\pi}{2}\to 0$

$\left(-\frac{3\pi}{2},-\frac{\pi}{2}\right)\to -1$

$-\frac{\pi}{2}\to 0$

$\left(-\frac{\pi}{2},\frac{\pi}{2}\right)\to 1$

$\frac{\pi}{2}\to 0$

$\left(\frac{\pi}{2},\frac{3\pi}{2}\right)\to -1$

$\frac{3\pi}{2}\to 0$

discontinuous at $-\frac{5\pi}{2},-\frac{3\pi}{2},-\frac{\pi}{2},\frac{\pi}{2},\frac{3\pi}{2}$ five points

B $\to$ P, Q, S, T

(C) $f_3(x)=\max(\{x+1\},\{5-x\})$

$5\to$ points discontinuous –2, –1, 0, 1, 2

$9\to$ pints N.D. $\left[\begin{matrix}-2,-1,0,1,2\\ -\frac{3}{2},-\frac{1}{2},\frac{1}{2},\frac{3}{2}\end{matrix}\right.$

C $\to$ P, T

(D) $f_4(x)=\sqrt{x^2}+[x]^2$

$=|x|+[x]^2$

$$f_4(x)=\begin{cases}-x+4 & x\in[-2,-1)\\ -x+1 & x\in[-1,0)\\ x & x\in[0,1)\\ x+1 & x\in[1,2)\\ 6 & x=2\end{cases}$$

discontinuous at x = –1,0,1,2

N.D. = –1,0,1,2

D → P, Q, S, T

22.

	Column-I		**Column-II**
(A)	$\lim_{x\to\infty}\left(e^{\sqrt{x^4+1}}-e^{(x^2+1)}\right)$ is	(P)	e
(B)	For $a>0$ let $f(x)=\begin{cases}\frac{a^x+a^{-x}-2}{x^2} & \text{if } x>0\\ 3\ln(a-x)-2, & \text{if } x\le 0\end{cases}$	(Q)	e^2
	If f is continuous at x = 0 then 'a' equals	(R)	1/e
(C)	Let $L=\lim_{x\to a}\frac{x^x-a^a}{x-a}$ and $M=\lim_{x\to a}\frac{x^x-a^x}{x-a}$ $(a>0)$.		
	If L = 2M then the value of 'a' is equal to	(S)	non existent

	स्तम्भ-I		**स्तम्भ-II**
(A)	$\lim_{x\to\infty}\left(e^{\sqrt{x^4+1}}-e^{(x^2+1)}\right)$ is	(P)	e
(B)	$a>0$ के लिए माना $f(x)=\begin{cases}\frac{a^x+a^{-x}-2}{x^2} & \text{if } x>0\\ 3\ln(a-x)-2, & \text{if } x\le 0\end{cases}$	(Q)	e^2
	यदि फलन f, x = 0 पर सतत् है तो 'a' का मान होगा	(R)	1/e
(C)	माना $L=\lim_{x\to a}\frac{x^x-a^a}{x-a}$ तथा $M=\lim_{x\to a}\frac{x^x-a^x}{x-a}$ $(a>0)$.		
	यदि L = 2M, तब 'a' का मान होगा	(S)	अस्तित्व विहीन

Ans. A → S; B → P,Q; C → P

Sol. **(A)** $l=\lim_{x\to\infty}e^{x^2+1}\left[e^{\sqrt{x^4+1}-(x^2+1)}-1\right]$

consider $\lim_{x\to\infty}\left[\sqrt{x^4+1}-(x^2+1)\right]=\lim_{x\to\infty}\frac{x^4+1-(x^4+1+2x^2)}{\sqrt{x^4+1}+(x^2+1)}$

$$=\lim_{x\to\infty}\frac{-2x^2}{x^2\left(\sqrt{1+\left(1/x^4\right)}+1+\left(1/x^2\right)\right)}=-1$$

now as $x\to\infty$

$$\sqrt{x^2+1}-(x^2+1)\to -1$$

$$\infty \times \left(\frac{1}{e}-1\right) \to -\infty \quad \text{and} \quad \text{hence limit does not exist}$$

(B) $f(0^+) = \underset{h\to 0}{\text{Lim}} \dfrac{a^{2h}-2a^h+1}{h^2} = \underset{h\to 0}{\text{Lim}} \left(\dfrac{a^h-1}{h}\right)^2 = l\text{n}^2 a$

$f(0^-) = \underset{h\to 0}{\text{Lim}} \, 3\,l\text{n}(a+h)-2 = 3 \ln a - 2 = f(0)$

for continuous, $l\text{n}^2 a = 3\, l\text{n}\, a - 2 \quad \Rightarrow \quad l\text{n}^2 a - 3\, l\text{n}\, a + 2 = 0 \quad \Rightarrow \quad (l\text{n}\, a - 2)(l\text{n}\, a - 1) = 0$

$a = e^2$ or $a = e \quad \Rightarrow$ **(P), (Q)**

(C) $L = a^a\, l\text{n}\, ae$

$M = a^a$

$\therefore \quad a^a\, l\text{n}\, ae = 2a^a$

$\therefore \quad l\text{n}\, ae = 2$

$ae = e^2 \quad \Rightarrow \quad \boxed{a = e}$

23. Match the following column :

Column -I	**Column -II**
(A) $\lim\limits_{x\to\infty}\left(\dfrac{x^2+2x-1}{2x^2-3x-2}\right)^{\frac{2x+1}{2x-1}} =$	(P) $\dfrac{1}{2}$
(B) $\lim\limits_{x\to 0}\dfrac{\log_{\sec x/2}\cos x}{\log_{\sec x}\cos\frac{x}{2}} =$	(Q) 2
(C) Let f(x) = max. (cos x, x, 2x – 1) where $x \ge 0$ then number of points of non-differentiability of f(x) is	(R) 5
(D) If $f(x) = [2 + 3 \sin x]$, $0 < x < \pi$ then number of points at which the function is discontinuous, is	(S) 16

निम्न कॉलम को सुमेलित कीजिए :

Column -I	**Column -II**
(A) $\lim\limits_{x\to\infty}\left(\dfrac{x^2+2x-1}{2x^2-3x-2}\right)^{\frac{2x+1}{2x-1}} =$	(P) $\dfrac{1}{2}$
(B) $\lim\limits_{x\to 0}\dfrac{\log_{\sec x/2}\cos x}{\log_{\sec x}\cos\frac{x}{2}} =$	(Q) 2
(C) माना f(x) = max. (cos x, x, 2x – 1) जहां $x \ge 0$ है तो f(x) जहां पर अवकलनीय नहीं है, के बिन्दुओं की संख्या होगी	(R) 5
(D) यदि $f(x) = [2 + 3 \sin x]$, $0 < x < \pi$ है तो बिन्दुओं की संख्या जहां	(S) 16

पर फलन असतत है, होगी

Ans. $A \to P,\ B \to S,\ C \to Q,\ D \to R$

Sol. (A) $\lim\limits_{x\to\infty}\left(\dfrac{x^2+2x-1}{2x^2-3x-2}\right)^{\frac{2x+1}{2x-1}} = \dfrac{1}{2}$

$A \to P$

(B) $\lim\limits_{x\to 0}\dfrac{\ell n\cos x\cdot \ell n\sec x}{\ell n\sec\dfrac{x}{2}\cdot\ell n\cos\dfrac{x}{2}}$

$$= \lim_{x\to 0}\left(\frac{\ell n\cos x}{\ell n\cos\frac{x}{2}}\right)^2$$

L' Hospital Rule

$$= \lim_{x\to 0}\left(\frac{\tan x}{\tan\frac{x}{2}}\cdot 2\right)^2 = \lim_{x\to 0}\left(\frac{\tan x}{x}\cdot\frac{\frac{x}{2}}{\tan\frac{x}{2}}\cdot 4\right)^2 = 16$$

$B \to S$